Probability Theory - Problem Book

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0.1 Introduction

In this document, I solve problems and theoretical exercises in probability thoery with a focus on mastery. Problems are selected by me from the chapters I had studied in my probability theory course from Ross' textbook: *A First Course in Probabiliy Theory.* Additional self-selected problems from the textbook *A Practical Introduction to Quantitative Finance*, or the "Green Book", are included.

Many exercises are labeled based on their source or descriptions of their main results. Exercises selected from Ross' textbook are labeled by number and whether they are problems or theoretical exercises as in the following examples:

- Ross' TE 1.1: Theoretical Exercise 1 in Chapter 1 in Ross' textbook.
- Ross' P 1.1: Problem 1 in Chapter 1 in Ross' textbook.
- Ross' Self-test 1.1: problem or exercise 1 in the Self-test Problems and Exercises section in Chapter 1 in Ross' textbook.

1 Combinatorial Analysis

Exercise 1.1 (Ross' TE 1.1: The generalized basic counting principle or the generalized multiplication rule)**.** *If there are n experiments to be performed such that for each positive integer $1 \leq i \leq n$, for each possible outcome of the experiments $1, 2, \ldots, i-1$, experiment i can result in any of m_i outcomes, then there are $\prod_{i=1}^{n} m_i$ possible outcomes of experiments $1, 2, \ldots, n$.*

Solution. We prove the generalized basic counting principle by induction on n.

Base case: If $n = 2$, this is the simple basic counting principle.

Inductive hypothesis (IH): Assume for some positive integer $n \geq 2$ that the generalized basic counting principle holds.

Inductive step: We prove the generalized basic counting principle for $n+1$. Consider the $(n+1)$-th experiment. For each of its m_{n+1} outcomes, there are, by IH, $\prod_{i=1}^{n} m_i$ possible outcomes of experiments $1, 2, \ldots, n$. By the basic counting principle, there are $m_{n+1} \prod_{i=1}^{n} m_i$ possible outcomes for experiments $1, 2, \ldots, n, n+1$

□

Exercise 1.2 (Ross' TE 1.2)**.** *Two experiments are to be performed. The first can result in any one of m possible outcomes. If the first experiment results in outcome i, then the second experiment can result in any of n_i possible outcomes, $i = 1, 2, \ldots, m$. What is the the number of possible outcomes of the two experiments.*

Solution. The first experiment either results in outcome 1, or results in outcome 2, ..., or results in outcome m. These are disjoint cases, and they are exhaustive. By the addition rule, there are $n_1 + n_2 + \cdots + n_m$ outcomes of the two experiments. □

Exercise 1.3 (Ross' TE 1.4)**.** *There are $\binom{n}{r}$ different linear arrangements of n balls of which r are black and $n-r$ are white. Give a combinatorial explanation of this fact.*

Solution. In every linear arrangement, there are n positions. Of the n positions, choose r positions, fill them with black balls, and fill the remaining $n-r$ positions with white balls. (Linear arrangements correspond to functions $f : [n] \to \{0, 1\}$ where 0 represents white and 1 represents black. There are $\binom{n}{r}$ distinct r-subsets A of $[n]$. Letting $f^{-1}(\{1\}) = A$ forces $f^{-1}(\{0\}) = [n] - A$ and defines f completely.) □

Exercise 1.4 (Ross' TE 1.5)**.** *Determine the number of vectors $(x_1, \ldots, x_n)$ such that each x_i is either 0 or 1 and*

$$\sum_{i=1}^{n} x_i \geq k$$

Solution. For each positive integer m such that $k \leq m \leq n$, the number of vectors such that $\sum_{i=1}^{n} x_i = m$ is the number of linear arrangements of n balls of which m are black and $n-m$ are white. Since each x_i is either 0 or 1, then at most, $\sum_{i=1}^{n} x_i = n$. Then, the number of ways such that $\sum_{i=1}^{n} x_i \geq k$ is the number of ways such that $\sum_{i=1}^{n} x_i = k, k+1, \ldots,$ or n. By Exercise (1.3), it follows that there are $\sum_{m=k}^{n} \binom{n}{m}$ ways. □

Exercise 1.5 (Ross' TE 1.8)**.** *Prove that*

$$\binom{n+m}{r} = \sum_{k=0}^{r} \binom{n}{k} \binom{m}{r-k}$$

Solution. Let A be a set of size $n+m$. Write $A = B \cup C$ as a union of two disjoint sets B of size n and C of size m. Then, Each r-subset D of A either has 0 elements in B and r elements in C, or has 1 element in B and $r-1$ elements in C, ..., or has r elements in B and 0 elements in C. These are disjoint, exhaustive cases. D has k elements in B if and only if D is a union of two disjoint sets X of size k contained in B and Y of size $r-k$ contained in C. If D has k elements in B, then there are $\binom{n}{k}$ ways to choose X and, for any X, $\binom{m}{r-k}$ ways to choose Y; by the basic principle of counting, there are $\binom{n}{k}\binom{m}{r-k}$ ways to choose D. Therefore, in total, by the addition rule, there are $\sum_{k=0}^{r} \binom{n}{k}\binom{m}{r-k}$ ways to choose D. □

Exercise 1.6 (Ross' TE 1.9)**.** *Use Theoretical Exercise 8 to prove that*

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2$$

.

Solution. By Exercise (1.5),

$$\binom{2n}{n} = \binom{n+n}{n} = \sum_{k=0}^{n} \binom{n}{k}\binom{n}{n-k}.$$

The result follows by an identity, which is derived as follows:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n!}{(n-k)!k!} = \frac{n!}{(n-k)!(n-(n-k))!} = \binom{n}{n-k}$$

□

Exercise 1.7. *Prove Fermat's combinatorial identity:*

$$\binom{n}{k} = \sum_{i=k}^{n} \binom{i-1}{k-1}, \quad n \geq k.$$

Give a combinatorial argument to establish this identity.

Solution. By Exercise (1.2), interpret $\binom{n}{k}$ as the number of linear arrangements of n balls of which k are black and $n-k$ are white. Let i be the position of the last black ball. Since there are k black balls and n balls in total, then $k \leq i \leq n$. Then, of the $i-1$ balls that are positioned before the k-th black ball, exactly $k-1$ of them must be black. There are $\binom{i-1}{k-1}$ possible ways to arrange the balls before the k-th black ball. The remaining balls must of course be white. Therefore, there are $\sum_{i=k}^{n} \binom{i-1}{k-1}$ linear arrangements of n balls of which k are black and $n-k$ are white, hence the identity holds. □

Exercise 1.8 (Ross' TE 1.12)**.** *Consider the following combinatorial identity:*

$$\sum_{k=1}^{n} k\binom{n}{k} = n \cdot 2^{n-1}.$$

(a) *Present a combinatorial argument for this identity by considering a set of n people and determining, in two ways, the number of possible selections of a committee of any size and a chairperson for the committee.*

(b) *Verify the following identity for $n = 1, 2, 3, 4, 5$:*

$$\sum_{k=1}^{n} \binom{n}{k} k^2 = 2^{n-2} n(n+1)$$

Solution. Okay.

(a) On the LHS, choose the size of the committee k; then, with k fixed, of n people, choose a committee of size k; then, of the k committee members, make one of them a chairperson. On the RHS, of n people, choose the chairperson; then, for each person of the remaining $n-1$ people, determine whether that person is a member of the committee. These are two ways of counting the same set, hence the identity holds.

(b) (1)

$$LHS = \binom{1}{1} 1^2 = 1$$
$$RHS = 2^{1-2}(1)(1+1) = (1/2)(2) = 1.$$

(2)

$$LHS = 2(1^2) + 1(2^2) = 6,$$
$$RHS = 2^0(2)(3) = 6.$$

(3)

$$LHS = 3(1^2) + 3(2^2) + 1(3^2) = 3 + 12 + 9 = 24,$$
$$RHS = 2(3)(4) = 24.$$

(4)

$$\begin{aligned} LHS &= 4(1^2) + 6(2^2) + 4(3^2) + 1(4^2) = 4 + 24 + 36 + 16 = 80 \\ RHS &= 2^2(4)(5) = 80. \end{aligned}$$

(5)

$$\begin{aligned} LHS &= 5(1^2) + 10(2^2) + 10(3^2) + 5(4^2) + 1(5^2) = 5 + 40 + 90 + 80 + 25 = 240, \\ RHS &= 2^3(5)(6) = 240. \end{aligned}$$

□

Exercise 1.9 (Ross' TE 1.17)**.** *Present a combinatorial explanation of why*

$$\binom{n}{r} = \binom{n}{r, n-r}.$$

Solution. On the LHS, we are counting the number of ways to choose r distinct objects out of n distinct objects. On the RHS, we are counting the number of ways to order n balls of which r are red and $n-r$ are blue, and by Exercise (1.3), this number counting linear arrangements equals $\binom{n}{r}$, which is the RHS. □

Exercise 1.10. *Define the multinomial coefficient*

$$\binom{n}{n_1, n_2, \ldots, n_r}$$

as the number of ways to permute n balls n_i of which are of color i for distinct colors $i = 1, 2, \ldots, r$. Use the generalized product rule to show that

$$\binom{n}{n_1, n_2, \ldots, n_r} = \frac{n!}{n_1! n_2! \ldots n_r!}.$$

Proof. The multinomial coefficient can be computed as follows. Of n available positions, choose n_1 positions for balls of color 1. Then, of $n - n_1$ available positions, choose n_2 for balls of color 2, and so forth until you choose positions for all balls.

Thus, we arrive at a nice algebraic cancelation:

$$\begin{aligned} &\binom{n}{n_1, n_2, \ldots, n_r} \\ &= \binom{n}{n_1}\binom{n-n_1}{n_2} \cdots \binom{n_{r-1}+n_r}{n_{r-1}}\binom{n_r}{n_r} \\ &= \frac{n!}{n_1!(n-n_1)!}\frac{(n-n_1)!}{n_2!(n-n_1-n_2)!} \cdots \frac{(n_{r-1}+n_r)!}{n_{r-1}!n_r!}\frac{n_r!}{n_r!} \\ &= \frac{n!}{n_1!\cancel{(n-n_1)!}}\frac{\cancel{(n-n_1)!}}{n_2!\cancel{(n-n_1-n_2)!}} \cdots \frac{\cancel{(n-n_1-n_2)!}}{\cdots} \cdots \frac{\cdots}{n_{r-2}!\cancel{(n_{r-1}+n_r)!}}\frac{\cancel{(n_{r-1}+n_r)!}}{n_{r-1}!\cancel{n_r!}}\frac{\cancel{n_r!}}{n_r!} \\ &= \frac{n!}{n_1!n_2! \cdots n_{r-1}!n_r!} \end{aligned}$$

□

Exercise 1.11 (Ross' TE 1.19)**.** *Prove the multinomial theorem.*

Solution. But what is the multinomial theorem?

My guess, a generalization of the binomial theorem as follows: $(x_1 + x_2 + \cdots + x_m)^n = \sum$

From Ross':

$$(x_1 + x_2 + \cdots + x_r)^n = \sum_{\substack{(n_1, n_2, \ldots, n_r): \\ n_1 + n_2 + \cdots + n_r = n}} \binom{n}{n_1, n_2, \ldots, n_r} x_1^{n_1} x_2^{n_2} \cdots x_r^{n_r}$$

Well, write

$$(x_1 + \cdots + x_r)^n = \underbrace{(x_1 + \cdots + x_r)(x_1 + \cdots + x_r) \cdots (x_1 + \cdots + x_r)}_{\text{n times}}.$$

Note that each term in the expansion is a product of one term from each of the r factors $(x_1 + \cdots + x_r)$. Then, for every r-tuple $(n_1, , \ldots, n_r)$ such that $n_1 + \cdots + n_r = n$, the coefficient of the term $x_1^{n_1} x_2^{x_2} \cdots x_r^{n_r}$ is the number of permutations of $\underbrace{x_1 x_1 \cdots x_1}_{n_1 \text{ times}} \underbrace{x_2 x_2 \cdots x_2}_{n_2 \text{ times}} \cdots \underbrace{x_r x_r \cdots x_r}_{n_r \text{ times}}$, which is $\binom{n}{n_1, n_2, \ldots, n_r}$.

□

1.1 Enrichment: Set-theoretic Combinatorial Definitions

Definition 1.1 (coefficients as functions)**.** *The following set of functions*

$$\{f : [k] \to [n] \mid f \text{ is injective}\}$$

is called the set of <u>*k-permutations of* $[n]$</u>*. Its size is the falling factorial* $(n)_k = n \cdot (n-1) \cdots (n-k+1)$.

The following <u>*set of functions*</u>

$$\{f : [n] \to \{0, 1\} \mid \left|f^{-1}(\{1\})\right| = k\}$$

is called the set of <u>*k-combinations of* $[n]$</u>*. Its size is the binomial coefficient* $\binom{n}{k}$.

Let $n = n_1 + n_2 + \cdots + n_r$ *where* $n, r, n_1, n_2, \ldots, n_r$ *are positive integers. The size of the following set of functions*

$$\{f : [n] \to [r] \mid \left|f^{-1}(\{i\})\right| = n_i\}$$

is the multinomial coefficient $\binom{n}{n_1, n_2, \ldots, n_r}$.

2 Axioms of Probability

Exercise 2.1 (Ross' P 2.14)**.** *The following data were given in a study of a group of* 1000 *subscribers to a certain magazine: In reference to job, marital, status, and education, there were* 312 *professionals,* 470 *married persons,* 525 *college graduates,* 42 *professional college graduates,* 147 *married college graduates,* 86 *married professionals, and* 25 *married professional college graduates. Show that the numbers reported in the study must be incorrect.*

Solution. Let $S = P \cup M \cup C$ where P denotes the set of professionals, M denotes the set of married persons, C denotes the set of college graduates, and S denotes the set of subscribers. By the inclusion-exclusion principle, we must have

$$\begin{aligned} 1000 = |S| &= |P \cup M \cup C| \\ &= |P| + |M| + |C| - |PM| - |PC| - |MC| + |PMC| \\ &= 312 + 470 + 525 - 86 - 42 - 147 + 25 = 1057, \end{aligned}$$

which is a contradiction. Therefore, the reported data must be incorrect. □

Exercise 2.2 (Ross' P 2.16)**.** *Poker dice is played by simultaneously rolling 5 dice. Show that*

(a) $P\{\textit{no two alike}\} = .0926$

(b) $P\{\textit{one pair}\} = .4630$

(c) $P\{\textit{two pair}\} = .2315$

(d) $P\{\textit{three alike}\} = .1543$

(e) $P\{\textit{full house}\} = .0386$

(f) $P\{\textit{four alike}\} = .0193$

(g) $P\{\textit{five alike}\} = .0008$

Solution. [Initial Attempt] Since the dice are rolled simultaneously, order is irrelevant, so we count combinations, not permutations. First, we count the number of possible outcomes of Poker dice. If order matters, then assuming fair 6-sided dice, the number of possible outcomes is 6^5, but order matters. We split into cases. Every number from 1 to 6 can appear once up to 5 times. Note that the "hand" is composed of "categories" (numbers present) and "frequencies" (how many times each number appears). If split into cases based on the pattern of frequencies, the possible patterns are the integer partitions of 5:

(5) There are 6 ways to get five alike; choose a number from 1 to 6 and the hand is fully determined as that number with frequency 5. Given the number, the probability that it appears 5 times is $(1/6)^5$. Therefore, $P\{\text{five alike}\} = 6/6^5 = 1/6^4$.

(4 1) To count the number of ways to get (exactly) four alike, first choose which number from 1 to 6 appears four times, then choose one of the five remaining numbers to appear once. By the basic counting principle, there are $6 \cdot 5 = 30$ ways. Given the two numbers,

(3 2) To count the number of ways to get a full house, first choose which number from 1 to 6 appears thrice, then choose which number from the remaining 5 numbers appears twice.

(2 2 1)

(2 1 1 1)

(1 1 1 1 1)

Okay wait a second! Order doesn't matter in the sense that getting a 1 on the first die and a 2 in the second die is the same as getting a 2 in the first and a 1 in the second. However, order does matter in the sense that, if say we roll two dice, then to get 1 once and 2 once you either get 1 in the first die and 2 in the other or vice versa, and although you consider them the same outcome, still, due to the number of permutations possible to get a 1 and a 2, it has a higher chance than getting two 1's as this corresponds to only 1 permutation. □

Solution. [Correct Attempt] Assuming fair six-sided dice, the sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}^5$, and each 5-tuple is assigned a probability of $\frac{1}{6^5}$. We treat each case as an event $E \subseteq \Omega$, and we count the number of 5-tuples in E and multiply by $1/6^5$.

(a) The number of 5-tuples with no two numbers alike is 6!. The probability is $\frac{6!}{6^5}$. Using a calculator and rounding to the fourth decimal, this evaluates to .0926.

(b) The number of 5-tuples with one pair is counted as follows. First, choose which numbers are present with which frequencies; there are 6 ways to choose the pair and $\binom{5}{3}$ ways to choose 3 distinct numbers different other than the number of which there is a pair, hence there are $6\binom{5}{3}$ ways in this step. Then, there are $\binom{5}{2,1,1,1} = \frac{5!}{2!}$ ways to order the chosen numbers. In total, there are $6\binom{5}{3}\frac{5!}{2!} = \frac{6!5!}{3!2!2!} = \frac{6!5!}{4!} = 5 \cdot 6!$ 5-tuples with one pair, hence the probability is $\frac{5 \cdot 6!}{6^5}$. Using a calculator and rounding to the fourth decimal, this evaluates to .4630.

(c) The number of 5-tuples with two pairs is counted as follows. First, there are $\binom{6}{2}$ ways to choose two numbers from 1 to 6, which will form two pairs, and there are 4 ways to choose a number from 1 to 6 different from the numbers corresponding to the two pairs. Then, there are $\binom{5}{2,2,1} = \frac{5!}{2!2!}$ ways to order the chosen numbers. In total, there are $\binom{6}{2}4\frac{5!}{2!2!} = \frac{6!5!4}{2!4!2!2!} = \frac{6!5}{2}$ 5-tuples with two pairs, hence the probability is $\frac{6!5}{6^5 2}$. Using a calculator and rounding to the fourth decimal, this evaluates to .2315.

(d) The number of 5-tuples with three alike is counted as follows. First, choose the number from 1 to 6 that appears thrice and choose two distinct numbers from 1 to 6 different from the firstly chosen number that appear once; there are $6\binom{5}{2}$ ways to choose the numbers. Then, order the chosen numbers; there are $\binom{5}{3,1,1} = \frac{5!}{3!} = 5 \cdot 4 = 20$ ways to do so. Thus, there are $6 \cdot \binom{5}{2} \cdot 20$ 5-tuples with three alike, hence the probability is $\frac{6 \cdot \binom{5}{2} \cdot 20}{6^5}$. Using a calculator and rounding to the fourth decimal, this evaluates to .1543.

(e) The number of 5-tuples with a full house is counted as follows. First, choose one number from 1 to 6 that appears thrice and a different number from 1 to 6 that appears twice; there are $6 \cdot 5 = 30$ ways to do so. Then, order the chosen numbers; there are $\binom{5}{3,2} = \frac{5!}{3!2!} = \frac{5 \cdot 4}{2} = 10$ ways to do so. Thus, there are $30 \cdot 10 = 300$ 5-tuples with a full house, hence the probability is $\frac{300}{6^5}$. Using a calculator and rounding to the fourth decimal, this evaluates to .0386.

(f) The number of 5-tuples with four alike is counted as follows. First, choose one number from 1 to 6 that appears four times and a different number from 1 to 6 that appears once; there are $6 \cdot 5 = 30$ ways to do so. Then, order the chosen numbers; there are $\binom{5}{4,1} = \frac{5!}{4!1!} = 5$ ways to do so. Thus, there are $30 \cdot 5 = 150$ 5-tuples with four alike, hence the probability is $\frac{150}{6^5}$. Using a calculator and rounding to the fourth decimal, this evaluates to .0193.

(g) The number of 5-tuples with five alike is 6, hence the probability is $\frac{6}{6^5} = \frac{1}{6^4}$. Using a calculator and rounding to the fourth decimal, this evaluates to .0008.

The previous attempts shows that these cases are exhaustive, and they are of course disjoint, hence the probabilities should add up to 1 up to rounding error. Let's check:

$$0.0926 + 0.4630 + 0.2315 + 0.1543 + 0.0386 + 0.0193 + 0.0008 = 1.0001 \approx 1.$$

□

Exercise 2.3 (Ross' P 2.17)**.** *If 8 rooks (castles) are randomly placed on a chess-board, compute the probability that none of the rooks can capture any of the others. That is, compute the probability that no row or file contains more than one rook.*

Solution. Assume that the rooks are indistinguishable, and assume that each square is equally likely to hold a rook. Well, two rooks can capture each other if and only if they are placed in the same row or in the same column. Therefore, if for every two rooks, they are placed in different rows and different columns, then no two rooks can capture each other. With 8 columns, 8 rows, and 8 rooks, there is exactly one rook in each row, represented by which column it is in, and there is exactly one rook in each column, represented by which row it is in. The number of ways this happens is the number of ways we can assign each row from 1 to 8 a single column from 1 to 8 such that each column gets assigned to exactly one row. That is the number of bijections from [8] to [8], which is 8!. In total, there are $\binom{64}{8}$ ways to place 8 rooks in a chessboard. Since each chessboard position is equally likely to hold a rook, the desired probability is $\frac{8!}{\binom{64}{8}} = .000009$. □

Exercise 2.4 (Ross' P 2.21)**.** *Consider Example 5o, which is concerned with the number of runs of wins obtained when n wins and m losses are randomly permuted. Now consider the total number of runs–that is, win runs plus loss runs–and show that*

$$P\{2k \text{ runs}\} = 2\frac{\binom{m-1}{k-1}\binom{n-1}{k-1}}{\binom{m+n}{n}}, \tag{1}$$

$$P\{2k+1 \text{ runs}\} = \frac{\binom{m-1}{k-1}\binom{n-1}{k} + \binom{m-1}{k}\binom{n-1}{k-1}}{\binom{m+n}{n}}. \tag{2}$$

Solution. In total, there are $\binom{m+n}{n,m} = \binom{m+n}{n}$ ways to permute n wins and m losses.

Note that for every win run, if there is a next win run, then they are separated by a loss run, for otherwise they form a single run; this holds for loss runs. Thus, if there are j win runs, there are either $j-1$ loss runs, j loss runs, or $j+1$ loss runs. That is, either there are as many win runs as there are loss runs, or there are one more loss or win runs, and all other cases are impossible.

Now, either the total number of runs is even, or it is odd.

- Suppose the total number of runs is even, say $2k$. Then, we must have that there are k win runs; if there are j win runs, then we cannot have $j-1$ loss runs, nor $j+1$ loss runs, hence there are j loss runs as well, and so $2k = j + j$, hence $j = k$. Either the first run is a win run, or the first run is a loss run, and the two cases are symmetric. WLOG, suppose the first run is a win run. Then, we have the sequence $W_1L_1W_2L_2 \dots W_kL_k$ where for each integer $1 \le i \le k$, W_i is a run containing of 1 or more wins, and similarly for losses. Consider the win runs $W_1, W_2, \dots, W_k$. The number of ways to obtain them is the composition of n into k; there are $\binom{n-1}{k-1}$ ways. Similarly, there are $\binom{m-1}{k-1}$ ways to obtain the k loss runs. Therefore, there are $\binom{n-1}{k-1}\binom{m-1}{k-1}$ ways to permute n wins and m losses such that there are $2k$ runs and the first run is a win run. If we relax the condition that the first run is a win run, then by symmetry of the situation, there are $2\binom{n-1}{k-1}\binom{m-1}{k-1}$ ways. Since each permutation is equally likely, we conclude that
$$P\{2k \text{ runs}\} = \frac{2\binom{m-1}{k-1}\binom{n-1}{k-1}}{\binom{m+n}{n}}.$$

- Suppose the total number of runs is odd, say $2k+1$. Thus, either we have k win runs and $k+1$ loss runs, in which case the sequence the sequence must be of the form $L_1W_1L_2W_2 \dots L_kW_kL_{k+1}$ and there are $\binom{m-1}{k+1-1}\binom{n-1}{k-1}$ such sequences, or there are $k+1$ win runs and k loss runs, in which case the sequence must be of the form $W_1L_1W_2L_2 \dots W_kL_kW_{k+1}$ and there are $\binom{n-1}{k+1-1}\binom{m-1}{k-1}$.

The two cases are disjoint, so by the addition rule, there are $\binom{m-1}{k}\binom{n-1}{k-1} + \binom{n-1}{k}\binom{m-1}{k-1}$ ways to permute n wins and m losses such that the total number of runs is odd. Therefore, since each permutation is equally likely, we conclude that

$$P\{2k+1 \text{ runs}\} = \frac{\binom{m-1}{k}\binom{n-1}{k-1} + \binom{n-1}{k}\binom{m-1}{k-1}}{\binom{m+n}{n}}.$$

□

Exercise 2.5 (Ross' P 2.46)**.** *How many people have to be in a room in order that the probability that at least two of them celebrate their birthday in the same month is at least $\frac{1}{2}$? Assume that all possible monthly outcomes are equally likely.*

Solution. Assume there are n different people in a room. We compute the probability of the complement event E^c: that people in the room celebrate their birthdays in distinct months. The sample space is $\Omega = [12]^n$. Note that $|\Omega| = 12^n$. The number of n-tuples in Ω without repetition is the number of n-permutations of $[12]$; therefore, $|E^c| = (12)_n$, so $P\{E^c\} = \frac{(12)_n}{12^n}$. Note that $1/2 \leq P\{E\} = 1 - P\{E^c\} \iff P\{E^c\} \leq 1/2$. Thus, we find the number of people to be in the room in order that the probability that they celebrate their birthdays in distinct months is at most $1/2$.

- If $n = 1$, then $P\{E^c\} = \frac{12}{12} = 1 \not\leq 1/2$.
- If $n = 2$, then $P\{E^c\} = \frac{12 \cdot 11}{12 \cdot 12} = 11/12 \not\leq 1/2$.
- If $n = 3$, then $P\{E^c\} = \frac{12 \cdot 11 \cdot 10}{12 \cdot 12 \cdot 12} = 55/72 \not\leq 1/2$.
- If $n = 4$, then $P\{E^c\} = \frac{12 \cdot 11 \cdot 10 \cdot 9}{12^4} = 55/95 \not\leq 1/2$.
- If $n = 5$, then $P\{E^c\} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{12^5} = 55/144 \leq 1/2$.

Therefore, 5 people have to be in the room. □

Exercise 2.6 (Ross' P 47)**.** *If there are 12 strangers in a room, what is the probability that no two of them celebrate their birthday in the same month.*

Solution. Assuming all possible monthly outcomes are equally likely, that probability is $\frac{12!}{12^{12}}$. □

Exercise 2.7 (Green book: N points on a circle)**.** *Given N points drawn randomly on the circumference of a circle, what is the probability that they are all within a semicircle.*

Solution. Let's see.

- (Initial thoughts.) I have the following idea. For simplicity, assume the circle has circumference 1. The first point, a, can be anywhere on the circumference of the circle. The second point, b, can be anywhere too, for if the arc from a to b clockwise has length $> 1/2$, then the arc from b to a clockwise has length $\leq 1/2$. Now, let A be the arc joining a and b with length $x \leq 1/2$. Let c be a third point on the circumference of the circle. For a, b, c to lie in a semicircle, either c lies in A, with probability x, or c is the new initial point, with probability $1/2 - x$, or c is the new final point, with probability $1/2 - x$; thus, the probability of choosing c such that a, b, c lie in a semicircle is $x + 2(1/2 - x) = 1 - x$. To begin with, x could have been any value from 0 to $1/2$. Hmmm... It's not clear how to proceed.
- (Correct Solution; fill in the details) Draw a point x_1. Let $x_2, \ldots, x_N$ be the rest of the points drawn, going clockwise from x_1. For $1 \leq i \leq N$, let E_i be the event that the arc connecting all points and beginning with x_i is less than half the circumference in length.

  First, prove that the points are all within a semicircle if and only if one of the events E_i occurs.

  Then, prove that the events E_i are disjoint.

  Finally, compute
  $$P\left(\bigcup_{i=1}^{N} E_i\right) = \sum_{i=1}^{N} P(E_i) = N\left(\frac{1}{2}\right)^{N-1}.$$

□

Exercise 2.8 (Ross' TE 2.1-2.4)**.** *Prove the following relations:*

1. $EF \subseteq E \subseteq E \cup F$
2. $E \subseteq F \implies F^{\mathsf{c}} \subseteq E^{\mathsf{c}}$
3. $F = FE \cup FE^{\mathsf{c}}$ *and* $E \cup F = E \cup E^{\mathsf{c}}F$
4. $(\bigcup_1^\infty E_i)\, F = \bigcup_1^\infty E_i F$ *and* $(\bigcap_1^\infty E_i) \cup F = \bigcap_1^\infty (E_i \cup F)$

Solution. We prove the relations in order.

1. Let $a \in EF$. Then $a \in E \wedge a \in F$. In particular, $a \in E$. In general, $a \in E \vee a \in F$. Thus, $a \in E \cup F$.

2. Assume that $E \subseteq F$. Thus, (*) $a \in E \implies a \in F$. Let $a \in F^{\mathsf{c}}$. Thus, $a \notin F$. By the contrapositive of *, $a \notin E$, hence $a \in E^{\mathsf{c}}$.

3. We first prove $F = FE \cup FE^{\mathsf{c}}$. Let $a \in F$. Either $a \in E$, or $a \notin E$ and hence $a \in E^{\mathsf{c}}$. If $a \in E$, then since $a \in F$, we have $a \in EF$. If $a \in E^{\mathsf{c}}$, then since $a \in F$, we have $a \in EF^{\mathsf{c}}$. Therefore, either $a \in FE$ or $a \in FE^{\mathsf{c}}$, hence $a \in FE \cup FE^{\mathsf{c}}$. This proves that $F \subseteq FE \cup FE^{\mathsf{c}}$. On the other hand, if $a \in FE \cup FE^{\mathsf{c}}$, then $a \in FE \vee a \in FE^{\mathsf{c}}$, so $(a \in F \wedge a \in E) \vee (a \in F \wedge a \in E^{\mathsf{c}})$, so by the distributive law, $a \in F \wedge (a \in E \vee a \in E^{\mathsf{c}})$, so in particular $a \in F$. Therefore, $F = FE \cup FE^{\mathsf{c}}$.

   We then prove $E \cup F = E \cup E^{\mathsf{c}}F$. Either $a \in E$ or $a \notin E$. If $a \in E$, then $a \in E \cup F$ and $a \in E \cup E^{\mathsf{c}}F$. Suppose $a \notin E$. Then, if $a \in F$, then $a \in E^{\mathsf{c}}F$, and so $a \in E \cup F$ and $a \in E \cup E^{\mathsf{c}}F$. On the other hand, if $a \notin F$, then $a \notin E^{\mathsf{c}}F$, and so since $a \notin E$, then $a \notin E \cup F$ and $a \notin E \cup E^{\mathsf{c}}F$. In every case, either a lies in both LHS and RHS, or a lies in neither. Therefore, equality holds.

4. We first prove $(\bigcup_1^\infty E_i)\, F = \bigcup_1^\infty E_i F$. If $a \in (\bigcup_1^\infty E_i)\, F$, then $a \in (\bigcup_1^\infty E_i) \wedge a \in F$, so $a \in F \wedge \exists i \in \mathbb{Z}^+, a \in E_i$, hence $a \in E_i F$, and so in general, $a \in \bigcup_1^\infty E_j F$. If $a \in \bigcup_1^\infty E_i F$, then $\exists i \in \mathbb{Z}^+, a \in E_i F$, so $a \in E_i \wedge a \in F$, so in general, $a \in \bigcup_1^\infty E_j \wedge a \in F$, hence $a \in (\bigcup_1^\infty E_i)\, F$.

   We then prove $(\bigcap_1^\infty E_i) \cup F = \bigcap_1^\infty (E_i \cup F)$. Suppose $a \in (\bigcap_1^\infty E_i) \cup F$. Then $a \in \bigcap_1^\infty E_i$ or $a \in F$. If $a \in F$, then $\forall i \in \mathbb{Z}^+$, we deduce in general that $a \in E_i \cup F$; hence, $a \in \bigcap_1^\infty (E_i \cup F)$. If $a \in \bigcap_1^\infty E_i$, then $\forall i \in \mathbb{Z}^+, a \in E_i$, so in general, $\forall i \in \mathbb{Z}^+, a \in E_i \cup F$; hence, $a \in \bigcap_1^\infty (E_i \cup F)$. On the other hand, suppose $a \in \bigcap_1^\infty (E_i \cup F)$. Then, $\forall i \in \mathbb{Z}^+, a \in E_i \cup F$. If $a \in F$, then in general, $a \in (\bigcap_1^\infty E_i) \cup F$. Suppose $a \notin F$. Let $i \in \mathbb{Z}^+$. Then, $a \in E_i \cup F$, and since $a \notin F$, then $a \in E_i$. Thus, $\forall \mathbb{Z}^+, a \in E_i$. It follows that $a \in \bigcap_1^\infty E_i$. In general, $a \in (\bigcap_1^\infty E_i) \cup F$.

□

Exercise 2.9 (Ross' Self-test 2.14)**.** *Prove Boole's Inequality:*

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i).$$

Solution. We use induction to prove that for every positive integer n, we have

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i).$$

- Base case: if $n = 2$, then $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1A_2) \leq P(A_1) + P(A_2)$.
- Inductive hypothesis (IH): assume for some positive integer $n \geq 2$ that inequality holds.

  Then,

$$\begin{aligned} P\left(\bigcup_{i=1}^{n+1} A_i\right) &= P\left(A_{n+1} \cup \bigcup_{i=1}^{n} A_i\right) \\ &= P(A_{n+1}) + P\left(\bigcup_{i=1}^{n} A_i\right) - P\left(A_{n+1} \cap \left(\bigcup_{i=1}^{n} A_i\right)\right) \end{aligned}$$

$$\leq P(A_{n+1}) + P\left(\bigcup_{i=1}^{n} A_i\right),$$

hence by IH, we have

$$\begin{aligned} P\left(\bigcup_{i=1}^{n+1} A_i\right) &\leq P(A_{n+1}) + \sum_{i=1}^{n} P(A_i) \\ &= \sum_{i=1}^{n+1} P(A_i). \end{aligned}$$

Therefore, by induction, for every integer $n \geq 2$, we have

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i).$$

Taking the limit as $n \to \infty$, we obtain

$$\begin{aligned} \sum_{i=1}^{\infty} P(A_i) &\geq \lim_{n\to\infty} P\left(\bigcup_{i=1}^{n} A_i\right) \\ &= P\left(\lim_{n\to\infty} \bigcup_{i=1}^{n} A_i\right) \end{aligned}$$

by continuity of the probability function. This proves Boole's inequality. □

3 Conditional Probability

Exercise 3.1 (Ross' P 3.1)**.** *Two fair dice are rolled. What is the conditional probability that at least one lands on* 6 *given that the dice land on different numbers.*

Solution. The sample space is $[6]^2$, and all outcomes are equally likely. Let S be the event that at least one die lands on 6. Let I be the event that the dice land on the same number. We compute $P(S \mid I^c)$. By the law of total probability, we have

$$\begin{aligned} P(S) &= P(S \mid I)P(I) + P(S \mid I^c)P(I^c) \\ \iff P(S \mid I^c) &= \frac{P(S) - P(S \mid I)P(I)}{1 - P(I)} \end{aligned}$$

Well, we compute the probabilities on the RHS:

- First, we have $P(S) = 1 - P(S^c)$, and $P(S^c)$, the probability that neither die lands on 6, equals $(5/6)^2 = 25/36$, hence $P(S) = 11/36$.
- Second, out of 36 outcomes for this experiment, exactly 6 outcomes make the event I: $(1,1)$, $(2,2)$, $\ldots$, $(6,6)$. Hence, $P(I) = 6/36 = 1/6$.
- Third, out of all outcomes in event I, only the outcome $(6,6)$ is an element of S. Therefore,

$$P(S \mid I) = \frac{P(S \cap I)}{P(I)} = \frac{1/36}{6/36} = 1/6$$

.

Thus, we obtain

$$P(S \mid I^c) = \frac{11/36 - (1/6)(1/6)}{1 - 1/6} = \frac{10/36}{5/6} = \boxed{\frac{1}{3}}$$

□

Exercise 3.2 (Ross' P 3.2)**.** *If two fair dice are rolled, what is the conditional probability that the first one lands on* 6 *given that their sum is* i*. Compute this probability for all values of* i *from* 2 *to* 12*.*

Solution. We solve case by case. For each i, we enumerate outcomes in which the sum is i; if we got n outcomes, then in only one of those does the first die land on 6 because the second die has to be exactly $i-6$, hence since the dice are fair, the desired probability would be $1/n$.

(2,3,4,5,6) If $i \leq 6$, in none of the possible outcomes does the first die land on 6, hence the probability is 0.

(7) The outcomes in which the sum is 7 are $(6,1),(1,6),(5,2),(2,5),(4,3),(3,4)$. These are 6 possible outcomes. Of these 6 outcomes, only in the outcome $(6,1)$ does the first die land on 6. The desired probability is $1/6$.

(8) The outcomes in which the sum is 8 are $(6,2),(2,6),(5,3),(3,5),(4,4)$. These are 5 possible outcomes. Of these 5 outcomes, only in the outcome $(6,2)$ does the first die land on 6. The desired probability is $1/5$.

(9) The outcomes in which the sum is 9 are $(6,3),(3,6),(5,4),(4,5)$. These are 4 outcomes. The desired probability is $1/4$.

(10) The outcomes in which the sum is 10 are $(6,4),(4,6),(5,5)$. The desired probability is $1/3$.

(11) The outcomes in which the sum is 11 are $(6,5),(5,6)$. The desired probability is $1/2$.

(12) The outcomes in which the sum is 12 are $(6,6)$. The desired probability is 1.

Alternatively, make a table of sums for each possible outcome.

First die \ Second die	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Then, observe that when the first die lands on 6, the possible sums are $7,8,9,10,11,12$, and observe all outcomes with a certain sum i lie on the diagonal line from $(6,i-6)$ to $(i-6,6)$. Therefore, the number of outcomes with sum i is $6-(i-6)+1 = 13-i$. Thus, for $i = 7,8,\ldots,12$, the number of outcomes with sum i is $6,5,\ldots,1$, respectively, and hence probability that the first die lands on 6 given that the sum is i is $1/6, 1/5, \ldots, 1$, respectively. □

Exercise 3.3 (Ross' P 3.7)**.** *The king comes from a family of* 2 *children. What is the probability that the other child is his sister?*

Solution. Realistically, a family of two children consists of an older child and a younger child. Let M denote a male child and F denote a female child. Then, there are four possible outcomes for a family: $(M,M),(M,F),(F,M),(F,F)$. For a family from which comes a king, there are three possible outcomes: $(M,M),(M,F),(F,M)$. In exactly two out of those three outcomes is the other child of the king's family his sister: $(M,F),(F,M)$. Therefore, assuming all outcomes are equally likely, the probability that the other child of the king's family is his sister is $\boxed{\frac{2}{3}}$. □

Exercise 3.4 (Ross' P 3.8)**.** *A couple has* 2 *children. What is the probability that both are girls if the older of the two is a girl?*

Solution. There are four possible outcomes for a family of 2 children: $(B,B),(B,G),(G,B),(G,G)$ where B denotes a boy and G denotes a girl. In exactly two of those four outcomes is the older child a girl: $(G,B),(G,G)$. In exactly one of those two outcomes are both children girls: (G,G). Therefore, assuming all possible outcomes are equally likely, the probability that both children are girls given that the older of the two is a girl is $\boxed{\frac{1}{2}}$. □

Exercise 3.5 (Ross' P 3.9)**.** *Consider* 3 *urns. Urn* A *contains* 2 *white and* 4 *red balls, urn* B *contains* 8 *white and* 4 *red balls, and urn* C *contains* 1 *white ball and* 3 *red ball. If* 1 *balls is selected from each urn, what is the probability that the ball chosen from urn* A *was white given that exactly* 2 *white balls were selected.*

Solution. Let $\Omega = \{w, r\}^3$ be the sample space. Let W_A, W_B, W_C be the events of selecting a white ball from urn A, from urn B, and from urn C, respectively. Let E denote the event of selecting exactly 2 white balls. Note that $E = W_A W_B W_C^{\mathsf{c}} \cup W_A W_B^{\mathsf{c}} W_C \cup W_A^{\mathsf{c}} W_B W_C$. Because the constituent events are disjoint, we have

$$\begin{aligned} P(E) &= P(W_A W_B W_C^{\mathsf{c}} \cup W_A W_B^{\mathsf{c}} W_C \cup W_A^{\mathsf{c}} W_B W_C) \\ &= P(W_A W_B W_C^{\mathsf{c}}) + P(W_A W_B^{\mathsf{c}} W_C) + P(W_A^{\mathsf{c}} W_B W_C). \end{aligned}$$

Because W_A, W_B, W_C are independent and in each urn, each ball is equally likely to be chosen, we have

$$\begin{aligned} P(W_A W_B W_C^{\mathsf{c}}) &= P(W_A)P(W_B)P(W_C^{\mathsf{c}}) \\ &= (2/6)(8/12)(3/4) = 48/288. \\ P(W_A W_B^{\mathsf{c}} W_C) &= (2/6)(4/12)(1/4) = 8/288. \\ P(W_A^{\mathsf{c}} W_B W_C) &= (4/6)(8/12)(1/4) = 32/288. \end{aligned}$$

Therefore,

$$P(E) = \frac{48 + 8 + 32}{288} = \frac{88}{288} = \frac{11}{36}.$$

Then, the event that the ball chosen from urn A was white given that exactly 2 white balls were selected is the complement of the event that the ball chosen from urn A was not white given that exactly 2 white balls were selected. Then, the desired probability is

$$\begin{aligned} P(W_A \mid E) &= 1 - P(W_A^{\mathsf{c}} \mid E) \\ P(W_A^{\mathsf{c}} \mid E) &= \frac{P(W_A^{\mathsf{c}} E)}{P(E)} = \frac{P(W_A^{\mathsf{c}} W_B W_C)}{P(E)} \\ &= \frac{32/288}{88/288} = \frac{32}{88} = \frac{4}{11}. \end{aligned}$$

Therefore,

$$P(W_A \mid E) = 1 - \frac{4}{11} = \boxed{\frac{7}{11}}$$

□

Exercise 3.6 (Ross' P 3.15)**.** *An ectopic pregnancy is twice as likely to develop when the pregnant is a smoker as it is when she is a nonsmoker. If* 32 *percent of women of childbearing age are smokers, what percentage of women having ectopic pregnancies are smokers.*

Solution. Let E denote the event that a pregnant woman develops an ectopic pregnancy. Let S denote the event that a pregnant woman is a smoker. We compute $P(S \mid E)$. First, we have

$$\begin{aligned} P(E) &= P(E \mid S)P(S) + P(E \mid S^{\mathsf{c}})P(S^{\mathsf{c}}) \\ &= P(E \mid S)P(S) + \frac{1}{2}P(E \mid S)P(S^{\mathsf{c}}) \\ &= P(E \mid S)P(S) + \frac{1}{2}P(E \mid S)(1 - P(S)) \\ &= \frac{1}{2}P(E \mid S)(1 + P(S)). \end{aligned}$$

Then,

$$P(S \mid E) = \frac{P(E \mid S)P(S)}{P(E)}$$

$$
\begin{aligned}
&= \frac{P(E \mid S)P(S)}{(P(E \mid S)/2)(1+P(S))} \\
&= \frac{2P(S)}{1+P(S)} \\
&= \frac{2(0.32)}{1+0.32} = \frac{0.64}{1.32} = 0.\overline{48} \approx \boxed{48.5\%}.
\end{aligned}
$$

□

Exercise 3.7 (Ross' P 3.33)**.** *On rainy days, Joe is late to work with probability .3; on nonrainy days, he is late with probability .1. With probability .7 it will rain tomorrow.*

(a) Find the probability that Joe is early tomorrow.

(b) Given that Joe was early, what is the conditional probability that it rained?

Solution. Let R denote the event that it rained, and let L denote the event that Joe is late. We are given

$$P(L \mid R) = .3,\ P(L \mid R^c) = .1,\ P(R) = .7.$$

(a) We compute $P(L^c)$. We have

$$
\begin{aligned}
P(L) &= P(L \mid R)P(R) + P(L \mid R^c)(1 - P(R)) \\
&= (.3)(.7) + (.1)(.3) = .21 + .03 = .24. \\
P(L^c) &= 1 - P(L) = 1 - .24 = \boxed{.76}.
\end{aligned}
$$

(b) We compute $P(R \mid L^c)$. Well we have

$$
\begin{aligned}
P(R \mid L^c) &= \frac{P(L^c \mid R)P(R)}{P(L^c)} = \frac{(1 - P(L \mid R))P(R)}{P(L^c)} \\
&= \frac{(1-.3)(.7)}{.76} = \frac{.49}{.76} = .64447 \approx \boxed{65.4\%}.
\end{aligned}
$$

□

Exercise 3.8 (Green book: Fair probability from an unfair coin)**.** *If you have an unfair coin, which may bias toward either heads or tails at an unknown probability, can you generate even odds using this coin?*

Solution. I will propose my solution then use the classical solution proposed by von Neumann.

- Toss the coin twice. In the first toss, read the coin at face value; that is, read heads as heads and tails as tails. In the second toss, read the coin inversely; that is, read heads as tails and tails as heads. Let getting two heads be a win, getting two tails be a loss, and repeat otherwise. Suppose that the unfair coin lands on heads with probability p. Then, $P(HH) = p(1-p),\ P(TT) = (1-p)p$, hence it is equally likely to win or lose.

  In general, I claim that the following works as well: toss the coin $2n$ times, reading at face value in the odd tosses and inversely in the even tosses, and let getting more heads than tails be a win, let getting more tails than heads be a loss, and repeat in case of a tie.

- von Neuman's method: let p_H and p_T denote the probabilities that a coin lands on heads and that a coin lands on tails, respectively; toss the coin twice; and let getting a head then a tail be a win, let getting a tail then a head be a loss, and repeat otherwise. Observe that $P(HT) = p_H p_T,\ P(TH) = p_T p_H$.

□

Exercise 3.9 (Green book: Dart Game 1)**.** *Jason throws two darts at a dartboard, aiming for the center. The second dart lands farther from the center than the first dart. If Jason throws a third dart, what is the probability that it is farther from the center than the first dart, assuming that Jason's skill is the same each time he throws a dart?*

Solution. Assuming that there can be no ties, there are 6 possible outcomes with how close the darts are relative to each other: $(1,2,3),(1,3,2),(2,1,3),(2,3,1),(3,1,2),(3,2,1)$. Of those, there are 3 possible outcomes with the second dart farther than the first dart: $(1,2,3),(1,3,2),(3,1,2)$. Given that the second dart landed farther than the first dart, there are 2 possible outcomes with the third dart farther than the first dart: $(1,2,3),(1,3,2)$. Therefore, the probability that the third dart lands farther than the first dart given that the second dart landed farther from the first dart is $\boxed{\frac{2}{3}}$. □

Exercise 3.10 (Green book: Dart Game 2)**.** *Similarly to Exercise* (3.9)*, Jason throws n darts at a dartboard. Given that every throw except the first throw landed farther from the center than the first, if Jason throws an $(n+1)$th dart, What is the probability that it lands farther from the center than the first dart?*

Solution. Either the $(n+1)$th dart is closer than the first, or it is farther. If it is closer, then given that every other dart is farther than the first, it follows that the closest dart is the $(n+1)$th and the second closest is the first, and the remaining $n-1$ darts can be in any order; there are $(n-1)!$ possible outcomes in this case. If the $(n+1)$th dart is farther than the first, then given that every other dart is farther from the first, the first dart is the closest dart, and the remaining n darts can be in any order; there are $n!$ possible outcomes in this case. Therefore, there are $n!+(n-1)!$ possible outcomes given that the darts $2,3,\ldots,n$ are farther from the first, and the probability that the $(n+1)$th dart is farther than the first given that the darts $2,3,\ldots,n$ are farther than the first is $\frac{n!}{n!+(n-1)!} = \boxed{\frac{n}{n+1}}$. □

Exercise 3.11 (Green book: Birthday Line)**.** *At a moview theater, a whimsical manager announces that she will give a free ticket to the first person in line whose birthday is the same as someone who has already bought a ticket. You are given the opportunity to choose any position in line. Assuming that you don't know anyone else's birthday and all birthdays are distributed randomly throughout the year (assuming 365 days in a year), what position in line gives you the largest change of getting the free ticket?*

Solution. For every positive integer, let $p(n)$ denote the probability that you get the free ticket if you choose position n.

- First, by the pigeonhole principle, among the first 366 people must be two people with the same birthday, hence in any position farther than the 366th, it's impossible to get the free ticket. Therefore,
$$p(n) = 0,\ n \geq 367.$$
- If you choose position 366, then if nobody among the first 365 people in line gets the free ticket, which is equivalent to that each one of them has a unique birthday of the 365 possible birthdays, hence whatever your birthday is, there is a person before in you line whose birthday that is, hence you certainly get the ticket. Therefore, $p(366)$ equals the probability that the first 365 people in line have distinct birthdays. In general, note that among the first $n \leq 365$ people, there are 365^n possible outcomes, and of those outcomes, there are $(365)_n = 365 \cdot 364 \cdots (365-n+1)$ possible outcomes in which the first n people have distinct birthdays, hence the probability that the first n people have distinct birthdays is $\frac{(365)_n}{365^n}$. Therefore,
$$p(366) = \frac{365!}{365^{365}}.$$
- If you choose position $n \leq 365$, then you get the free ticket if and only if the first $n-1$ people have distinct birthdays and your birthday matches that of one of the first $n-1$ people. (Of course, if you are the first in line, you don't get the free ticket, so $p(1)=0$.) Therefore,
$$p(n) = \frac{n-1}{365} \cdot \frac{(365)_{n-1}}{365^{n-1}} = \frac{(n-1)(365)_{n-1}}{365^n}, \quad 1 \leq n \leq 365.$$

We find $\arg\max_{n\in[366]} p(n)$. Observe for $2 \leq n \leq 364$ that
$$p(n+1)/p(n) = \frac{(n)(365)_n/365^{n+1}}{(n-1)(365)_{n-1}/365^n} = \frac{n}{n-1}\frac{(365-n+1)}{365}.$$

Observe that

- $0 = p(1) < p(2)$.
- $p(3)/p(2) = (2/1)(364/365) > 1 \implies p(2) < p(3)$.
- As n increases, $p(n+1)/p(n)$ decreases as both of its factors $n/(n-1)$ and $(365-n+1)/365$ get smaller.
- $p(365)/p(364) = (364/363)(2/365) < 1 \implies p(364) > p(365)$.
- $p(365) = \frac{364(365)!}{365^{365}} > \frac{365!}{365^{365}} = p(366)$

Therefore, for some $2 \leq n \leq 364$, we have

$$p(1) < p(2) < p(3) < \cdots < p(n-1) < p(n) > p(n+1) > \cdots > p(364) > p(365) > p(366).$$

Observe that

$$\begin{aligned}
p(n) > p(n-1) &\iff p(n)/p(n-1) > 1 \iff \frac{n-1}{n-2}\frac{365-(n-2)}{365} > 1 \\
&\iff -(n-1)(n-2) + 365(n-1) > 365(n-2) \\
&\iff -(n-1)(n-2) > -365 \\
&\iff (n-1)(n-2) < 365. \\
p(n) > p(n+1) &\iff p(n+1)/p(n) < 1 \iff \frac{n}{n-1}\frac{365-(n-1)}{365} < 1 \\
&\iff -n(n-1) + 365n < (n-1)365 \\
&\iff -n(n-1) < -365 \\
&\iff n(n-1) > 365.
\end{aligned}$$

Observe that the second condition implies $n^2 > 365$, which implies $n > 19$. Let's try $n = 20$. We have

$$\begin{aligned}
19(18) < 19^2 = 361 < 365 &\implies p(20) > p(19) \\
20(19) = 380 > 365 &\implies p(20) > p(21).
\end{aligned}$$

Therefore, the position that gives you the largest chance of getting the free ticket is position $\boxed{20}$.

□

Exercise 3.12 (Green Book: Dice Order)**.** *If we throw* 3 *dice in order, what is the probability of landing numbers in strictly increasing order?*

Solution. We can solve this in 2 ways.

- **Counting method:** Assuming the dice are six-sided, fair, and independent, each outcome is equally likely with a probability of $(1/6)^3 = 1/216$. The number of desired outcomes is the number of strictly increasing 3-permutations of 6 is equal to the number of 3-combinations of 6; observe that for every combination, there is exactly one way to order it in increasing order, and that each strictly increasing permutation gives a unique combination. Therefore, the desired probability is $\binom{6}{3}(1/216) = 20/216 = \boxed{5/54}$.
- **Conditional method:** Let A denote the event of landing distinct numbers, B denote the event of landing numbers in non-decreasing order, and C denote the event of landing numbers in strictly increasing order. We want $P(C)$. Well,

$$\begin{aligned}
P(C) &= P(AB) = P(A)P(B \mid A) \\
&= (1 \cdot (5/6) \cdot (4/6)) \cdot ((1/3)(1/2)(1)) = (5 \cdot 4)/(6 \cdot 6 \cdot 3 \cdot 2) = 20/216 = \boxed{5/54}.
\end{aligned}$$

□

Exercise 3.13 (Green Book: Monty Hall Problem)**.** *Monty Hall problem is a probability puzzle based on an old American show Let's Make a Deal. The problem is named after the show's host. Suppose you're on the show now, and you're given the choice of* 3 *doors. Behind one door is a car; behind the other two, goats. You don't know ahead of time what is behind each of the doors.*

You pick one of the doors and announce it. As soon as you pick the door, Monty opens one of the other two doors that he knows has a goat behind it. Then he gives you the option to either keep your original choice or switch to the third door. Should you switch? What is the probability of winning a car if you switch?

Solution. Let D be an indicator variable denoting that you initially choose the door behind which there is the car. Thus, $P\{D = 1\} = 1/3$ and $P\{D = 0\} = 2/3$. Let S be the indicator variable that your strategy is to switch. Let W be the indicator variable that you win the car.

- **Keep strategy:** You win if and only if you initially announce a door behind which there is the car, hence

$$P\{W = 1 \mid S = 0\} = P\{D = 1\} = 1/3.$$

- **Switch strategy:** If you initially announce a door behind which there is a goat, then Monty will open a different door, behind which there is a goat, and you will switch to yet a different door, behind which there must be the car; in this case, you win. If you initially announce a door behind which there is the car, then upon switching to a different door, you open a door behind which there is a goat; in this case you lose. Therefore, you win if and only if you initially announce a door behind which there is a goat, hence

$$P\{W = 1 \mid S = 1\} = P\{D = 0\} = \boxed{2/3}.$$

Therefore, you are **twice as likely to win** if you intend to switch than you are if you intend to keep, hence, **yes, should intend to switch**. □

Remark 3.14. *The Monty Hall problem can be solved using only events and probability without any random variable. I introduced indicator variables for my convenience.*

Exercise 3.15 (Green Book: Amoeba Population)**.** *There is one amoeba in a pond. After every minute, the amoeba either dies, or stays the same, or splits into two, or splits into three, with equal probability. All its offspring, if any, will behave the same, independently of other amoebas. What is the probability the amoeba population will die out.*

Solution. Let $p \in [0, 1]$ be the probability that an amoeba and all its offspring die out. Starting with one amoeba, the probability that the amoeba population will die out equals p. Then, using the law of total probability, we have

$$\begin{aligned} p &= (1/4) + (1/4)p + (1/4)p^2 + (1/4)p^3 \\ \iff 4p &= 1 + p + p^2 + p^3 \\ \iff 0 &= p^3 + p^2 - 3p + 1. \end{aligned}$$

The roots of the cubic equation above are $1, \sqrt{2} - 1, -\sqrt{2} - 1$. Since $p \geq 0$, then $p \neq -\sqrt{2} - 1$.

Since, in expectation, one amoeba grows to $(1 + 2 + 3)/4 = 1.5 > 1$ amoebas, then $p < 1$ [USING A RESULT I CAN'T PROVE YET, WHICH IS THE EXTINCTION CRITERION FOR A GALTON-WATSON PROCESS].

Therefore, $p = \sqrt{2} - 1 \approx \boxed{0.414}$. □

Exercise 3.16 (Green Book: Candies in a Jar)**.** *You are taking out candies one by one from a jar that has* 10 *red candies,* 20 *blue candies, and* 30 *green candies in it. What is the probability that there are at least* 1 *blue candy and* 1 *green candy left in the jar when you have taken out all the red candies?*

Solution. Let T_r, T_b, T_g be random variables denoting the step at which the last red candy is taken, the step at which the last blue candy is taken, and the step at which the last green candy is taken. Then,

$$\begin{aligned} p &:= P(\text{at least 1 blue candy and 1 green candy remain after all red candies are taken}) \\ &= P\{\{T_r < T_b\} \cap \{T_r < T_g\}\} \\ &= P\{\{T_r < T_b < T_g\} \cup \{T_r < T_g < T_b\}\} \\ &= P\{T_r < T_b < T_g\} + P\{T_r < T_g < T_b\} \\ &= P\{T_g = 60\}P\{T_r < T_b \mid T_g = 60\} + P\{T_b = 60\}P\{T_r < T_g \mid T_b = 60\}. \end{aligned}$$

Each ball is equally likely to be the last ball taken out.

It follows that with probability $30/60 = 1/2$, the last ball taken out is green. Therefore,

$$P\{T_g = 60\} = 1/2.$$

Similarly, with probability $20/60 = 1/3$, the last ball taken out is blue. Therefore,

$$P\{T_b = 60\} = 1/3.$$

Suppose the last ball taken out is green. Conditioned on that, the probability that the $T_r < T_b$ is computed as follows: fix an arbitrary choice of when all the green balls are taken out; of the 30 red and blue balls remaining, each is equally likely to be the last one taken out, so the probability that the last one taken out is blue is $20/30 = 2/3$. Therefore,

$$P\{T_r < T_b \mid T_g = 60\} = 2/3.$$

Suppose the last ball taken out is blue. Similarly, we find that

$$P\{T_r < T_g \mid T_b = 60\} = 30/40 = 3/4.$$

Thus, we obtain

$$p = (1/2)(2/3) + (1/3)(3/4) = (1/3) + (1/4) = (4+3)/12 = \boxed{\frac{7}{12}}.$$

□

Exercise 3.17 (Green Book: Coin Toss Game)**.** *Two players, A and B alternately toss a fair coin, with A the first to toss. When a head is tossed followed by a tail, the game ends, and the one who tossed the tail wins. What is the probability that A wins the game?*

Solution.

$$\begin{aligned}
&P(A) = \frac{1}{2}P(A \mid H) + \frac{1}{2}P(A \mid T)\\
&P(A \mid T) = 1 - P(B \mid T) = 1 - P(A)\\
&P(A \mid H) = \frac{1}{2}P(A \mid HH) + \frac{1}{2}\underbrace{P(A \mid HT)}_{0}\\
&P(A \mid HH) = P(B \mid H) = 1 - P(A \mid H)\\
&\implies P(A \mid H) = \frac{1}{2}(1 - P(A \mid H)) \implies \frac{3}{2}P(A \mid H) = \frac{1}{2} \implies P(A \mid H) = \frac{1}{3}\\
&\implies P(A) = \frac{1}{2}\frac{1}{3} + \frac{1}{2}(1 - P(A))\\
&\implies \frac{3}{2}P(A) = \frac{1}{6} + 12 = \frac{4}{6} = \frac{2}{3}\\
&\implies P(A) = \frac{2}{3}\frac{2}{3} = \boxed{\frac{4}{9}}.
\end{aligned}$$

□

Exercise 3.18 (Ross' TE 3.1)**.** *Show that if $P(A) > 0$, then*

$$P(AB|A) \geq P(AB|A \cup B)$$

Solution. Since $P(A \cup B) \geq P(A)$, then $\frac{1}{P(A \cup B)} \leq \frac{1}{P(A)}$. Well,

$$P(AB|A) = \frac{P(ABA)}{P(A)} = \frac{P(AB)}{P(A)} \geq \frac{P(AB)}{P(A \cup B)} = \frac{P((AB) \cap (A \cup B))}{P(A \cup B)} = P(AB|A \cup B)$$

□

Exercise 3.19 (Ross' TE 3.2)**.** *Let $A \subseteq B$. Express the following probabilities as simply as possible:*

$$P(A|B), \quad P(A|B^c), \quad P(B|A), \quad P(B|A^c)$$

Solution. Since $A \subseteq B$, then $A \cap B = A$ and $A \cap B^{\mathsf{c}} = \emptyset$. We express the probabilities as simply as possible, aiming to express them in terms of $P(A)$, $P(B)$, and constants:

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A)}{P(B)} \tag{3}$$

$$P(A|B^{\mathsf{c}}) = \frac{P(AB^{\mathsf{c}})}{P(B^{\mathsf{c}})} = 0 \tag{4}$$

$$P(B|A) = \frac{P(BA)}{P(A)} = \frac{P(A)}{P(A)} = 1 \tag{5}$$

$$P(B|A^{\mathsf{c}}) = \frac{P(A^{\mathsf{c}}B)}{P(A^{\mathsf{c}})} = \frac{P(B) - P(A)}{1 - P(A)} \tag{6}$$

□

Exercise 3.20 (Ross' TE 3.5)**.** *(a) Prove that if E and F are mutually exclusive, then*

$$P(E|E \cup F) = \frac{P(E)}{P(E) + P(F)}$$

(b) Prove that if $E_i, i \geq 1$ are mutually exclusive, then

$$P\left(E_j \mid \bigcup_{i=1}^{\infty} E_i\right) = \frac{P(E_j)}{\sum_{i=1}^{\infty} P(E_i)}$$

Solution. We mainly use the axiom of countable additivity of probability.

(a) Since E, F are mutually exclusive, then $P(E \cup F) = P(E) + P(F)$. Then,

$$P(E \mid E \cup F) = \frac{P(E \cap (E \cup F))}{P(E \cup F)} = \frac{P(E)}{P(E) + P(F)}.$$

(b) Since $E_1, E_2, \ldots$ are mutually exclusive, then $P(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} P(E_i)$. Let $j \geq 1$ be an integer. Then,

$$P\left(E_j | \bigcup_{i=1}^{\infty} E_i\right) = \frac{P(E_j \cap (\bigcup_{i=1}^{\infty} E_i))}{P(\bigcup_{i=1}^{\infty} E_i)} = \frac{P(E_j)}{\sum_{i=1}^{\infty} P(E_i)}$$

□

Exercise 3.21 (Ross' TE 3.6)**.** *Prove that if $E_1, E_2, \ldots, E_n$ are independent events, then*

$$P\left(\bigcup_{i=1}^{n} E_i\right) = 1 - \prod_{i=1}^{n} [1 - P(E_i)].$$

Lemma 3.22. *If $E_1, E_2, \ldots, E_n$ are independent events, then so are the complement events $E_1^{\mathsf{c}}, E_2^{\mathsf{c}}, \ldots, E_n^{\mathsf{c}}$.*

Proof 1. By definition, $E_1^{\mathsf{c}}, E_2^{\mathsf{c}}, \ldots, E_n^{\mathsf{c}}$ are independent if for every integer $1 \leq k \leq n$, for every subset $A = \{i_1, i_2, \ldots, i_k\} \subseteq \{1, 2, \ldots, n\}$ of size k, the following condition holds:

$$P(E_{i_1}^{\mathsf{c}} E_{i_2}^{\mathsf{c}} \cdots E_{i_k}^{\mathsf{c}}) = \prod_{j=1}^{k} P(E_{i_j}^{\mathsf{c}}). \tag{7}$$

We prove that by induction on k.

Base Case: If $k = 1$, then for every subset $\{i\} \subseteq \{1, 2, \ldots, n\}$ we have that $P(E_i^{\mathsf{c}}) = P(E_i^{\mathsf{c}})$.

Inductive Hypothesis (IH): Assume for some integer $1 \leq k \leq n - 1$ that for every subset $A = \{i_1, i_2, \ldots, i_k\} \subseteq \{1, 2, \ldots, n\}$ with size k that the condition (7) holds on A.

Inductive Step: Let $B = \{i_1, i_2, \ldots, i_k, i_{k+1}\} \subseteq \{1, 2, \ldots, n\}$ be a subset with size $k+1$. Let $A = B \setminus \{i_1\}$. Then, $|A| = k$. Therefore, by IH, condition (7) holds on A. Let $F = E_{i_2}^{\mathsf{c}} E_{i_3}^{\mathsf{c}} \cdots E_{i_k}^{\mathsf{c}} E_{i_{k+1}}^{\mathsf{c}}$. Thus,

$$P(F) = P(E_{i_2}^{\mathsf{c}} \cdots E_{i_{k+1}}^{\mathsf{c}}) = P(E_{i_2}^{\mathsf{c}}) \cdots P(E_{i_{k+1}}^{\mathsf{c}})$$

Hence, to prove that condition (7) holds on B, it suffices to prove

$$\begin{aligned}
P(E_{i_1}^{\mathsf{c}} F) &= P(E_{i_1}^{\mathsf{c}}) P(F) \\
&= [1 - P(E_{i_1})][1 - P(F^{\mathsf{c}})] \\
&= 1 - P(E_{i_1}) - P(F^{\mathsf{c}}) + P(E_{i_1}) P(F^{\mathsf{c}}) \\
&= P(E_{i_1}^{\mathsf{c}}) - P(F^{\mathsf{c}}) + P(E_{i_1}) P(F^{\mathsf{c}})
\end{aligned}$$

First, we show that $P(E_{i_1})P(F^{\mathsf{c}}) = P(E_{i_1} F^{\mathsf{c}})$. Well,

$$\begin{aligned}
P\left(E_{i_1} F^{\mathsf{c}}\right) &= P\left(E_{i_1} \left[\bigcap_{j=2}^{k+1} E_{i_j}^{\mathsf{c}}\right]^{\mathsf{c}}\right) \\
&= P\left(E_{i_1} \bigcup_{j=2}^{k+1} E_{i_j}\right) \\
&= P\left(\bigcup_{j=2}^{k+1} E_{i_1} E_{i_j}\right) \\
&= \sum_{l=1}^{k} (-1)^{l-1} \sum_{\substack{\{j_1,\ldots,j_l\} \\ \subseteq \{i_2,\ldots,i_{k+1}\}}} P(E_{i_1} E_{j_1} E_{j_2} \cdots E_{j_l}),
\end{aligned}$$

by the inclusion-exclusion principle. Then, by independence of the events $E_1, \ldots, E_n$, we have

$$\begin{aligned}
P(E_{i_1} F^{\mathsf{c}}) &= \sum_{l=1}^{k} (-1)^{l-1} \sum_{\substack{\{j_1,\ldots,j_l\} \\ \subseteq \{i_2,\ldots,i_{k+1}\}}} P(E_{i_1}) P(E_{j_1}) P(E_{j_2}) \cdots P(E_{j_l}) \\
&= P(E_{i_1}) \left[\sum_{l=1}^{k} (-1)^{l-1} \sum_{\substack{\{j_1,\ldots,j_l\} \\ \subseteq \{i_2,\ldots,i_{k+1}\}}} P(E_{j_1}) P(E_{j_2}) \cdots P(E_{j_l}))\right] \\
&= P(E_{i_1}) \left[\sum_{l=1}^{k} (-1)^{l-1} \sum_{\substack{\{j_1,\ldots,j_l\} \\ \subseteq \{i_2,\ldots,i_{k+1}\}}} P(E_{j_1} E_{j_2} \cdots E_{j_l})\right] \\
&= P(E_{i_1}) P\left(\bigcup_{j=2}^{k+1} E_{i_j}\right) \\
&= P(E_{i_1}) P(F^{\mathsf{c}}).
\end{aligned}$$

Now, it suffices to prove

$$P(E_{i_1}^{\mathsf{c}} F) = P(E_{i_1}^{\mathsf{c}}) - P(F^{\mathsf{c}}) + P(E_{i_1} F^{\mathsf{c}}). \tag{8}$$

Well, note that

$$P(E_{i_1}^{\mathsf{c}}) = P(E_{i_1}^{\mathsf{c}} F) + P(E_{i_1}^{\mathsf{c}} F^{\mathsf{c}}) \tag{9}$$

$$P(F^{\mathsf{c}}) = P(E_{i_1} F^{\mathsf{c}}) + P(E_{i_1}^{\mathsf{c}} F^{\mathsf{c}}). \tag{10}$$

Plugging into equation (8), we obtain

$$\begin{aligned}
RHS &= [P(E_{i_1}^{\mathsf{c}} F) + \cancel{P(E_{i_1}^{\mathsf{c}} F^{\mathsf{c}})}] - [\cancel{P(E_{i_1} F^{\mathsf{c}})} + \cancel{P(E_{i_1}^{\mathsf{c}} F^{\mathsf{c}})}] + \cancel{P(E_{i_1} F^{\mathsf{c}})} \\
&= P(E_{i_1}^{\mathsf{c}} F) = LHS,
\end{aligned}$$

hence condition (7) holds on B. This completes the inductive step.

□

Proof 2. We present an alternative proof of Lemma (3.22). For every integer $1 \leq i \leq n$, define the indicator variable of the event E_i as follows:

$$I_i = \begin{cases} 1 & \text{if } E_i \text{ occurs} \\ 0 & \text{otherwise.} \end{cases}$$

We use the fact that $E_1, E_2, \ldots, E_n$ are independent if and only if the corresponding indicator variables $I_1, I_2, \ldots, I_n$ are independent (PROVE THIS!). Assume independence. Observe that the indicator variables for $E_1^{\mathsf{c}}, E_2^{\mathsf{c}}, \ldots, E_n^{\mathsf{c}}$ can be defined as $(I_1 - 1)^2, (I_2 - 1)^2, \ldots, (I_n - 1)^2$. Then, for every $\{i_1, i_2, \ldots, i_k\} \subseteq \{1, 2, \ldots, n\}$, observe that the indicator random variable for $E_{i_1}^{\mathsf{c}} \cdots E_{i_k}^{\mathsf{c}}$ can be defined as $\prod_{j=1}^{k} (I_j - 1)^2$, and so we have

$$\begin{aligned} P(E_{i_1}^{\mathsf{c}} E_{i_2}^{\mathsf{c}} \cdots E_{i_k}^{\mathsf{c}}) &= \mathrm{E}\left[\prod_{j=1}^{k} (I_j - 1)^2\right] \\ &= \prod_{j=1}^{k} \mathrm{E}\left[(I_j - 1)^2\right] \\ &= \prod_{j=1}^{k} P(E_{i_j}^{\mathsf{c}}) \end{aligned}$$

since functions of independent random variables are independent. Therefore, $E_1^{\mathsf{c}}, E_2^{\mathsf{c}}, \ldots, E_n^{\mathsf{c}}$ are independent. □

Solution. Assume that $E_1, E_2, \ldots, E_n$ are independent.

$$\begin{aligned} P\left(\bigcup_{i=1}^{n} E_i\right) &= P\left(\left(\bigcap_{i=1}^{n} E_i^{\mathsf{c}}\right)^{\mathsf{c}}\right) \\ &= 1 - P\left(\bigcap_{i=1}^{n} E_i^{\mathsf{c}}\right). \end{aligned}$$

By Lemma (3.22), we have that $E_1^{\mathsf{c}}, E_2^{\mathsf{c}}, \ldots, E_n^{\mathsf{c}}$ are independent. Therefore,

$$\begin{aligned} P\left(\bigcup_{i=1}^{n} E_i\right) &= 1 - \prod_{i=1}^{n} P(E_i^{\mathsf{c}}) \\ &= 1 - \prod_{i=1}^{n} [1 - P(E_i)]. \end{aligned}$$

□

Exercise 3.23. *Give an example of three events that are pairwise-independent but not mutually independent.*

Exercise 3.24. *Give an example of n events that are k-wise independent but, for every $k' > k$, not k'-wise independent.*

4 Random Variables

Let me state the pmf (with support) and derive the expectation and variance for each of the discrete random variables I learned. To that end, let me derive, in general, the n-th moment for each.

It will remain to prove that the stated pmf is actually a pmf (and later to derive the moment-generating function) for each.

4.1 Common Discrete Distributions

RV	PMF	Support	$\mathrm{E}[X]$	$\mathrm{Var}(X)$
$\mathrm{Bernoulli}(p)$	$p^k(1-p)^{1-k}$	$k=0,1$	p	$p(1-p)$
$\mathrm{b}(n,p)$	$\binom{n}{k}p^k(1-p)^{n-k}$	$k=0,1,\ldots,n$	np	$np(1-p)$
$\mathrm{Geom}(p)$	$(1-p)^{k-1}p$	$k=1,2,\ldots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
$\mathrm{nb}(r,p)$	$\binom{k-1}{r-1}(1-p)^{k-r}p^r$	$k=r,r+1,\ldots$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$
$\mathrm{Poisson}(\lambda)$	$e^{-\lambda}\frac{\lambda^k}{k!}$	$k=0,1,\ldots$	λ	λ
$\mathrm{hg}(N,M,n)$	$\frac{\binom{M}{m}\binom{N-M}{n-m}}{\binom{N}{n}}$	$m=0,1,\ldots,n$	$\frac{nM}{N}$	$\frac{nM}{N}\left(\frac{(n-1)(M-1)}{N-1}+1-\frac{nM}{N}\right)$
$\mathrm{Zeta}(s)$	$\frac{1}{k^s}\frac{1}{\zeta(s)},\quad s>1$	$k=1,2,\ldots$	$\frac{\zeta(s-1)}{\zeta(s)},\quad s>2$	$\frac{\zeta(s)\zeta(s-2)-\zeta(s-1)^2}{\zeta(s)^2},\quad s>3$

Remark 4.1. *The parameters are:*

- $p \in \mathbb{R}, 0 < p < 1$.
- $n \in \mathbb{Z}^+$, $p \in \mathbb{R}, 0 < p < 1$.
- $p \in \mathbb{R}, 0 < p < 1$.
- $r \in \mathbb{Z}^+$, $p \in \mathbb{R}, 0 < p < 1$.
- $\lambda > 0$.
- $N, M, n \in \mathbb{Z}^+$, $n, M \leq N$.
- $s > 1$.

Remark 4.2. *For the hypergeometric random variable, the support is*

$$\{m \in \mathbb{Z} : \max\{0, n-(N-M)\} \leq m \leq \min\{n, M\}\},$$

but under the convention that $\binom{n}{k} = 0$ *if* $n < k$*, the support in the table works and is more convenient.*

Exercise 4.3. *Verify that the pmf for the Bernoulli random variable is actually a pmf.*

Solution. $1 = (1-p) + p = p^0(1-p)^1 + p^1(1-p)^0$. □

Exercise 4.4. *Verify that the pmf for the binomial random variable is actually a pmf.*

Solution. By Binomial Theorem, we have

$$1 = 1^n = (p + (1-p))^n = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k}.$$

□

Exercise 4.5. *Derive the* l*-th moment for the binomial random variable.*

Solution. Let $X \sim b(n,p)$. Then,

$$\begin{aligned}\mathrm{E}\left[X^l\right] &= \sum_{k=0}^{n} k^l \binom{n}{k} p^k (1-p)^{n-k} \\ &= \sum_{k=1}^{n} k^{l-1} k \binom{n}{k} p^k (1-p)^{n-k}.\end{aligned}$$

By Lemma (4.14), we have

$$k\binom{n}{k} = n\binom{n-1}{k-1},$$

hence

$$\begin{aligned}
\mathrm{E}\left[X^l\right] &= \sum_{k=1}^{n} k^{l-1} n \binom{n-1}{k-1} p^k (1-p)^{n-k} \\
&= \sum_{k=0}^{n-1} (k+1)^{l-1} n \binom{n-1}{k} p^{k+1} (1-p)^{n-(k+1)} \\
&= np \sum_{k=0}^{n-1} (k+1)^{l-1} \binom{n-1}{k} p^k (1-p)^{n-1-k} \\
&= \boxed{np\mathrm{E}\left[(Y+1)^{l-1}\right], \quad \text{where } Y \sim b(n-1,p).}
\end{aligned}$$

□

Exercise 4.6. *Derive the expectation and variance of the binomial random variable.*

Solution. Let $X \sim b(n,p)$. By Exercise (4.5), with $Y \sim b(n-1,p)$, we obtain

$$\begin{aligned}
\mathrm{E}[X] &= np\mathrm{E}\left[(Y+1)^0\right] = np\mathrm{E}[1] = \boxed{np} \\
\mathrm{Var}(X) &= \mathrm{E}\left[X^2\right] - \mathrm{E}[X]^2 \\
&= np\mathrm{E}[Y+1] - (np)^2 = np(\mathrm{E}[Y]+1) - (np)^2 = np((n-1)p+1) - (np)^2 = \boxed{np(1-p)}.
\end{aligned}$$

□

Exercise 4.7. *Derive the expectation and variance of the Bernoulli random variable.*

Solution. Equivalently, let $X \sim b(1,p)$. By Exercise (4.6), we obtain

$$\mathrm{E}[X] = 1 \cdot p = \boxed{p}, \quad \mathrm{Var}(X) = 1 \cdot p(1-p) = \boxed{p(1-p)}.$$

□

Exercise 4.8. *Verify that the pmf for the geometric random variable is actually a pmf.*

Solution. Using geometric series, we have

$$\begin{aligned}
\frac{1}{p} &= \frac{1}{1-(1-p)} = \sum_{k=0}^{\infty} (1-p)^k = \sum_{k=1}^{\infty} (1-p)^{k-1} \\
\iff 1 &= \sum_{k=1}^{\infty} (1-p)^{k-1} p.
\end{aligned}$$

□

Exercise 4.9. *Derive the first moment for the geometric random variable.*

Solution. Let $X \sim Geometric(p)$. Thus,

$$\begin{aligned}
\mathrm{E}[X] &= \sum_{k=1}^{\infty} k(1-p)^{k-1} p = p \sum_{k=1}^{\infty} k(1-p)^{k-1} \\
&= p \sum_{k=1}^{\infty} \frac{d}{dp}\left[-(1-p)^k\right] = -p\left(\frac{d}{dp}\sum_{k=1}^{\infty}(1-p)^k\right) \\
&= -p\left(\frac{d}{dp}\left[\frac{1}{p}\sum_{k=1}^{\infty}(1-p)^k p\right]\right) = -p\left(\frac{d}{dp}\left[\frac{1}{p}\sum_{k=2}^{\infty}(1-p)^{k-1} p\right]\right)
\end{aligned}$$

$$= -p\left(\frac{d}{dp}\left[\frac{1}{p}\left(\underbrace{\sum_{k=1}^{\infty}(1-p)^{k-1}p}_{1} - p\right)\right]\right) = -p\frac{d}{dp}\frac{1-p}{p}$$

$$= -p\frac{-p-(1-p)}{p^2} = \boxed{\frac{1}{p}}.$$

□

Exercise 4.10. *Derive the n-th moment for the geometric random variable.*

Solution. Let $X \sim Geometric(p)$. We derive a recursive formula. Let n be a positive integer. Thus,

$$\begin{aligned}
\mathrm{E}\left[X^n\right] &= \sum_{k=1}^{\infty} k^n(1-p)^{k-1}p = p\sum_{k=1}^{\infty} k^n(1-p)^{k-1} \\
&= p\sum_{k=1}^{\infty}\frac{d}{dp}\left[-k^{n-1}(1-p)^k\right] = -p\frac{d}{dp}\sum_{k=1}^{\infty} k^{n-1}(1-p)^k \\
&= -p\frac{d}{dp}\left[\frac{1-p}{p}\sum_{k=1}^{\infty} k^{n-1}(1-p)^{k-1}p\right] = p\frac{d}{dp}\left[\frac{p-1}{p}\mathrm{E}\left[X^{n-1}\right]\right] \\
&= p\left[\frac{p-(p-1)}{p^2}\mathrm{E}\left[X^{n-1}\right] + \frac{p-1}{p}\frac{d}{dp}\mathrm{E}\left[X^{n-1}\right]\right] \\
&= \boxed{\frac{1}{p}\mathrm{E}\left[X^{n-1}\right] - (1-p)\frac{d}{dp}\mathrm{E}\left[X^{n-1}\right]}.
\end{aligned}$$

□

Exercise 4.11. *Derive the variance for the geometric random variable.*

Solution. Let $X \sim Geometric(p)$. We compute $\mathrm{Var}\left(X\right) = \mathrm{E}\left[X^2\right] - \mathrm{E}\left[X\right]^2$. Knowing that $\mathrm{E}\left[X\right] = \frac{1}{p}$, we have $\mathrm{Var}\left(X\right) = \mathrm{E}\left[X^2\right] - \frac{1}{p^2}$. Using the recursive formula for the n-th moment by Exercise (4.10), we compute

$$\begin{aligned}
\mathrm{E}\left[X^2\right] &= \frac{1}{p}\mathrm{E}\left[X\right] - (1-p)\frac{d}{dp}\mathrm{E}\left[X\right] \\
&= \frac{1}{p}\frac{1}{p} - (1-p)\frac{-1}{p^2} \\
&= \frac{2-p}{p^2}.
\end{aligned}$$

Thus, we obtain

$$\mathrm{Var}\left(X\right) = \frac{2-p}{p^2} - \frac{1}{p^2} = \boxed{\frac{1-p}{p^2}}$$

□

Exercise 4.12. *Verify that the pmf for the negative binomial random variable is actually a pmf.*

Solution. Write

$$\begin{aligned}
\frac{1}{p^r} &= \left(\frac{1}{1-(1-p)}\right)^r = \left(\sum_{k=0}^{\infty}(1-p)^k\right)^r \\
&= \underbrace{\left[1+(1-p)+(1-p)^2+\dots\right]\cdots\left[1+(1-p)+(1-p)^2+\dots\right]}_{r \text{ times}}
\end{aligned}$$

Now, we reason combinatorially to find the coefficients for the terms of the series in the RHS. Each term on the right is of the form $(1-p)^k$ where $k \geq 0$ is an integer. The coefficient of a term $(1-p)^k$ is the number of ways we can write

$$(1-p)^k = (1-p)^{k_1+\dots+k_r} = \prod_{i=1}^{r}(1-p)^{k_i}$$

where $k_i \geq 0$ for each $i = 1, 2, \ldots, r$. This is the weak composition of k into r; there are $\binom{k+r-1}{r-1}$ ways to do so. Therefore, we obtain

$$\begin{aligned}
\frac{1}{p^r} &= \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} (1-p)^k \\
&= \sum_{k=r}^{\infty} \binom{k-1}{r-1} (1-p)^{k-r} \\
\iff 1 &= \sum_{k=r}^{\infty} \binom{k-1}{r-1} (1-p)^{k-r} p^r.
\end{aligned}$$

□

Exercise 4.13. *Derive the first moment for the negative binomial random variable.*

Lemma 4.14.

$$k\binom{n}{k} = n\binom{n-1}{k-1}$$

Proof.

$$k\binom{n}{k} = k\frac{n!}{k!(n-k)!} = k\frac{n(n-1)!}{k(k-1)!(n-k)!} = n\frac{(n-1)!}{(k-1)!(n-k)!} = n\binom{n-1}{k-1}$$

□

Solution. Let $X \sim nb(r, p)$. Thus,

$$\mathrm{E}[X] = \sum_{k=r}^{\infty} k\binom{k-1}{r-1}(1-p)^{k-r}p^r.$$

By Lemma (4.14) the following:

$$k\binom{k-1}{r-1} = r\binom{k}{r}.$$

Using Pascal's identity, expand into the following:

$$k\binom{k-1}{r-1} = r\left[\binom{k-1}{r} + \binom{k-1}{r-1}\right]. \tag{11}$$

Plugging this into the first moment sum, we have

$$\begin{aligned}
\mathrm{E}[X] &= \sum_{k=r}^{\infty} r\left[\binom{k-1}{r} + \binom{k-1}{r-1}\right](1-p)^{k-r}p^r \\
&= r\left[\sum_{k=r}^{\infty}\binom{k-1}{r}(1-p)^{k-r}p^r + \sum_{k=r}^{\infty}\binom{k-1}{r-1}(1-p)^{k-r}p^r\right] \\
&= r\left[\frac{1-p}{p}\sum_{k=r}^{\infty}\binom{k-1}{(r+1)-1}(1-p)^{k-(r+1)}p^{r+1} + \sum_{k=r}^{\infty}\binom{k-1}{r-1}(1-p)^{k-r}p^r\right].
\end{aligned}$$

Since both sums are pmfs, we obtain

$$\mathrm{E}[X] = r\left(\frac{1-p}{p} + 1\right) = \boxed{\frac{r}{p}}$$

□

Exercise 4.15. *Derive the n-th moment for the negative binomial random variable.*

Solution. Let $X \sim nb(r,p)$. Thus,

$$\mathrm{E}\left[X^n\right] = \sum_{k=r}^{\infty} k^n \binom{k-1}{r-1}(1-p)^{k-r}p^r.$$

Using the identity (11), we have

$$\begin{aligned}
\mathrm{E}\left[X^n\right] &= \sum_{k=r}^{\infty} k^{n-1} r \left[\binom{k-1}{r} + \binom{k-1}{r-1}\right](1-p)^{k-r}p^r \\
&= r\left[\sum_{k=r}^{\infty} k^{n-1}\binom{k-1}{r}(1-p)^{k-r}p^r + \sum_{k=r}^{\infty} k^{n-1}\binom{k-1}{r-1}(1-p)^{k-r}p^r\right] \\
&= r\left[\frac{1-p}{p}\sum_{k=r}^{\infty} k^{n-1}\binom{k-1}{(r+1)-1}(1-p)^{k-(r+1)}p^{r+1} + \sum_{k=r}^{\infty} k^{n-1}\binom{k-1}{r-1}(1-p)^{k-r}p^r\right] \\
&= \boxed{r\left[\frac{1-p}{p}\mathrm{E}\left[Y^{n-1}\right] + \mathrm{E}\left[X^{n-1}\right]\right], \quad \text{where } Y \sim nb(r+1,p)}.
\end{aligned}$$

□

Exercise 4.16. *Derive the variance for the negative binomial random variable.*

Solution. Let $X \sim nb(r,p)$. We compute $\mathrm{Var}\left(X\right) = \mathrm{E}\left[X^2\right] - \mathrm{E}\left[X\right]^2$. By Exercise (4.13), we have $\mathrm{Var}\left(X\right) = \mathrm{E}\left[X^2\right] - \frac{r^2}{p^2}$. We compute $\mathrm{E}\left[X^2\right]$ using the recursive formula in Exercise (4.15). Let $Y \sim nb(r+1,p)$. Thus,

$$\begin{aligned}
\mathrm{E}\left[X^2\right] &= r\left[\frac{1-p}{p}\mathrm{E}\left[Y\right] + \mathrm{E}\left[X\right]\right] \\
&= r\left[\frac{1-p}{p}\frac{r+1}{p} + \frac{r}{p}\right] \\
&= r\left[\frac{r+1-pr-p+pr}{p^2}\right] \\
&= \frac{r^2 + r(1-p)}{p^2}.
\end{aligned}$$

Plugging back into the variance equation, we have

$$\mathrm{Var}\left(X\right) = \frac{r^2 + r(1-p)}{p^2} - \frac{r^2}{p^2} = \boxed{\frac{r(1-p)}{p^2}}.$$

□

Exercise 4.17. *Verify that the pmf for the Poisson random variable is actually a pmf.*

Solution. Expand the exponential function into its Maclaurin series:

$$e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}.$$

Then, divide both sides by e^{λ} and obtain

$$1 = \sum_{k=0}^{\infty} e^{-\lambda}\frac{\lambda^k}{k!}.$$

□

Exercise 4.18. *Derive the n-th moment for the Poisson random variable.*

Solution. Let $X \sim Poisson(\lambda)$. Let $n \in \mathbb{Z}^+$. Then,

$$\begin{aligned}
\mathrm{E}\left[X^n\right] &= \sum_{k=0}^{\infty} k^n e^{-\lambda} \frac{\lambda^k}{k!} = \sum_{k=1}^{\infty} k^n e^{-\lambda} \frac{\lambda^k}{k!} \\
&= \lambda \sum_{k=1}^{\infty} k^{n-1} e^{-\lambda} \frac{\lambda^{k-1}}{(k-1)!} \\
&= \lambda \sum_{k=0}^{\infty} (k+1)^{n-1} e^{-\lambda} \frac{\lambda^k}{k!} \\
&= \boxed{\lambda \mathrm{E}\left[(X+1)^{n-1}\right]}.
\end{aligned}$$

□

Exercise 4.19. *Derive the expectation and variance of the Poisson random variable.*

Solution. Let $X \sim Poisson(\lambda)$. By Exercise (4.18), we obtain

$$\begin{aligned}
\mathrm{E}\left[X\right] &= \lambda \mathrm{E}\left[(X+1)^0\right] = \boxed{\lambda}. \\
\mathrm{Var}\left(X\right) &= \mathrm{E}\left[X^2\right] - \mathrm{E}\left[X\right]^2 = \lambda \mathrm{E}\left[X+1\right] - \lambda^2 = \lambda(\lambda+1) - \lambda^2 = \boxed{\lambda}.
\end{aligned}$$

□

Exercise 4.20. *Verify that the pmf for the hypergeometric random variable is actually a pmf.*

Solution. Using Exercise (1.4), we obtain

$$\binom{N}{n} = \binom{M + (N-M)}{n} = \sum_{m=0}^{n} \binom{M}{m} \binom{N-M}{n-m}.$$

Dividing both sides by $\binom{N}{n}$, we have

$$1 = \sum_{m=0}^{n} \frac{\binom{M}{m}\binom{N-M}{n-m}}{\binom{N}{n}}.$$

□

Exercise 4.21. *Derive the k-th moment for the hypergeometric distribution.*

Solution. Let $X \sim hg(N, M, n)$. Then,

$$\mathrm{E}\left[X^k\right] = \sum_{m=0}^{n} m^k \frac{\binom{M}{m}\binom{N-M}{n-m}}{\binom{N}{n}} = \sum_{m=1}^{n} m^{k-1} n \frac{m\binom{M}{m}\binom{N-M}{n-m}}{n\binom{N}{n}}.$$

By Lemma (4.14), we have

$$m\binom{M}{m} = M\binom{M-1}{m-1}, \quad n\binom{N}{n} = N\binom{N-1}{n-1}.$$

Thus, we obtain

$$\begin{aligned}
\mathrm{E}\left[X^k\right] &= \sum_{m=1}^{n} m^{k-1} n \frac{M\binom{M-1}{m-1}\binom{N-M}{n-m}}{N\binom{N-1}{n-1}} \\
&= \frac{nM}{N} \sum_{m=1}^{n} m^{k-1} \frac{\binom{M-1}{m-1}\binom{N-M}{n-m}}{\binom{N-1}{n-1}} \\
&= \frac{nM}{N} \sum_{m=0}^{n-1} (m+1)^{k-1} \frac{\binom{M-1}{m}\binom{N-M}{n-1-m}}{\binom{N-1}{n-1}} \\
&= \boxed{\frac{nM}{N} \mathrm{E}\left[(Y+1)^{k-1}\right], \quad Y \sim hg(N-1, M-1, n-1)}.
\end{aligned}$$

□

Exercise 4.22. *Derive the expectation and variance for the hypergeometric distribution.*

Solution. Let $X \sim hg(N, M, n)$. Then by Exercise (4.21) we obtain

$$\begin{aligned}
\mathrm{E}[X] &= \boxed{\frac{nM}{N}} \\
\mathrm{Var}(X) &= \mathrm{E}\left[X^2\right] - \mathrm{E}[X]^2 \\
&= \frac{nM}{N}\mathrm{E}[Y+1] - \left(\frac{nM}{N}\right)^2 \quad (\text{where } Y \sim hg(N-1, M-1, n-1)) \\
&= \frac{nM}{N}\left(\frac{(n-1)(M-1)}{N-1} + 1\right) - \left(\frac{nM}{N}\right)^2 \\
&= \boxed{\frac{nM}{N}\left(\frac{(n-1)(M-1)}{N-1} + 1 - \frac{nM}{N}\right)}.
\end{aligned}$$

□

Exercise 4.23. *Verify that the pmf for the zeta random variable is actually a pmf.*

Solution. Let $s > 1$. Recall that

$$\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s}.$$

Dividing both sides by $\zeta(s)$, we have

$$1 = \sum_{k=1}^{\infty} \frac{1}{k^s}\frac{1}{\zeta(s)}.$$

□

Exercise 4.24. *Derive the n-th moment for the zeta distribution.*

Solution. Let $X \sim Zeta(s)$ where $s > 1$. Then,

$$\begin{aligned}
\mathrm{E}[X^n] &= \sum_{k=1}^{\infty} k^n \frac{1}{k^s}\frac{1}{\zeta(s)} \\
&= \sum_{k=1}^{\infty} \frac{1}{k^{s-n}}\frac{1}{\zeta(s)} \\
&= \frac{\zeta(s-n)}{\zeta(s)} \sum_{k=1}^{\infty} \frac{1}{k^{s-n}}\frac{1}{\zeta(s-n)}
\end{aligned}$$

If $n < s - 1$ (so that $s - n > 1$), then the sum is a pmf, hence we obtain

$$\mathrm{E}[X^n] = \boxed{\frac{\zeta(s-n)}{\zeta(s)}}.$$

□

Exercise 4.25. *Derive the expectation and variance for the zeta distribution.*

Solution. Let $X \sim Zeta(s)$ where $s > 1$. By Exercise (4.24), then if $s > 2$, then

$$\mathrm{E}[X] = \frac{\zeta(s-1)}{\zeta(s)}.$$

As for the variance, if $s > 3$, then

$$\begin{aligned}
\mathrm{Var}(X) &= \mathrm{E}\left[X^2\right] - \mathrm{E}[X]^2 \\
&= \frac{\zeta(s-2)}{\zeta(s)} - \frac{\zeta(s-1)^2}{\zeta(s)^2} \\
&= \boxed{\frac{\zeta(s)\zeta(s-2) - \zeta(s-1)^2}{\zeta(s)^2}}.
\end{aligned}$$

□

4.2 Exercises

Exercise 4.26 (Ross' P 4.5)**.** *Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. What are the possible values of X?*

Solution. If n is even, the possible values are $0, 2, 4, \ldots, n$. If n is odd, the possible values are $1, 3, \ldots, n$. □

Exercise 4.27 (Ross' P 4.26)**.** *One of the numbers* 1 *through* 10 *is randomly chosen. You are to try to guess the number chosen by asking questions with "yes-no" answers. Compute the expected number of questions you will need to ask in each of the following two cases:*

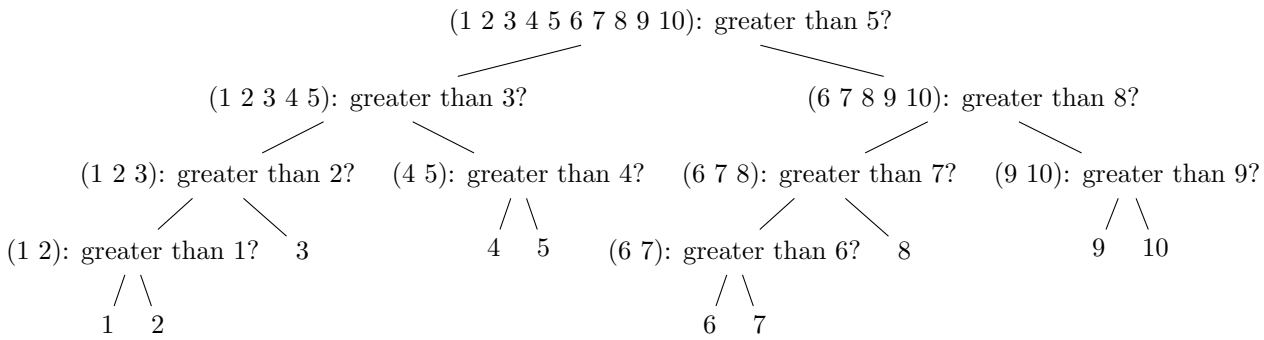
(a) Your ith question is to be "Is it i?" ($i = 1, 2, , 3, \ldots, 10$)

(b) With each question you try to eliminate one-half of the remaining numbers, as nearly as possible.

Solution. We solve somewhat informally.

(a) The expected number of questions is $\sum_{i=1}^{10}(1/10)i = (1/10)\frac{10(11)}{2} = 5.5$.

(b) The following decision tree eliminates one-half of the remaining numbers as nearly as possible with each question.



With probability 6/10, the chosen number is $3, 4, 5, 8, 9$, or 10, in which case you need exactly 3 questions. With probability 4/10, the chosen number is $1, 2, 6$, or 7, in which case you need exactly 4 questions. The expected number of questions is hence $\frac{3(6)+4(4)}{10} = \frac{34}{10} = 3.4$.

□

Exercise 4.28 (Ross' P 4.58)**.** *Compare the Poisson approximation with the correct binomial probability for the following cases:*

(a) $P\{X = 2\}$ when $n = 8, p = .1$;

(b) $P\{X = 9\}$ when $n = 10, p = .95$;

(c) $P\{X = 0\}$ when $n = 10, p = .1$;

(d) $P\{X = 4\}$ when $n = 9, p = .2$.

Solution. We make a table with a row for Poisson and a row for binomial and with a column for each case.

dist. / case	n=8,p=.1 $P\{X$=2$\}$	n=10,p=.95 $P\{X$=9$\}$	n=10,p=.1 $P\{X$=0$\}$	n=9,p=.2 $P\{X$=4$\}$
Poisson(np)	$e^{-0.8}(0.8^2/2)$ =0.144	$e^{-9.5}(9.5^9/9!)$ =0.130	e^{-1} =0.368	$e^{-1.8}(1.8^4/4!)$ =0.072
Binomial(n, p)	$\binom{8}{2}(0.1)^2(0.9)^6$ =0.149	$\binom{10}{9}(0.95)^9(0.05)$ =0.315	0.9^{10} =0.349	$\binom{9}{4}(0.2)^4(0.8)^5$ =0.066
Error of Approx.	3.4%	58.7%	5.4%	9.1%

□

Exercise 4.29 (Ross' P 4.74)**.** *An interviewer is given a list of people she can interview. If the interviewer needs to interview* 5 *people, and if each person (independently) agrees to be interviewed with probability* $\frac{2}{3}$*, what is the probability that her list of people will enable her to obtain her necessary number of interviews if the list consists of (a)* 5 *people and (b)* 8 *people? For part (b), what is the probability that the interviewer will speak to exactly (c)* 6 *people and (d)* 7 *people on the list?*

Solution. Assume that the interviewer will attempt to interview every person on the list regardless of whether the interviewer already obtained her necessary number.

(a) If her list consists of 5 people, the probability is $(2/3)^5 = 32/243$.

(b) If her list consists of 8 people, let $X \sim b(8, \frac{2}{3})$. Then, the desired probability is

$$P\{X \geq 5\} = \sum_{k=5}^{8} P\{X = k\} = \sum_{k=5}^{8} \binom{8}{k} (2/3)^k (1/3)^{8-k}.$$

(c) For part (b), the probability that the interviewer will speak to exactly 6 people is

$$\binom{8}{6} (2/3)^6 (1/3)^2 = \frac{1792}{6561}.$$

(d) For part (b), the probability that the interviewer will speak to exactly 7 people is

$$\binom{8}{7} (2/3)^7 (1/3) = \frac{1024}{6561}.$$

□

Exercise 4.30 (Ross' P 4.75)**.** *A fair coin is continually flipped until heads appears for the* 10*th time. Let* X *denote the number of tails that occur. Compute the probability mass function of* X*.*

Solution. Let Y denote the number of trials needed to get 10 heads. Thus, $Y \sim nb(10, \frac{1}{2})$. Observe that $X = Y - 10$. Therefore,

$$P\{X = k\} = P\{Y - 10 = k\} = P\{Y = k + 10\} = \binom{k+9}{9} (1/2)^{(k+10)-10} (1/2)^{10} = \binom{k+9}{9} (1/2)^{k+10}$$

□

Exercise 4.31 (Ross' P 4.78)**.** *An urn contains* 4 *white and* 4 *black balls. We randomly choose* 4 *balls. If* 2 *of them are white and* 2 *are black, we stop. If not, we replace the balls in the urn and again randomly select* 4 *balls. This continues until exactly* 2 *of the* 4 *chosen are white. What is the probability that we shall make exactly* n *selection?*

Solution. Let $X \sim hg(8, 4, 4)$. Let $p = P\{X = 2\}$. That is,

$$p = \frac{\binom{4}{2}\binom{4}{2}}{\binom{8}{4}} = \frac{36}{70} = \frac{18}{35}.$$

Then, let $Y \sim Geometric(p)$. Then,

$$P\{Y = n\} = (1-p)^{n-1} p = (17/35)^{n-1} (18/35) = \frac{17^{n-1} \cdot 18}{35^n}$$

□

Exercise 4.32 (Ross' P 4.11)**.** *(a) An integer* N *is to be selected at random from* $\{1, 2, \ldots, 10^3\}$ *in the sense that each integer has the same probability of being selected. What is the probability that* N *will be divisible by* 3*? by* 5*? by* 7*? by* 15*? by* 105*? How would your answer change if* 10^3 *is replaced by* 10^k *as* k *becomes larger and larger?*

(b) An important function in number theory–one whose properties can be shown to be related to what is probably the most important unsolved problem of mathematics, the Riemann hypothesis–is the Mobius function $\mu(n)$*, defined for all positive integral values* n *as follows: Factor* n *into its prime factors. If there is a repeated prime factor, as in* $12 = 2 \cdot 2 \cdot 3$ *or* $49 = 7 \cdot 7$*, then* $\mu(n)$ *is defined to equal* 0*. Now let* N *be chosen at random from* $\{1, 2, \ldots, 10^k\}$*, where* k *is large. Determine* $P\{\mu(N) = 0\}$ *as* $k \to \infty$*.*

Solution. (a) Since each integer has the same probability of being selected, then the probability that N will be divisible by d is the fraction where the numerator is the number of integers from 1 to 10^3 divisible by d, $\lfloor 10^3/d \rfloor$, and the denominator is 10^3. Thus, first, $P\{3 \mid N\} = \frac{\lfloor 10^3/3 \rfloor}{10^3} = \frac{333}{1000} = 0.333$. Second, $P\{5 \mid N\} = \frac{\lfloor 10^3/5 \rfloor}{10^3} = \frac{200}{1000} = 0.2$. Third, $P\{7 \mid N\} = \frac{\lfloor 10^3/7 \rfloor}{10^3} = \frac{142}{1000} = 0.142$. Fourth, $P\{15 \mid N\} = \frac{\lfloor 10^3/15 \rfloor}{10^3} = \frac{66}{1000} = 0.066$. Fifth, $P\{105 \mid N\} = \frac{\lfloor 10^3/105 \rfloor}{10^3} = \frac{9}{1000} = 0.009$.

TBC.

□

Exercise 4.33 (Ross' TE 4.1)**.** *There are N distinct types of coupons, and each time one is obtained it will, independently of past choices, be of type i with probability p_i, $i = 1, 2, \ldots, N$. Let T denote the number one needs to select to obtain at least one of each type. Compute $P\{T = n\}$.*

Solution. We compute $P\{T = n\} = P\{T > n-1\} - P\{T > n\}$. Note that $T > n$ if and only if in the first n coupons some type is "missing". For each positive integer $1 \le i \le N$, let $A_{n,i}$ denote the event that there is no coupon of type i in the first n coupons. Thus, $P\{T > n\} = P(\bigcup_{i=1}^N A_{n,i})$. By the inclusion-exclusion principle, we obtain

$$\begin{aligned}
P\{T > n\} = P(\bigcup_{i=1}^N A_{n,i}) &= \sum_{j=1}^N (-1)^{j+1} \sum_{\{i_1, i_2, \ldots, i_j\} \subseteq [N]} P(A_{n,i_1} A_{n,i_2} \cdots A_{n,i_j}) \\
&= \sum_{j=1}^N (-1)^{j+1} \sum_{\{i_1, i_2, \ldots, i_j\} \subseteq [N]} (1 - \sum_{k=1}^j p_{i_k})^n \\
P\{T > n-1\} = P(\bigcup_{i=1}^N A_{n-1,i}) &= \sum_{j=1}^N (-1)^{j+1} \sum_{\{i_1, i_2, \ldots, i_j\} \subseteq [N]} (1 - \sum_{k=1}^j p_{i_k})^{n-1}.
\end{aligned}$$

Thus, we obtain

$$\begin{aligned}
P\{T = n\} &= P\{T > n-1\} - P\{T > n\} \\
&= \sum_{j=1}^N (-1)^{j+1} \sum_{\{i_1, i_2, \ldots, i_j\} \subseteq [N]} (1 - \sum_{k=1}^j p_{i_k})^{n-1} - (1 - \sum_{k=1}^j p_{i_k})^n \\
&= \sum_{j=1}^N (-1)^{j+1} \sum_{\{i_1, i_2, \ldots, i_j\} \subseteq [N]} (1 - \sum_{k=1}^j p_{i_k})^{n-1} (1 - (1 - \sum_{k=1}^j p_{i_k})) \\
&= \boxed{\sum_{j=1}^N (-1)^{j+1} \sum_{\{i_1, i_2, \ldots, i_j\} \subseteq [N]} (1 - \sum_{k=1}^j p_{i_k})^{n-1} (\sum_{k=1}^j p_{i_k})}.
\end{aligned}$$

□

Exercise 4.34 (Ross' TE 4.2)**.** *If X has a distribution function F, what is the distribution of e^X?*

Solution. Let $Y = e^X$. Since e^X is positive, then $F_Y(x) = 0$ if $x \le 0$. Suppose $x > 0$. Then $F_Y(x) = P\{e^X \le x\} = P\{X \le \ln x\}$ since e^X is monotonically increasing, and so $F_Y(x) = F(\ln x)$. □

Exercise 4.35 (Ross' TE 4.3)**.** *If X has distribution function F, what is the distribution function of the random variable $\alpha X + \beta$, where α and β are constants with $\alpha \neq 0$?*

Solution. Let $Y = \alpha X + \beta$. If $\alpha > 0$, then $F_Y(x) = P\{\alpha X + \beta \le x\} = P\{X \le \frac{x-\beta}{\alpha}\} = F(\frac{x-\beta}{\alpha})$. If $\alpha < 0$, then $F_Y(x) = P\{\alpha X + \beta \le x\} = P\{X \ge \frac{x-\beta}{\alpha}\} = 1 - P\{X < \frac{x-\beta}{\alpha}\} = 1 - P\{X \le \frac{x-\beta}{\alpha}\} + P\{X = \frac{x-\beta}{\alpha}\} = 1 - F(\frac{x-\beta}{\alpha}) + P\{X = \frac{x-\beta}{\alpha}\}$; if X is a continuous random variable, then $P\{X = \frac{x-\beta}{\alpha}\} = 0$, hence $F_Y(x) = 1 - F(\frac{x-\beta}{\alpha})$. Therefore,

$$F_Y(x) = \begin{cases} F(\frac{x-\beta}{\alpha}), & \alpha > 0 \\ 1 - F(\frac{x-\beta}{\alpha}) - P\{X = \frac{x-\beta}{\alpha}\}, & \alpha < 0 \end{cases}$$

$$= \begin{cases} F(\frac{x-\beta}{\alpha}), & \alpha > 0 \\ 1 - F(\frac{x-\beta}{\alpha}), & \alpha < 0 \end{cases} \qquad \text{if X is continuous.}$$

□

Exercise 4.36 (Ross' TE 4.4)**.** *The random variable X is said to have the Yule-Simons distribution if*

$$P\{X = n\} = \frac{4}{n(n+1)(n+2)}, \ n \geq 1.$$

(a) Show that the preceding is actually a probability mass function.

(b) Show that $\mathrm{E}[X] = 2$.

(c) Show that $\mathrm{E}\left[X^2\right] = \infty$.

Solution. We show those.

(a) First, observe that

$$\frac{1}{n(n+1)(n+2)} = \frac{(n+2)-(n+1)}{n(n+1)(n+2)} = \frac{1}{n(n+1)} - \frac{1}{n(n+2)}.$$

Then, observe that, in general, we have

$$\begin{aligned} \frac{k}{n(n+k)} &= \frac{A}{n} + \frac{B}{n+k} \\ &= (A+B)n + Ak \\ &\iff A = 1 \wedge B = -1 \\ \iff \frac{k}{n(n+k)} &= \frac{1}{n} - \frac{1}{n+k} \\ \implies \frac{1}{n(n+1)} &= \frac{1}{n} - \frac{1}{n+1} \\ \wedge \frac{1}{n(n+2)} &= \frac{1}{2}(\frac{1}{n} - \frac{1}{n+2}). \end{aligned}$$

Therefore, we have

$$\begin{aligned} \sum_{n=1}^{\infty} P\{X = n\} &= \sum_{n=1}^{\infty} \frac{4}{n(n+1)(n+2)} \\ &= 4\sum_{n=1}^{\infty} \left[\frac{1}{n(n+1)} - \frac{1}{n(n+2)}\right] \\ &= 4\sum_{n=1}^{\infty} \left[(\frac{1}{n} - \frac{1}{n+1}) - (1/2)(\frac{1}{n} - \frac{1}{n+2})\right] \\ &= 4\sum_{n=1}^{\infty} \left[(1/2)(\frac{1}{n}) - \frac{1}{n+1} + (1/2)\frac{1}{n+2}\right] \\ &= 2\sum_{n=1}^{\infty} \frac{1}{n} - 4\sum_{n=1}^{\infty} \frac{1}{n+1} + 2\sum_{n=1}^{\infty} \frac{1}{n+2} \\ &= (2/1) + (2/2) - (4/2) + (2-4+2)\sum_{n=3}^{\infty} \frac{1}{n} \\ &= 1, \end{aligned}$$

hence the given pmf is actually a pmf.

(b) We proceed from the definition of expectation as follows:

$$\mathrm{E}[X] = \sum_{n=1}^{\infty} nP\{X = n\} = \sum_{n=1}^{\infty} n\frac{4}{n(n+1)(n+2)}$$

$$= 4\sum_{n=1}^{\infty} n\left[\frac{1}{n(n+1)} - \frac{1}{n(n+2)}\right] = 4\sum_{n=1}^{\infty}\left[\frac{1}{n+1} - \frac{1}{n+2}\right]$$
$$= 4\left(\sum_{n=1}^{\infty}\frac{1}{n+1} - \sum_{n=1}^{\infty}\frac{1}{n+2}\right) = 4(1/2) = 2.$$

(c) We obtain

$$\begin{aligned}\mathrm{E}\left[X^2\right] &= \sum_{n=1}^{\infty}\frac{n}{(n+1)(n+2)} \\ &> \sum_{n=2}^{\infty}\frac{n}{(n+1)(n+2)},\end{aligned}$$

and we use the integral test. Let $f(x) = x/[(x+1)(x+2)]$. We know that f is positive on $[2,\infty)$. If f is decreasing on $[2,\infty)$ and $\int_2^{\infty} f(x)dx$ diverges, then $\sum_{n=2}^{\infty} n/[(n+1)(n+2)]$ diverges by the integral test, hence $\sum_{n=1}^{\infty} n/[(n+1)(n+2)]$ diverges. It can be verified that

$$\int_2^{\infty}\frac{x}{(x+1)(x+2)}dx = \int_2^{\infty}\left[\frac{-1}{x+1} + \frac{2}{x+2}\right]dx,$$

and so

$$\begin{aligned}\int_2^{\infty}\frac{x}{(x+1)(x+2)}dx &= \lim_{M\to\infty}\left[-\ln(x+1) + 2\ln(x+2)\right]\Big|_2^M \\ &= \lim_{M\to\infty}[\ln\left(\frac{(M+2)^2 2}{(M+1)16}\right)] \\ &= \ln\left(\lim_{M\to\infty}\left(\frac{(M+2)^2 2}{(M+1)16}\right)\right) \\ &= \ln(\infty) = \infty.\end{aligned}$$

We take the derivative of f to show that f is decreasing on $[2,\infty)$, and that completes the proof. We have

$$\begin{aligned}\frac{d}{dx}\left[\frac{x}{(x+1)(x+2)}\right] &= \frac{(x+1)(x+2) - x[(x+2)+(x+1)]}{(x+1)^2(x+2)^2} \\ &= \frac{-x^2+2}{(x+1)^2(x+2)^2},\end{aligned}$$

and the critical points are $\pm\sqrt{2}$. Testing the region $(\sqrt{2},\infty) \supseteq [2,\infty)$ using the point $x = 2$, we have $f'(2) = \frac{-2}{3^2 4^2} < 0$, hence f is indeed decreasing on $[2,\infty)$.

□

Exercise 4.37 (Ross' TE 4.7)**.** *Let X be a random variable having expectation μ and variance σ^2. Find the expectation and variance of*

$$Y = \frac{X-\mu}{\sigma}.$$

Solution.

$$\begin{aligned}\mathrm{E}\left[Y\right] &= \mathrm{E}\left[X/\sigma - \mu/\sigma\right] = (1/\sigma)\mathrm{E}\left[X\right] - \mu/\sigma = \mu/\sigma - \mu/\sigma = 0. \\ \mathrm{Var}\left(Y\right) &= \mathrm{Var}\left(X/\sigma - \mu/\sigma\right) = (1/\sigma)^2\mathrm{Var}\left(X\right) = \sigma^2/\sigma^2 = 1.\end{aligned}$$

□

Exercise 4.38 (Ross' TE 4.10)**.** *Let X be a binomial random variable with parameters n and p. Show that*

$$\mathrm{E}\left[\frac{1}{X+1}\right] = \frac{1-(1-p)^{n+1}}{(n+1)p}.$$

Solution. Let $X \sim b(n,p)$. Then,

$$\begin{aligned}
\mathrm{E}\left[\frac{1}{X+1}\right] &= \sum_{k=0}^{n} \frac{1}{k+1}\binom{n}{k} p^k (1-p)^{n-k} \\
&= \sum_{k=0}^{n} \frac{1}{n+1}\binom{n+1}{k+1} p^k (1-p)^{n-k} \\
&= \frac{1}{(n+1)p} \sum_{k=0}^{n} \binom{n+1}{k+1} p^{k+1} (1-p)^{(n+1)-(k+1)} \\
&= \frac{1}{(n+1)p} \sum_{k=1}^{n+1} \binom{n+1}{k} p^{k} (1-p)^{(n+1)-k} \\
&= \frac{1}{(n+1)p} \left[\sum_{k=0}^{n+1} \binom{n+1}{k} p^{k} (1-p)^{(n+1)-k} - (1-p)^{n+1}\right] \\
&= \frac{1-(1-p)^{n+1}}{(n+1)p}.
\end{aligned}$$

□

Exercise 4.39 (Ross' TE 4.15)**.** *Suppose that n independent tosses of a coin having probability p of coming up heads are made. Show that the probability that an even number of heads results is $\frac{1}{2}\left[1+(q-p)^n\right]$, where $q = 1-p$. Do this by proving and then utilizing the identity*

$$\sum_{i=0}^{\lfloor n/2 \rfloor} \binom{n}{2i} p^{2i} q^{n-2i} = \frac{1}{2}\left[(p+q)^n + (q-p)^n\right].$$

Solution. Let p, q be real numbers. Observe

$$\begin{aligned}
(q+p)^n = \sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k}, \quad (q-p)^n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} p^k q^{n-k}, \\
\implies (p+q)^n + (q-p)^n = \sum_{i=0}^{\lfloor n/2 \rfloor} 2\binom{n}{2i} p^{2i} q^{n-2i}.
\end{aligned}$$

Dividing both sides by 2 proves the identity. Now, let $X \sim b(n,p)$. The probability of getting an even number of heads is

$$\begin{aligned}
\sum_{i=0}^{\lfloor n/2 \rfloor} P\{X = 2i\} &= \sum_{i=0}^{\lfloor n/2 \rfloor} \binom{n}{2i} p^{2i} (1-p)^{n-2i} \\
&= \frac{1}{2}\left[((1-p)+p)^n + ((1-p)-p)^n\right] = \frac{1}{2}\left[1+(q-p)^n\right], \quad \text{where } q = 1-p.
\end{aligned}$$

□

Exercise 4.40 (Ross' TE 4.16)**.** *Let X be a Poisson random variable with parameter λ. Show that $P\{X = i\}$ increases monotonically and then decreases monotonically as i increases, reaching its maximum when i is the largest integer not exceeding λ.*

Solution. Equivalently, we prove that for every positive integer i, the following holds:

$$\frac{P\{X=i\}}{P\{X=i-1\}} \geq 1 \iff i \leq \lambda.$$

Well, recalling the pmf of Poisson, we have

$$\frac{P\{X=i\}}{P\{X=i-1\}} = \frac{e^{-\lambda}(\lambda^i/i!)}{e^{-\lambda}(\lambda^{i-1}/(i-1)!)} = \frac{\lambda}{i} \geq 1 \iff \lambda \geq i.$$

□

Exercise 4.41 (Ross' TE 4.17)**.** *Let X be a Poisson random variable with parameter λ.*

(a) *Show that* $P\{X \text{ is even}\} = \frac{1}{2}\left[1 + e^{-2\lambda}\right]$ *by using the result of Ross' TE 4.15 and the relationship between Poisson and binomial random variables.*

(b) *Verify the formula in part (a) directly by making use of the expansion* $e^{-\lambda} + e^{\lambda}$.

Solution. Let $X \sim Poisson(\lambda)$.

(a) Let $Y_n \sim b(n, \lambda/n)$. We know that $Y_n \xrightarrow{d} X$. Then,

$$\begin{aligned}
P\{X \text{ is even}\} &= \sum_{i=0}^{\infty} P\{X = 2i\} = \sum_{i=0}^{\infty} \lim_{n\to\infty} P\{Y_n = 2i\} \\
&= \lim_{n\to\infty} \sum_{i=0}^{\infty} P\{Y_n = 2i\} \quad \text{(THIS IS NONTRIVIAL! JUSTIFY.)} \\
&= \lim_{n\to\infty} P\{Y_n \text{ is even}\} \\
&= \lim_{n\to\infty} \frac{1}{2}\left[(1 + ((1 - \frac{\lambda}{n}) - \frac{\lambda}{n})^n\right] \quad \text{(using the identity in Exercise (4.39))} \\
&= \frac{1}{2}\left[1 + \lim_{n\to\infty}\left(1 + \frac{-2\lambda}{n}\right)^n\right] \quad \text{(knowing that the limit exists)} \\
&= \frac{1}{2}\left[1 + e^{-2\lambda}\right].
\end{aligned}$$

(b) Using Maclaurin series expansions, we have

$$\begin{aligned}
e^{\lambda} + e^{-\lambda} &= \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} + \sum_{k=0}^{\infty} (-1)^k \frac{\lambda^k}{k!} = \sum_{i=0}^{\infty} 2\frac{\lambda^{2i}}{(2i)!}. \\
\implies \frac{1}{2}\left[1 + e^{-2\lambda}\right] &= e^{-\lambda}\frac{1}{2}\left[e^{\lambda} + e^{-\lambda}\right] = \sum_{i=0}^{\infty} e^{-\lambda}\frac{\lambda^{2i}}{(2i)!} = P\{X \text{ is even}\}.
\end{aligned}$$

□

Exercise 4.42 (Ross' TE 4.18)**.** *Let X be a Poisson random variable with parameter λ. What value of λ maximizes $P\{X = k\},\ k \geq 0$?*

Solution. Let $f(\lambda) = e^{-\lambda}\frac{\lambda^k}{k!}$. If $k = 0$, then $f(\lambda) = e^{-\lambda}$ is maximized at $\lambda = 0$. Assume $k \geq 1$. Then,

$$\frac{d}{d\lambda} f(\lambda) = -e^{-\lambda}\frac{\lambda^k}{k!} + e^{-\lambda}\frac{k\lambda^{k-1}}{k!} = e^{-\lambda}\frac{\lambda^{k-1}}{k!}(k - \lambda), \text{ hence } \frac{d}{d\lambda} f(\lambda) = 0 \iff \lambda = k.$$

At $\lambda = k$, $f'(\lambda) = 0$, and $\lambda < k \implies f'(\lambda) > 0$, and $\lambda > k \implies f'(\lambda) < 0$. Therefore, $\lambda = k$ maximizes $f(\lambda) = P\{X = k\}$ where $X \sim Poisson(\lambda)$. □

Exercise 4.43 (Ross' TE 4.26)**.** *Prove*

$$\sum_{i=0}^{n} e^{-\lambda}\frac{\lambda^i}{i!} = \frac{1}{n!}\int_{\lambda}^{\infty} e^{-x}x^n dx.$$

Solution. We use induction on n to prove that for every nonnegative integer n, we have

$$\int_{\lambda}^{\infty} e^{-x}x^n dx = \sum_{i=0}^{n} n! e^{-\lambda}\frac{\lambda^i}{i!}$$

In the base case, we have $\int_{\lambda}^{\infty} e^{-x}x^0 dx = -e^{-x}\big|_{\lambda}^{\infty} = e^{-\lambda} = 0! e^{-\lambda}\frac{\lambda^0}{0!}$.
Suppose for some integer $n \geq 1$ that

$$\int_{\lambda}^{\infty} e^{-x}x^{n-1} dx = \sum_{i=0}^{n-1} (n-1)! e^{-\lambda}\frac{\lambda^i}{i!}.$$

Then, we use integration by parts. Let $u = x^n$ and $dv = e^{-x}dx$. Then, $du = nx^{n-1}dx$ and $v = -e^{-x}$. Thus,

$$\begin{aligned}
\int_\lambda^\infty e^{-x}x^n dx &= [-e^{-x}x^n]\big|_\lambda^\infty - \int_\lambda^\infty -e^{-x}nx^{n-1}dx \\
&= e^{-\lambda}\lambda^n + n\int_\lambda^\infty e^{-x}x^{n-1}dx \\
&= e^{-\lambda}\lambda^n + n\sum_{i=0}^{n-1}(n-1)!e^{-\lambda}\frac{\lambda^i}{i!} \\
&= n!e^{-\lambda}\frac{\lambda^n}{n!} + \sum_{i=0}^{n-1} n!e^{-\lambda}\frac{\lambda^i}{i!} \\
&= \sum_{i=0}^{n} n!e^{-\lambda}\frac{\lambda^i}{i!}.
\end{aligned}$$

□

Exercise 4.44 (Ross' TE 4.28)**.** *Let $X \sim nb(r,p)$, and let $Y \sim b(n,p)$. Show that*

$$P\{X > n\} = P\{Y < r\}$$

by proving and utilizing the identity

$$\sum_{k=n+1}^{\infty}\binom{k-1}{r-1}p^r(1-p)^{k-r} = \sum_{k=0}^{r-1}\binom{n}{k}p^k(1-p)^{n-k}.$$

Solution. Let $q = 1-p$. Well, we know

$$\begin{aligned}
P\{X > n\} &= \sum_{k=n+1}^{\infty} P\{X = k\} = \sum_{k=n+1}^{\infty}\binom{k-1}{r-1}p^r q^{k-r} =: L_n, \\
P\{Y < r\} &= \sum_{k=0}^{r-1} P\{Y = k\} = \sum_{k=0}^{r-1}\binom{n}{k}p^k q^{n-k} =: R_n.
\end{aligned}$$

We prove for every integer $n \geq 0$ that $L_n = R_n$ by induction on n. In the base case, we have

$$L_0 = \sum_{k=1}^{\infty}\binom{k-1}{r-1}p^r q^{k-r} = \sum_{k=r}^{\infty}\binom{k-1}{r-1}p^r q^{k-r} = 1$$

since by convention $\binom{k-1}{r-1} = 0$ when $k < r$, and we have

$$R_0 = \sum_{k=0}^{r-1}\binom{0}{k}p^k q^{-k} = \binom{0}{0}p^0 q^{-0} = 1 = L_0$$

since by convention $\binom{0}{k} = 0$ when $k > 0$.

Suppose for some integer $n \geq 0$ that $L_{n-1} = R_{n-1}$. We show that $L_n = R_n$. By the inductive hypothesis, we equivalently show that $L_{n-1} - L_n = R_{n-1} - R_n$. To that end, we show that both sides equal the same quantity.

$$\begin{aligned}
L_{n-1} - L_n &= \sum_{k=n}^{\infty}\binom{k-1}{r-1}p^r q^{k-r} - \sum_{k=n+1}^{\infty}\binom{k-1}{r-1}p^r q^{k-r} \\
&= \binom{n-1}{r-1}p^r q^{n-r}. \\
R_{n-1} - R_n &= \sum_{k=0}^{r-1}\binom{n-1}{k}p^k q^{n-1-k} - \sum_{k=0}^{r-1}\binom{n}{k}p^k q^{n-k} \\
&= \sum_{k=0}^{r-1}\binom{n-1}{k}p^k q^{n-1-k} - \sum_{k=0}^{r-1}\left[\binom{n-1}{k} + \binom{n-1}{k-1}\right]p^k q^{n-k}
\end{aligned}$$

$$
\begin{aligned}
&= \sum_{k=0}^{r-1} \binom{n-1}{k} p^k q^{n-1-k} - \left[\sum_{k=0}^{r-1} \binom{n-1}{k} p^k q^{n-k} + \sum_{k=0}^{r-1} \binom{n-1}{k-1} p^k q^{n-k} \right] \\
&= \sum_{k=0}^{r-1} \binom{n-1}{k} p^k q^{n-1-k} - q \sum_{k=0}^{r-1} \binom{n-1}{k} p^k q^{n-k} - \sum_{k=0}^{r-1} \binom{n-1}{k-1} p^k q^{n-k} \\
&= \underbrace{(1-q)}_{p} \sum_{k=0}^{r-1} \binom{n-1}{k} p^k q^{n-1-k} - \sum_{k=0}^{r-1} \binom{n-1}{k-1} p^k q^{n-k} \\
&= p \sum_{k=1}^{r} \binom{n-1}{k-1} p^{k-1} q^{n-k} - p \sum_{k=0}^{r-1} \binom{n-1}{k-1} p^{k-1} q^{n-k} \\
&= p \binom{n-1}{r-1} p^{r-1} q^{n-r} + (p-p) \sum_{k=1}^{r-1} \binom{n-1}{k-1} p^{k-1} q^{n-k} - p \binom{n-1}{-1} p^{-1} q^{n+1} \\
&= \binom{n-1}{r-1} p^k q^{n-k} = L_{n-1} - L_n.
\end{aligned}
$$

□

Exercise 4.45 (Ross' TE 4.31)**.** *A jar contains $m+n$ chips, numbered $1, 2, \ldots, n+m$. A set of size n is drawn. If we let X denote the number of chips drawn having numbers that exceed each of the numbers of those remaining, compute the probability mass function of X.*

Solution. The sample space is

$$\Omega = \{A = \{x_1, x_2, \ldots, x_n\} \subseteq [n+m] : 1 \leq x_1 < x_2 < \cdots < x_n \leq n+m\}.$$

Note that

$$|\Omega| = \binom{n+m}{n}.$$

Observe that there is a bijection between Ω and the weak compositions of m into $n+1$ determined as follows:

- Given integers $1 \leq x_1 < x_2 < \cdots < x_n \leq n+m$, to get a weak composition $m = \sum_{i=0}^{n} m_i$, let

$$
\begin{aligned}
&x_0 = 0, \\
&x_{n+1} = n+m+1, \\
&m_i = x_{i+1} - x_i - 1, \ i = 0, 1, \ldots, n.
\end{aligned}
$$

- Given integers $m_0, m_1, \ldots, m_n$ such that $m = \sum_{i=0}^{n} m_i$, to get $1 \leq x_1 < \cdots < x_n \leq n+m$, let

$$x_i = \sum_{j=0}^{i-1} (m_j + 1), \quad i = 1, 2, \ldots, n.$$

Observe that

$$\{X = i\} = \{\text{weak compositions } m = \sum_{i=0}^{n} m_i \text{ such that } m_{n-j} = 0 \text{ for each } 0 \leq j < i \text{ and } m_{n-i} \geq 1\}.$$

Thus, the number of weak compositions of $m-1$ into $n+1-i$ counts the outcomes belonging to $\{X = i\}$. Therefore, assuming that each outcome is equally likely, the pmf is computed as follows:

$$
\begin{aligned}
P\{X = i\} &= \frac{|\{X = i\}|}{|\Omega|} \\
&= \frac{\binom{m-1+(n+1-i)-1}{(n+1-i)-1}}{\binom{n+m}{n}} \\
&= \frac{\binom{m+n-i-1}{n-i}}{\binom{n+m}{n}}.
\end{aligned}
$$

Observe a simpler approach to count the outcomes belonging to $\{X = i\}$: $\{X = i\}$ is the collection of sets in which exactly the top i chips are drawn but the chip $n+m-i$ is not drawn. Its size is the number of ways to draw $n-i$ chips from the chips numbered $1, 2, \ldots, n+m-i-1$ as the top i chips must be drawn anyways. It follows that $|\{X = i\}| = \binom{n+m-i-1}{n-i}$. □

Exercise 4.46 (Ross' Self-test 4.20)**.** *Show that if X is a geometric random variable with parameter p, then*

$$\mathrm{E}\left[1/X\right] = \frac{-p\ln(p)}{1-p}.$$

Solution. Let $X \sim Geometric(p)$. Write

$$\mathrm{E}\left[1/X\right] = \sum_{k=1}^{\infty}(1/k)(1-p)^{k-1}p = \frac{p}{1-p}\sum_{k=1}^{\infty}\frac{(1-p)^k}{k}.$$

And we have

$$\sum_{k=1}^{\infty}\frac{(1-p)^k}{k} = \sum_{k=1}^{\infty}\int_0^{1-p} x^{k-1}\,dx = \int_0^{1-p}\sum_{k=1}^{\infty} x^{k-1}\,dx = \int_0^{1-p}\frac{1}{1-x}\,dx = -\ln p.$$

□

5 Continuous Random Variables

For each of the continuous random variables I learned, let me state the pdf (with support) and derive the expectation and variance. I may derive the n-th moment to that end. It will remain to verify that the pmfs are actually pmfs (and later to derive the mgfs) for each.

5.1 Common Continuous Distributions

RV	PDF	Support	$\mathrm{E}\left[X\right]$	$\mathrm{Var}(X)$
$\mathrm{Uniform}(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
$N(\mu,\sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})}$	$-\infty < x < \infty$	μ	σ^2
$\mathrm{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0 \le x < \infty$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$\mathrm{Gamma}(\alpha,\lambda)$	$\frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}$	$0 \le x < \infty$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
$\chi^2(n)$	$\frac{e^{-x/2}x^{n/2-1}}{2^{n/2}\Gamma(n/2)}$	$0 \le x < \infty$	n	$2n$
$T(n)$				
$F(n_1,n_2)$				

Exercise 5.1. *Derive the n-th moment for the uniform distribution.*

Solution. Let $X \sim Uniform(a, b)$. Then,

$$\mathrm{E}\left[X^n\right] = \int_a^b x^n \frac{1}{b-a} dx = \frac{x^{n+1}}{(n+1)(b-a)}\Bigg|_a^b = \boxed{\frac{b^{n+1} - a^{n+1}}{(n+1)(b-a)}}.$$

□

Exercise 5.2. *Derive the expectation and variance for the uniform distribution.*

Solution. Let $X \sim Uniform(a, b)$. Using the nth moment formula in Exercise (5.1), we obtain

$$\begin{aligned}
\mathrm{E}\left[X\right] &= \frac{b^2 - a^2}{2(b-a)} = \boxed{\frac{b+a}{2}}. \\
\mathrm{Var}\left(X\right) &= \mathrm{E}\left[X^2\right] - \mathrm{E}\left[X\right]^2 = \frac{b^3 - a^3}{3(b-a)} - \frac{(b+a)^2}{4} \\
&= \frac{4(b^2 + ba + a^2) - 3(b^2 + 2ab + a^2)}{12} \\
&= \frac{b^2 - 2ab + a^2}{12} = \boxed{\frac{(b-a)^2}{12}}.
\end{aligned}$$

□

Exercise 5.3. *Verify that the pdf for the normal distribution is actually a pdf.*

Lemma 5.4 (Gaussian Integral)**.** *Verify that* $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.

Proof. Let $I = \int_{-\infty}^{\infty} e^{-x^2} dx$. It suffices to prove that $I^2 = \pi$; since the exponential function is strictly positive, then $I > 0$, hence $I \neq -\sqrt{\pi}$. We write

$$\begin{aligned}
I^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2} dy\right) \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} e^{-x^2} dx\right) e^{-y^2} dy \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx\right) dy \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \quad \text{(Justify! Use Fubini or Tonelli).}
\end{aligned}$$

Using the polar substitution $r^2 = x^2 + y^2,\ x = r\cos(\theta),\ y = r\sin(\theta)$ with Jacobian r, we have

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = \int_0^{2\pi} \frac{1}{2} \int_0^{\infty} e^{-u} du d\theta = \int_0^{2\pi} \frac{1}{2} d\theta = \pi.$$

□

Solution. Let $X \sim N(\mu, \sigma^2)$.

$$\begin{aligned}
\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}} e^{-v^2} dv \\
&= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-v^2} dv \\
&= 1,
\end{aligned}$$

where the last step follows by Lemma (5.4). □

Exercise 5.5. *Let* $Z \sim N(0, 1)$. *Prove that* $\sigma Z + \mu \sim N(\mu, \sigma^2)$ *where* μ *is any real number and* $\sigma > 0$.

Solution. Let $X = \sigma Z + \mu$. Then,

$$\begin{aligned} P\{X \le x\} &= P\{\sigma Z + \mu \le x\} \\ &= P\{Z \le \frac{x-\mu}{\sigma}\} \\ &= \int_{-\infty}^{(x-\mu)/\sigma} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz. \end{aligned}$$

Taking the derivative with respect to x, we obtain

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma^2})} \cdot \frac{1}{\sigma},$$

hence $X \sim N(\mu, \sigma^2)$. □

Exercise 5.6. *Derive the expectation and variance for the normal distribution.*

Solution. First, we derive for the standard normal case. Let $Z \sim N(0,1)$. Then,

$$\begin{aligned} \mathrm{E}[Z] &= \int_{-\infty}^{\infty} z \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-z^2/2} dz \\ &= 0. \end{aligned}$$

since $g(z) = ze^{-z^2/2}$ is an odd function; $g(-z) = (-z)e^{(-z)^2/2} = -ze^{z^2/2} = -g(z)$. Also,

$$\begin{aligned} \mathrm{Var}(Z) &= \mathrm{E}\left[Z^2\right] - \mathrm{E}[Z]^2 = \mathrm{E}\left[Z^2\right] \\ &= \int_{-\infty}^{\infty} z^2 \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2} dz. \end{aligned}$$

Integrating by parts, we let $u = z$ and $dv = ze^{-z^2/2}dz$, so $du = dz$ and $v = -e^{-z^2/2}$, and so we obtain

$$\begin{aligned} \mathrm{Var}(Z) &= \frac{1}{\sqrt{2\pi}} \left(-ze^{-z^2/2}\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} -e^{-z^2/2} dz \right) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} dz = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 1. \end{aligned}$$

Now, let $X \sim N(\mu, \sigma^2)$. By Exercise (5.5), write $X = \sigma Z + \mu$ where $Z \sim N(0,1)$. Thus, we obtain

$$\mathrm{E}[X] = \mathrm{E}[\sigma Z + \mu] = \sigma \mathrm{E}[Z] + \mu = \boxed{\mu}.$$
$$\mathrm{Var}(X) = \mathrm{Var}(\sigma Z + \mu) = \sigma^2 \mathrm{Var}(Z) = \boxed{\sigma^2}.$$

□

Exercise 5.7. *Derive the nth moment for the Gamma distribution.*

Lemma 5.8.

$$\Gamma(a+1) = a\Gamma(a), \quad a > 0.$$

Proof.

$$\Gamma(a+1) = \int_0^{\infty} e^{-y} y^a dy = [(y^a)(-e^{-y})]\big|_0^{\infty} - \int_0^{\infty} (-e^{-y})(ay^{a-1}) dy = a \int_0^{\infty} e^{-y} y^{a-1} dy = a\Gamma(a).$$

□

Solution. Let $X \sim Gamma(\alpha, \lambda)$, with pdf $f(x) = \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}$, $0 \le x < \infty$. Then,

$$\begin{aligned}
\mathrm{E}\left[X^n\right] &= \int_0^\infty x^n \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)} dx = \frac{1}{\lambda^n} \int_0^\infty \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha+n-1}}{\Gamma(\alpha)} \\
&= \frac{\Gamma(\alpha+n)}{\lambda^n \Gamma(\alpha)} \int_0^\infty \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha+n-1}}{\Gamma(\alpha+n)} dx \\
&= \frac{\Gamma(\alpha+n)}{\lambda^n \Gamma(\alpha)}.
\end{aligned}$$

Thus, using Lemma (5.8), we obtain

$$\begin{aligned}
\Gamma(\alpha+n) &= (\alpha+n-1)\Gamma(\alpha+n-1) \\
&= (\alpha+n-1)(\alpha+n-2)\Gamma(\alpha+n-2) = \cdots = (\alpha+n-1)_n \Gamma(\alpha),
\end{aligned}$$

hence

$$\mathrm{E}\left[X^n\right] = \frac{(\alpha+n-1)_n \Gamma(\alpha)}{\lambda^n \Gamma(\alpha)} = \boxed{\frac{(\alpha+n-1)_n}{\lambda^n}}.$$

□

Exercise 5.9. *Derive the expectation and variance for the Gamma distribution.*

Solution. Let $X \sim Gamma(\alpha, \lambda)$, with pdf $f(x) = \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}$, $0 \le x < \infty$. Then, using the nth moment formula in Exercise (5.7), we obtain

$$\begin{aligned}
\mathrm{E}\left[X\right] &= \boxed{\frac{\alpha}{\lambda}}. \\
\mathrm{Var}\left(X\right) &= \mathrm{E}\left[X^2\right] - \mathrm{E}\left[X\right]^2 = \frac{(\alpha+1)\alpha}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2 = \frac{(\alpha+1)\alpha - \alpha^2}{\lambda^2} = \boxed{\frac{\alpha}{\lambda^2}}.
\end{aligned}$$

□

Exercise 5.10. *Verify that the pdf for the exponential distribution is actually a pdf.*

Solution. Let $X \sim exp(\lambda)$. Thus,

$$\begin{aligned}
\Gamma(1) &= \int_0^\infty e^{-y} y^{1-1} dy = \int_0^\infty e^{-y} dy = -e^{-y}\Big|_0^\infty = (-0-(-1)) = 1 \\
\implies \lambda e^{-\lambda x} &= \frac{\lambda e^{-\lambda x}(\lambda x)^{1-1}}{\Gamma(1)} \text{ is the pdf of } Gamma(1, \lambda).
\end{aligned}$$

□

Exercise 5.11. *Derive the nth moment for the exponential distribution.*

Solution. Let $X \sim exp(\lambda)$. Thus, $X \sim Gamma(1, \lambda)$. Therefore,

$$\mathrm{E}\left[X^n\right] = \frac{1+n-1}{\lambda^n} = \boxed{\frac{n}{\lambda^n}}.$$

□

Exercise 5.12. *Derive the expectation and variance for the exponential distribution.*

Solution. Let $X \sim exp(\lambda)$. Therefore,

$$\mathrm{E}\left[X\right] = \boxed{\frac{1}{\lambda}}, \quad \mathrm{Var}\left(X\right) = \mathrm{E}\left[X^2\right] - \mathrm{E}\left[X\right]^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \boxed{\frac{1}{\lambda^2}}.$$

□

Exercise 5.13. *Verify the pdf of $X \sim \chi^2_n$ as follows. Let $Z_1, Z_2, \ldots, Z_n$ be iid $N(0,1)$. Let $X = \sum_{i=1}^n Z_i^2$. Prove that $X \sim Gamma(\frac{n}{2}, \frac{1}{2})$.*

Lemma 5.14. *Let X, Y be independent random variables with joint pdf $f(x, y)$. Prove that*

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx.$$

Proof. Note that

$$P\{X+Y \le z\} = P\{Y \le z - X\} = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f(x, y) dy dx.$$

Since X and Y are independent, then

$$\begin{aligned} P\{X+Y \le z\} &= \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_X(x) f_Y(y) dy dx = \int_{-\infty}^{\infty} f_X(x) \int_{-\infty}^{z-x} f_Y(y) dy dx \\ &= \int_{-\infty}^{\infty} f_X(x) F_Y(z-x) dx. \end{aligned}$$

Taking the derivative with respect to z, we have

$$\begin{aligned} f_{X+Y}(z) &= \frac{d}{dz} \int_{-\infty}^{\infty} f_X(x) F_Y(z-x) dx \\ &= \int_{-\infty}^{\infty} \frac{d}{dz} \left[f_X(x) F_Y(z-x) \right] dx \quad \text{(Justify!)} \\ &= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx. \end{aligned}$$

□

Lemma 5.15. *Let $X_i \sim Gamma(a_i, \lambda)$ for all $1 \le i \le n$. Suppose that $X_1, X_2, \ldots, X_n$ are independent random variables. Show that $\sum_{i=1}^n X_i \sim Gamma(\sum_{i=1}^n a_i, \lambda)$.*

Proof. We first prove the simple case when $n = 2$.

Let $X \sim Gamma(a, \lambda)$ and $Y \sim Gamma(b, \lambda)$ be independent random variables. By Exercise (5.14), we have

$$\begin{aligned} f_{X+Y}(z) &= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx = \int_0^z f_X(x) f_Y(z-x) dx \\ &= \int_0^z \frac{\lambda e^{-\lambda x} (\lambda x)^{a-1}}{\Gamma(a)} \frac{\lambda e^{-\lambda(z-x)} (\lambda(z-x))^{b-1}}{\Gamma(b)} dx \\ &= \frac{\lambda^{a+b} e^{-\lambda z}}{\Gamma(a)\Gamma(b)} \int_0^z x^{a-1} (z-x)^{b-1} dx \\ &= \frac{\lambda^{a+b} e^{-\lambda z} z^{a+b-1}}{\Gamma(a)\Gamma(b)} \int_0^z (x/z)^{a-1} (1-x/z)^{b-1} (1/z) dx \\ &= \frac{\lambda^{a+b} e^{-\lambda z} z^{a+b-1}}{\Gamma(a)\Gamma(b)} \int_0^1 u^{a-1} (1-u)^{b-1} du \\ &= \frac{\lambda^{a+b} e^{-\lambda z} z^{a+b-1}}{\Gamma(a)\Gamma(b)} B(a, b), \end{aligned}$$

where $B(a, b)$ is the beta function. Given that

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

it follows that

$$f_{X+Y}(z) = \frac{\lambda^{a+b} e^{-\lambda z} z^{a+b-1}}{\Gamma(a+b)} = \frac{\lambda e^{-\lambda z} (\lambda z)^{a+b-1}}{\Gamma(a+b)},$$

hence $X + Y \sim Gamma(a+b, \lambda)$.

Let $X_i \sim Gamma(a_i, \lambda)$ for each $1 \le i \le n$. Then, using the sample case, we have

$$X_1 + X_2 \sim Gamma(a_1 + a_2, \lambda)$$

$$
\begin{aligned}
&X_1 + X_2 + X_3 \sim Gamma(a_1 + a_2 + a_3, \lambda) \\
&\vdots \\
&\sum_{i=1}^{n} X_i \sim Gamma(\sum_{i=1}^{n} a_i, \lambda).
\end{aligned}
$$

□

Solution. First, we prove that each Z_i^2 has distribution $Gamma(\frac{1}{2}, \frac{1}{2})$. Then, we use the sum property of independent Gamma random variables to complete the proof.

1. Let $Z \sim N(0,1)$, and we prove $Z^2 \sim Gamma(\frac{1}{2}, \frac{1}{2})$.

$$
\begin{aligned}
P\{Z^2 \le z\} &= P\{-\sqrt{z} \le Z \le \sqrt{z}\} = 2\Phi(\sqrt{z}) - 1 \\
\implies f_{Z^2}(z) &= \frac{d}{dz}\left[2\Phi(\sqrt{z}) - 1\right] = 2\phi(\sqrt{z})\frac{1}{2\sqrt{z}} = \frac{\phi(\sqrt{z})}{\sqrt{z}} \\
&= z^{-1/2}(2\pi)^{-1/2}e^{-z/2} = \frac{\frac{1}{2}e^{-\frac{1}{2}z}(\frac{1}{2}z)^{-\frac{1}{2}}}{\sqrt{\pi}}.
\end{aligned}
$$

   Observe that

$$
\begin{aligned}
\Gamma\left(\frac{1}{2}\right) &= \int_0^\infty e^{-y}y^{-\frac{1}{2}}dy \\
&= 2\int_0^\infty e^{-u^2}du \quad \text{(using the substitution } u = y^{\frac{1}{2}}) \\
&= \int_{-\infty}^\infty e^{-u^2} = \sqrt{\pi} \quad \text{(by Exercise (5.4))}.
\end{aligned}
$$

   Thus, we have

$$
f_{Z^2}(z) = \frac{\frac{1}{2}e^{-\frac{1}{2}z}(\frac{1}{2}z)^{-\frac{1}{2}}}{\Gamma(\frac{1}{2})},
$$

   hence $Z^2 \sim Gamma(\frac{1}{2}, \frac{1}{2})$. Therefore, for each $1 \le i \le n$, we have $Z_i^2 \sim Gamma(\frac{1}{2}, \frac{1}{2})$.

2. By Lemma (5.15), it follows that

$$
\begin{aligned}
&Z_1^2 + Z_2^2 \sim Gamma(1, \frac{1}{2}) \\
&Z_1^2 + Z_2^2 + Z_3^2 \sim Gamma(\frac{3}{2}, \frac{1}{2}) \\
&\vdots \\
&\sum_{i=1}^{n} Z_i^2 \sim Gamma(\frac{n}{2}, \frac{1}{2}).
\end{aligned}
$$

□

Exercise 5.16. *Derive the kth moment for the χ^2 distribution.*

Solution. Let $X \sim \chi_n^2$. By Exercise (5.13), we know that $X \sim Gamma(\frac{n}{2}, \frac{1}{2})$. By Exercise (5.7), we obtain

$$
\mathrm{E}\left[X^k\right] = \frac{(\frac{n}{2} + k - 1)_k}{(1/2)^k} = \boxed{2^k(\frac{n}{2} + k - 1)_k}.
$$

□

Exercise 5.17. *Derive the expectation and variance for the χ^2 distribution.*

Solution. Let $X \sim \chi_n^2$. By Exercise (5.13), we know that $X \sim Gamma(\frac{n}{2}, \frac{1}{2})$. By Exercise (5.9), we obtain

$$
\mathrm{E}\left[X\right] = \frac{n/2}{1/2} = \boxed{n}, \quad \mathrm{Var}\left(X\right) = \frac{n/2}{(1/2)^2} = \boxed{2n}.
$$

□

5.2 Exercises

Exercise 5.18 (Ross' TE 5.2)**.** *Show that*

$$\mathrm{E}[Y] = \int_0^\infty P\{Y > y\}dy - \int_0^\infty P\{Y < -y\}dy.$$

Solution. Let Y be a continuous random variable with pdf $f(x)$. Then, write

$$\mathrm{E}[Y] = \int_{-\infty}^\infty yf(y)dy = \int_0^\infty yf(y)dy + \int_{-\infty}^0 yf(y)dy.$$

The result follows by observing the following:

$$\begin{aligned}
\int_0^\infty P\{Y > y\}dy &= \int_0^\infty \int_y^\infty f(x)dxdy \\
&= \int_0^\infty \int_0^x f(x)dydx \\
&= \int_0^\infty xf(x)dx, \\
-\int_0^\infty P\{Y < -y\}dy &= -\int_0^\infty \int_{-\infty}^{-y} f(x)dxdy \\
&= -\int_{-\infty}^0 \int_0^{-x} f(x)dydx \\
&= -\int_{-\infty}^0 -xf(x)dx = \int_{-\infty}^0 xf(x)dx.
\end{aligned}$$

□

Exercise 5.19 (Ross' TE 5.29)**.** *Let X be a continuous random variable with cdf F. Let $Y = F(X)$ be a random variable. Show that $Y \sim Uniform(0,1)$.*

Solution. Since X is a continuous random variable, then F is continuous, so if F is strictly monotonically increasing, the inverse function F^{-1} exists. Thus,

$$\begin{aligned}
&P\{Y \le y\} = P\{F(X) \le y\} = P\{X \le F^{-1}(y)\} = F(F^{-1}(y)) = y \\
&(P\{Y \le 0\} = 0 \implies \forall y \le 0,\ P\{Y \le y\} = 0, \\
&\ P\{Y \le 1\} = 1 \implies \forall y \ge 1, P\{Y \le y\} = 1).
\end{aligned}$$

Taking the derivative with respect to y, we have $f_Y(y) = 1,\ 0 \le y \le 1$, hence $Y \sim Uniform(0,1)$.

If F is not strictly monotonically increasing, the generalized inverse $F^{-1}(y) = \sup\{x : F(x) \le y\}$ should work. □

6 Joint Distributions

Exercise 6.1 (Green Book: Meeting Probability)**.** *A bank opens for one hour from 5:00 pm to 6:00 pm. Two bankers arrive independently and uniformly at some time in that hour, stay exactly 5 minutes or, if that's not possible, until the bank closes. How likely is it that they meet?*

Solution. Let banker A arrive X minutes after 5 pm and banker B arrive Y minutes after 6 pm. Note that X, Y are iid $Uniform(0,60)$, so their joint pdf is $1/60^2$. For banker B to meet banker A, banker B has to arrive at most 5 minutes before or after banker A. Now, the desired probability can be found in two ways.

- **Integration method** The desired event corresponds to the region $\{(x,y) : 0 \le x \le 60, 0 \le y \le 60, x-5 \le y \le x+5\} = \{(x,y) : 0 \le x \le 60, \max(0, x-5) \le y \le \min(60, x+5)\}$. Thus,

$$\begin{aligned}
P\{\text{two bankers meet}\} &= \int_0^{60} \int_{\max(0,x-5)}^{\min(60,x+5)} \frac{1}{60^2} dydx \\
&= \frac{1}{60^2}\left[\int_0^5 \int_0^{x+5} dydx + \int_5^{55} \int_{x-5}^{x+5} dydx + \int_{55}^{60} \int_{x-5}^{60} dydx\right]
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{60^2}\left[\int_0^5 (x+5)dx + 500 + \int_{55}^{60}(65-x)dx\right] \\
&= \frac{1}{60^2}\left[\frac{75}{2} + \frac{1000}{2} + \frac{650-(60^2-55^2)}{2}\right] \\
&= \frac{1}{60^2}\frac{1725-(60-55)(60+55)}{2} = \frac{575}{60^2} = \boxed{\frac{23}{144}}.
\end{aligned}
$$

- **Area method** Consider the following figure.

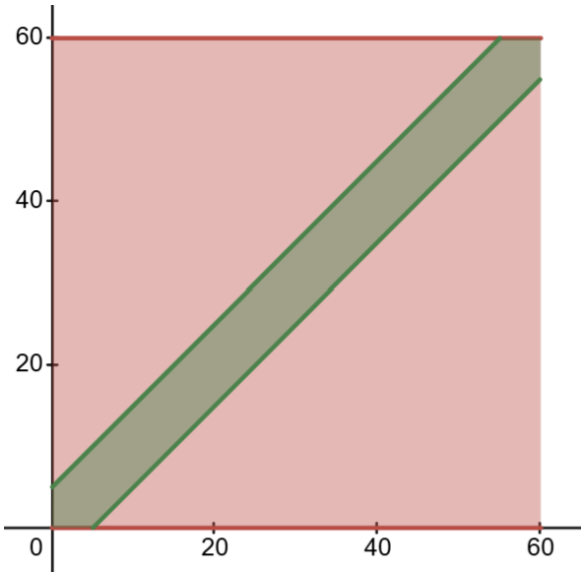


Figure 1: Meeting probability

The sample space is a square with side 60. The meeting probability is the proportion of the area of the region between the line $x-5$ and the line $x+5$ and inside the square; that is the green region. The area of the green region is the area of the square minus the area of the red region. The red region consists of two isosceles triangle, each with base 55 and height 55. Therefore,

$$
\begin{aligned}
P\{\text{two bankers meet}\} &= \frac{\text{area(green)}}{\text{area(square)}} = 1 - \frac{\text{area(red)}}{60^2} \\
&= \frac{60^2 - 2(55^2/2)}{60^2} = \frac{(60+55)(60-55)}{60^2} = \frac{575}{60^2} = \boxed{\frac{23}{144}}.
\end{aligned}
$$

□

Exercise 6.2 (Green Book: Sum of Random Variables)**.** *Let $X_1, X_2, \ldots, X_n$ be iid $Uniform(0,1)$ random variables. Let $S_n = X_1 + X_2 + \cdots + X_n$. Compute $P(S_n \le 1)$.*

Solution. Let's try.

- **Initial thoughts:** Represent the region using nested iterations: $\{S_n \le 1\} = \{0 \le x_1 \le 1, 0 \le x_2 \le 1 - x_1, \ldots, 0 \le x_n \le 1 - \sum_{i=1}^{n-1} x_i\}$. Thus,

$$
P\{S_n \le 1\} = \int_0^1 \int_0^{1-x_1} \cdots \int_0^{1-\sum_{i=1}^{n-1} x_i} 1 dx_n dx_{n-1} \cdots dx_1
$$

  since X_i's are iid each with pdf $f(x) = 1$. It's not clear to me how to evaluate this integral.

- **Solution:** We try to find an equation in terms of n based on simple cases.

  1. $P\{X_1 \le 1\} = 1$.
  2. $P\{X_1 + X_2 \le 1\} = \int_0^1 \int_0^{1-x_1} 1 dx_2 dx_1 = \int_0^1 (1-x_1)dx_1 = \int_0^1 y dy = \frac{1}{2}$.

3.

$$
\begin{aligned}
P\{X_1 + X_2 + X_3 \leq 1\} &= \int_0^1 \int_0^{1-x_1} \int_0^{1-x_1-x_2} 1 dx_3 dx_2 dx_1 \\
&= \int_0^1 \int_0^{1-x_1} (1 - x_1 - x_2) dx_2 dx_1 \\
&= \int_0^1 (1-x_1)^2 - \frac{(1-x)^2}{2} dx_1 = \int_0^1 \frac{(1-x_1)^2}{2} dx_1 \\
&= \frac{1}{2} \int_0^1 y^2 dy = \frac{1}{6}.
\end{aligned}
$$

Observe for $n = 1, 2, 3$ that $P\{S_n \leq 1\} = 1/n!$.

Guess: $\forall n \in \mathbb{Z}^+$, $P\{S_n \leq 1\} = 1/n!$.

Attempt to prove by induction:

It holds for $n = 1, 2$. Assume for some $n \geq 2$ that $P\{S_n \leq 1\} = 1/n!$. We prove $P\{S_{n+1} \leq 1\} = 1/(n+1)!$.

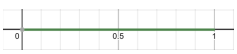
Conditioning on the value of X_{n+1} in order to use the induction hypothesis, note

$$
P\{S_{n+1} \leq 1\} = \int_0^1 P\{S_n \leq 1 - x_{n+1}\} dx_{n+1}
$$

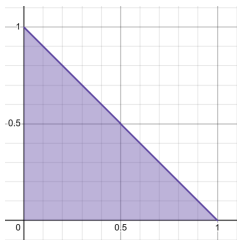
.

We know $P\{S_n \leq 1\} = 1/n!$, but what's the value of $P\{S_n \leq 1 - x_{n+1}\}$ for a fixed x_{n+1}?

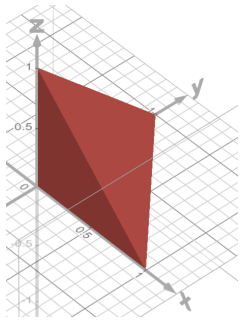
Insight: $P\{S_n \leq 1\}$ is the "area" under a polyhedron D, as shown in the following figures.



$\{S_1 \leq 1\}$



$\{S_2 \leq 1\}$



$\{S_3 \leq 1\}$

And $P\{S_n \leq 1 - x_{n+1}\}$ is the "area" under a polyhedron D' that is a dilation of D such that each of the n dimensions is dilated by a factor of $(1 - x_{n+1})/1$. Therefore, the volume of D' is $(1 - x_{n+1})^n$ times the volume of D. Therefore,

$$
P\{S_n \leq 1 - x_{n+1}\} = (1 - x_{n+1})^n P\{S_n \leq 1\} = \frac{(1 - x_{n+1})^n}{n!}.
$$

using the inductive hypothesis. Therefore,

$$
P\{S_{n+1} \leq 1\} = \int_0^1 \frac{(1 - x_{n+1})^n}{n!} dx_{n+1} = \frac{1}{n!} \int_0^1 y^n dy = \frac{1}{n!} \cdot \frac{1}{n+1} = \frac{1}{(n+1)!}.
$$

Indeed, by induction, our guess is correct.

Conclusion:

$$
\boxed{P\{S_n \leq 1\} = \frac{1}{n!}}.
$$

□

6.1 Transformation Method

Exercise 6.3 (Transformation Method)**.** *State and prove the bivariate transformation method.*

Solution. Let X_1, X_2 have joint pdf f_{X_1,X_2}. Let

$$\begin{aligned} Y_1 &= u_1(X_1, X_2), \\ Y_2 &= u_2(X_1, X_2). \end{aligned}$$

Suppose a solution to the system of equations above is

$$\begin{aligned} X_1 &= w_1(Y_1, Y_2), \\ X_2 &= w_2(Y_1, Y_2). \end{aligned}$$

Then, the joint pdf of Y_1 and Y_2 is obtained as follows:

$$f_{Y_1,Y_2}(y_1, y_2) = f_{X_1,X_2}(w_1, w_2) \cdot |J| \tag{12}$$

$$\text{where } J \text{ is the Jacobian: } \det \begin{pmatrix} \frac{\partial w_1}{\partial y_1} & \frac{\partial w_1}{\partial y_2} \\ \frac{\partial w_2}{\partial y_1} & \frac{\partial w_2}{\partial y_2} \end{pmatrix}. \tag{13}$$

□

Remark 6.4. *Suppose the original support is $(x_1, x_2) \in S$. To find the new support:*

1. *Set $(w_1(y_1, y_2), w_2(y_1, y_2)) \in S$.*
2. *Find S' such that $(w_1, w_2) \in S \iff (y_1, y_2) \in S'$.*

Remark 6.5. *If the system is underdetermined, for instance you are given X_1 and X_2 but asked for only the pdf of $Y_1 = u(X_1, X_2)$, do the following:*

1. *Introduce an appropriate dummy variable, for example $Y_2 = X_2$.*
2. *Find the joint pdf of Y_1, Y_2.*
3. *Marginalize over Y_2 to obtain the pdf of Y_1.*

Remark 6.6. *The multivariate transformation method is the intuitive extension of the bivariate method.*

Remark 6.7. *If the system is underdetermined, introduce dummy variables, for example in the form*

$$Y_i = X_i.$$

Exercise 6.8 (bivariate - want joint pdf)**.**

Exercise 6.9 (bivariate - underdetermined)**.** *Let X_1, X_2 have joint pdf f. Find the pdf of the quotient X_1/X_2. In particular, find the quotient pdf when $f(x_1, x_2) = \begin{cases} 10x_1x_2^2 & 0 < x_1 < x_2 < 1, \\ 0 & \text{otherwise.} \end{cases}$*

Solution.

- Let

$$\begin{aligned} Y_1 &= X_1/X_2, \\ Y_2 &= X_2. \end{aligned}$$

- Therefore,

$$\begin{aligned} X_2 &= Y_2, \\ X_1 &= Y_1Y_2. \end{aligned}$$

- Therefore,

$$J = \det \begin{pmatrix} \partial(y_1y_2)/\partial y_1 & \partial(y_1y_2)/\partial y_2 \\ \partial(y_2)/\partial y_1 & \partial(y_2)/\partial y_2 \end{pmatrix} = \det \begin{pmatrix} y_2 & y_1 \\ 0 & 1 \end{pmatrix} = y_2 \implies |J| = |y_2|.$$

- Using the transformation method,

$$f_{Y_1,Y_2}(y_1,y_2) = f(y_1y_2, y_2)\,|y_2|\,.$$

- Therefore,

$$\boxed{f_{Y_1}(y_1) = \int_{-\infty}^{\infty} f(y_1y_2, y_2)\,|y_2|\,dy_2}.$$

- In particular, given the joint pdf of X_1, X_2, we have

$$f_{Y_1,Y_2}(y_1,y_2) = \begin{cases} (10 \cdot y_1y_2 \cdot y_2^2)\,|y_2| & 0 < y_1y_2 < y_2 < 1, \\ 0 & \text{otherwise.} \end{cases}$$

- Since $0 < y_2$, then

$$0 < y_1y_2 < y_2 \iff 0 < y_1 < 1.$$

- Also, $|y_2| = y_2$. Therefore,

$$f_{Y_1,Y_2}(y_1,y_2) = \begin{cases} 10y_1y_2^4 & 0 < y_1 < 1 \wedge 0 < y_2 < 1, \\ 0 & \text{otherwise.} \end{cases}$$

- Marginalizing over Y_2, we have

$$\int_0^1 10y_1y_2^4 dy_2 = 2y_1.$$

- Therefore,

$$\boxed{f_{X_1/X_2}(x) = f_{Y_1}(x) = \begin{cases} 2x & 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}}$$

□

Exercise 6.10. *bivariate - verify independence*

Exercise 6.11 (multivariate - want joint pdf)**.**

Exercise 6.12 (multivariate - underdetermined)**.**

Exercise 6.13 (multivariate - verify independence)**.** *Let X_1, X_2, X_3 be iid, each with pdf*

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0, \\ 0 & \textit{otherwise.} \end{cases}$$

Let

$$\begin{aligned} Y_1 &= \frac{X_1}{X_1 + X_2 + X_3}, \\ Y_2 &= \frac{X_2}{X_1 + X_2 + X_3}, \\ Y_3 &= X_1 + X_2 + X_3. \end{aligned}$$

Are Y_1, Y_2, Y_3 independent?

Solution. We want to verify

$$f_{Y_1,Y_2,Y_3}(y_1,y_2,y_3) = f_{Y_1}(y_1)f_{Y_2}(y_2)f_{Y_3}(y_3).$$

We apply the transformation method to find the joint pdf of Y in terms of the joint pdf of X, then we marginalize and verify.

First, since X_1, X_2, X_3 are iid, the joint pdf of X is

$$f_{X_1,X_2,X_3}(x_1,x_2,x_3) = f(x_1)f(x_2)f(x_3) = \begin{cases} \lambda^3 e^{-\lambda(x_1+x_2+x_3)} & x_1 > 0 \wedge x_2 > 0 \wedge x_3 > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Now, we apply the transformation method.

- Note

$$\begin{aligned}
X_1 &= Y_1 Y_3, \\
X_2 &= Y_2 Y_3, \\
X_3 &= Y_3 - Y_1 Y_3 - Y_2 Y_3.
\end{aligned}$$

- Also,

$$\begin{aligned}
J &= \det \begin{pmatrix} y_3 & 0 & y_1 \\ 0 & y_3 & y_2 \\ -y_3 & -y_3 & 1 - y_1 - y_2 \end{pmatrix} = y_3 \det \begin{pmatrix} y_3 & y_2 \\ -y_3 & 1 - y_1 - y_2 \end{pmatrix} + y_1 \det \begin{pmatrix} 0 & y_3 \\ -y_3 & -y_3 \end{pmatrix} \\
&= y_3[y_3(1 - y_1 - y_2) - y_2(-y_3)] + y_1[0 - y_3(-y_3)] \\
&= y_3^2 - y_1 y_3^2 - y_2 y_3^2 + y_2 y_3^2 + y_1 y_3^2 \\
&= y_3^2. \\
|J| &= y_3^2.
\end{aligned}$$

Therefore,

$$\begin{aligned}
f_{Y_1,Y_2,Y_3}(y_1, y_2, y_3) &= f(y_1 y_3) f(y_2 y_3) f(y_3 - y_1 y_3 - y_2 y_3) y_3^2 \\
&= \begin{cases} \lambda^3 y_3^2 e^{-\lambda y_3} & y_1 y_3, y_2 y_3, y_3(1 - y_1 - y_2) > 0, \\ 0 & \text{otherwise.} \end{cases}
\end{aligned}$$

Note:

$$\begin{aligned}
&y_3 > 0 \iff y_1, y_2 > 0 \iff y_1 + y_2 < 1 \iff 0 < y_1 < 1 \wedge 0 < y_2 < 1 - y_1, \\
&y_3 < 0 \iff y_1, y_2 < 0 \implies y_3(1 - y_1 - y_2) \not> 0, \text{ so we must have } y_3 > 0.
\end{aligned}$$

Therefore,

$$f_{Y_1,Y_2,Y_3}(y_1, y_2, y_3) = \begin{cases} \lambda^3 y_3^2 e^{-\lambda y_3} & y_3 > 0 \wedge 0 < y_1 < 1 \wedge 0 < y_2 < 1 - y_1, \\ 0 & \text{otherwise.} \end{cases}$$

Using double integration by parts, obtain

$$\int_0^\infty \lambda^3 y_3^2 e^{-\lambda y_3} dy_3 = 2.$$

Now,

$$\begin{aligned}
f_{Y_1}(y_1) &= \int_0^{1-y_1} \int_0^\infty \lambda^3 y_3^2 e^{-\lambda y_3} dy_3 dy_2 = \int_0^{1-y_1} 2 dy_2 = 2(1 - y_1) \\
f_{Y_2}(y_2) &= \int_0^{1-y_2} \int_0^\infty \lambda^3 y_3^2 e^{-\lambda y_3} dy_3 dy_1 = 2(1 - y_2) \\
f_{Y_3}(y_3) &= \int_0^1 \int_0^{1-y_1} \lambda^3 y_3^2 e^{-\lambda y_3} dy_2 dy_1 = \int_0^1 (1 - y_1) \lambda^3 y_3^2 e^{-\lambda y_3} dy_1 = \frac{\lambda^3 y_3^2}{2} e^{-\lambda y_3}.
\end{aligned}$$

Therefore, within the support,

$$f_{Y_1}(y_1) f_{Y_2}(y_2) f_{Y_3}(y_3) = 2\lambda^3 (1 - y_1)(1 - y_2) y_3^2 e^{-\lambda y_3} \not\equiv \lambda^3 e^{-\lambda y_3} = f_{Y_1,Y_2,Y_3}(y_1, y_2, y_3).$$

Therefore, Y_1, Y_2, Y_3 are NOT independent.

□

Exercise 6.14. *Derive the pdf of the T distribution.*

Solution. Let $Z \sim N(0,1)$ and $V \sim \chi^2(n)$ be independent. We derive the pdf of the T random variable:

$$T = \frac{Z}{\sqrt{V/n}}.$$

Since Z and V are independent, the joint pdf of Z and V is

$$\begin{aligned} f_{Z,V}(z,v) &= f_Z(z) f_V(v) \\ &= \frac{e^{-\frac{1}{2}z^2}}{\sqrt{2\pi}} \cdot \frac{e^{-v/2} v^{n/2-1}}{2^{n/2}\Gamma(n/2)} \\ &= \frac{e^{-\frac{1}{2}(z^2+v)} v^{n/2-1}}{2^{(n+1)/2}\pi^{1/2}\Gamma(n/2)}, \ v \geq 0. \end{aligned}$$

With

$$T = \frac{Z}{\sqrt{V/n}}, \ U = \sqrt{V/n},$$

we have

$$Z = TU, \ V = nU^2.$$

As long the denominator $u \neq 0$, then tu is free to be any real number and $nu^2 \geq 0$ always holds, so the new support is

$$-\infty < t, u < \infty, \ u \neq 0.$$

Then, we compute the Jacobian:

$$|J| = \left| \det \begin{pmatrix} \frac{\partial z}{\partial t} & \frac{\partial z}{\partial u} \\ \frac{\partial v}{\partial t} & \frac{\partial v}{\partial u} \end{pmatrix} \right| = \left| \det \begin{pmatrix} u & t \\ 0 & 2u \end{pmatrix} \right| = \left|2u^2\right| = 2u^2.$$

Therefore, applying the transformation method, we have

$$\begin{aligned} f_{T,U}(t,u) &= 2u^2 f_{Z,V}(tu, nu^2) \\ &= \frac{2u^2 (nu^2)^{n/2-1} e^{-\frac{1}{2}((tu)^2+nu^2)}}{2^{(n+1)/2}\pi^{1/2}\Gamma(n/2)} \\ &= \frac{n^{\frac{n}{2}-1} u^n e^{-\frac{t+n}{2}u^2}}{2^{\frac{n-1}{2}}\pi^{\frac{1}{2}}\Gamma(\frac{n}{2})}, -\infty < t < \infty, \ -\infty < u < \infty, \ u \neq 0. \end{aligned}$$

Marginalizing over u, we obtain

$$\begin{aligned} f_T(t) &= \frac{n^{\frac{n}{2}-1}}{2^{\frac{n-1}{2}}\pi^{\frac{1}{2}}\Gamma(\frac{n}{2})} \left(\int_{-\infty}^{0} e^{-\frac{t+n}{2}u^2} u^n du + \int_0^{\infty} e^{-\frac{t+n}{2}u^2} u^n du \right) \\ &= TBC... \end{aligned}$$

□

Exercise 6.15. *Derive the pdf of the F distribution.*

Solution. Let $X_1 \sim \chi^2(n_1)$ and $X_2 \sim \chi^2(n_2)$ be independent. Thus,

$$\begin{aligned} f_{X_1}(x_1) &= \frac{e^{-\frac{x_1}{2}} x_1^{\frac{n_1}{2}-1}}{2^{\frac{n_1}{2}}\Gamma(\frac{n_1}{2})}, \ x_1 \geq 0 \\ f_{X_2}(x_2) &= \frac{e^{-\frac{x_2}{2}} x_2^{\frac{n_2}{2}-1}}{2^{\frac{n_2}{2}}\Gamma(\frac{n_2}{2})}, \ x_2 \geq 0 \end{aligned}$$

$$\Longrightarrow f_{X_1,X_2}(x_1,x_2) = f_{X_1}(x_1) f_{X_2}(x_2) = \frac{e^{-\frac{x_1+x_2}{2}} x_1^{\frac{n_1}{2}} x_2^{\frac{n_2}{2}}}{2^{\frac{n_1+n_2}{2}}\Gamma(\frac{n_1}{2})\Gamma(\frac{n_2}{2})}, \ x_1, x_2 \geq 0.$$

Let

$$Y_1 = \frac{X_1/n_1}{X_2/n_2}, \quad Y_2 = X_2.$$

Thus,

$$X_1 = \frac{n_1}{n_2} Y_1 Y_2, \; X_2 = Y_2.$$

For the support, note:

$$\frac{n_1}{n_2} y_1 y_2 \geq 0 \wedge y_2 \geq 0 \implies y_1 \geq 0$$
$$y_1 \geq 0 \wedge y_2 \geq 0 \implies \frac{n_1}{n_2} y_1 y_2 \geq 0 \wedge y_2 \geq 0.$$

Then, for the Jacobian, note:

$$|J| = \left| \det \begin{pmatrix} \frac{n_1}{n_2} y_2 & \frac{n_1}{n_2} y_1 \\ 0 & 1 \end{pmatrix} \right| = \left| \frac{n_1}{n_2} y_2 \right| = \frac{n_1}{n_2} y_2.$$

Therefore, applying the transformation method, we have

$$\begin{aligned} f_{Y_1,Y_2}(y_1, y_2) &= f_{X_1,X_2}(\frac{n_1}{n_2} y_1 y_2, y_2) \cdot \frac{n_1}{n_2} y_2 \\ &= \frac{e^{-\frac{(n_1/n_2)y_1 y_2 + y_2}{2}} (\frac{n_1}{n_2} y_1 y_2)^{\frac{n_1}{2}-1} y_2^{\frac{n_2}{2}-1}}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} \cdot \frac{n_1}{n_2} y_2 \\ &= \frac{\frac{n_1}{n_2} (\frac{n_1}{n_2} y_1)^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} \cdot (e^{-y_2})^{\frac{(n_1/n_2)y_1+1}{2}} y_2^{\frac{n_1+n_2}{2}-1}, \quad y_1, y_2 \geq 0, \quad 0 \text{ otherwise.} \end{aligned}$$

Marginalizing over y_2, we obtain

$$\begin{aligned} f_{Y_1}(y_1) &= \int_{-\infty}^{\infty} f_{Y_1,Y_2}(y_1, y_2) dy_2 \\ &= \frac{\frac{n_1}{n_2} (\frac{n_1}{n_2} y_1)^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} \int_0^{\infty} (e^{-y_2})^{\frac{(n_1/n_2)y_1+1}{2}} y_2^{\frac{n_1+n_2}{2}-1} dy_2. \end{aligned}$$

The idea now is to make the integral a gamma function evaluation. Let

$$c = \frac{(n_1/n_2)y_1 + 1}{2}.$$

Multiply and divide by

$$c^{\frac{n_1+n_2}{2}-1} = \left(\frac{(n_1/n_2)y_1 + 1}{2} \right)^{\frac{n_1+n_2}{2}-1}$$

to obtain

$$f_{Y_1}(y_1) = \frac{\frac{n_1}{n_2} (\frac{n_1}{n_2} y_1)^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} \cdot (1/c)^{\frac{n_1+n_2}{2}-1} \int_0^{\infty} (e^{-cy_2})(cy_2)^{\frac{n_1+n_2}{2}-1} dy_2.$$

Let $u = cy_2$, so $du = cdy_2$. Multiplying and dividing by c, we obtain

$$\begin{aligned} f_{Y_1}(y_1) &= \frac{\frac{n_1}{n_2} (\frac{n_1}{n_2} y_1)^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} \cdot (1/c)^{\frac{n_1+n_2}{2}} \int_0^{\infty} (e^{-u})(u)^{\frac{n_1+n_2}{2}-1} dy_2 \\ &= \frac{\frac{n_1}{n_2} (\frac{n_1}{n_2} y_1)^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} \cdot (1/c)^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1+n_2}{2}) \\ &= \frac{\Gamma(\frac{n_1+n_2}{2})}{\Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} 2^{-\frac{n_1+n_2}{2}} c^{-\frac{n_1+n_2}{2}} \left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} y_1^{\frac{n_1}{2}-1} \end{aligned}$$

Note that we can write $c^{-\frac{n_1+n_2}{2}}$ as

$$2^{\frac{n_1+n_2}{2}} \cdot (\frac{n_1}{n_2} y_1 + 1)^{-\frac{n_1+n_2}{2}}.$$

Thus, substituting c back, the term $2^{-\frac{n_1+n_2}{2}}$ cancels with the term $2^{\frac{n_1+n_2}{2}}$; substituting x for y_1, we obtain the final F pdf:

$$\boxed{f(x) = \frac{\Gamma(\frac{n_1+n_2}{2})}{\Gamma(\frac{n_1}{2})\Gamma(\frac{n_2}{2})} \cdot \left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} x^{\frac{n_1}{2}-1}(1+\frac{n_1}{n_2}x)^{-\frac{n_1+n_2}{2}},\ x \geq 0,\ 0 \text{ otherwise.}}$$

□

6.2 Order Statistics

Exercise 6.16. *Let $X_1, X_2, \ldots, X_n$ be i.i.d continuous random variables with density function f and distribution function F. Find the following:*

1. *The density function for* $\max\{X_1, \ldots, X_n\}$.
2. *The density function for* $\min\{X_1, \ldots, X_n\}$.
3. *The density function for $X_{(j)}$ where $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$ are the order statistics for $X_1, \ldots, X_n$.*
4. *The joint density function for the order statistics.*

Solution. In general, our strategy is to find the distribution function and then differentiate.

1.

$$\begin{aligned} F_{\max\{X_1,\ldots,X_n\}}(x) &= P\{\max\{X_1, \ldots, X_n\} \leq x\} = P\{\forall i,\ X_i \leq x\} \\ &= \prod_{i=1}^{n} P\{X_i \leq x\} = \prod_{i=1}^{n} F(x) \\ &= [F(x)]^n. \end{aligned}$$

Differentiating, we obtain

$$\boxed{f_{\max\{X_1,\ldots,X_n\}}(x) = n[F(x)]^{n-1} f(x)}. \tag{14}$$

2.

$$\begin{aligned} 1 - F_{\min\{X_1,\ldots,X_n\}}(x) &= P\{\min\{X_1, \ldots, X_n\} \geq x\} = P\{\forall i,\ X_i \geq x\} \\ &= \prod_{i=1}^{n} P\{X_i \geq x\} = \prod_{i=1}^{n} [1 - F(x)] \\ &= [1 - F(x)]^n \\ \implies F_{\min\{X_1,\ldots,X_n\}} &= 1 - [1 - F(x)]^n. \end{aligned}$$

Differentiating, we obtain

$$\boxed{f_{\min\{X_1,\ldots,X_n\}}(x) = n[1 - F(x)]^{n-1} f(x).} \tag{15}$$

3. By definition of the derivative, write $f_{X_{(j)}}(x) = \frac{d}{dx} F_{X_{(j)}}(x) = \lim_{h\to 0} \frac{F_{X_{(j)}}(x+h) - F_{X_{(j)}}(x)}{h} = \lim_{h\to 0} \frac{P\{x < X_{(j)} \leq x+h\}}{h}$, where $h > 0$.
(H): Assume only one value lies in the interval $(x, x+h]$. Under this assumption,

$$\begin{aligned} P\{x < X_{(j)} \leq x+h\} &= \binom{n}{j} j[F(x)]^{j-1}[F(x+h) - F(x)][1 - F(x+h)]^{n-j} \\ \implies \lim_{h\to 0} \frac{P\{x < X_{(j)} \leq x+h\}}{h} &= \binom{n}{j} j[F(x)]^{j-1} \lim_{h\to 0} \left(\frac{F(x+h) - F(x)}{h}[1 - F(x+h)]^{n-j}\right) \end{aligned}$$

Note that $f(x) = \lim_{h\to 0} \frac{F(x+h)-F(x)}{h}$ and, since F is right-continuous, that

$$\lim_{h\to 0} [1 - F(x+h)]^{n-j} = [1 - F(x)]^{n-j}.$$

It follows by the product property of limits that

$$f_{X_{(j)}}(x) = \binom{n}{j} j[F(x)]^{j-1} f(x)[1-F(x)]^{n-j} \tag{16}$$

$$= \boxed{\frac{n!}{(j-1)!(n-j)!}[F(x)]^{j-1}[1-F(x)]^{n-j} f(x)}. \tag{17}$$

Since the probability that exactly k variables lie in $(x, x+h]$ is $O(h^k)$, then as $h \to 0$, the probabilities given 2 or more variables lie in the interval are negligible. Therefore, relaxing assumption (H), the formula we found holds.

4. It can be shown that

$$f_{X_{(1)},\ldots,X_{(n)}}(x_1,\ldots,x_n) = n!\prod_{i=1}^{n} f(x_i) = n! f_{X_1,\ldots,X_n}(x_1,\ldots,x_n).$$

Let $\sigma : [n] \to [n]$ be any permutation. Let $x_1 < x_2 < \cdots < x_n$. Let $\epsilon > 0$ be small enough. Then,

$$P\{\forall i, x_i - \epsilon < X_{\sigma(i)} < x_i + \epsilon\} \approx \epsilon^n f(x_1) f(x_2) \ldots f(x_n).$$

Thus,

$$P\{\forall i, x_i - \epsilon < X_{(i)} < x_i + \epsilon\} \approx n!\epsilon^n f(x_1) f(x_2) \ldots f(x_n).$$

Dividing by ϵ^n and taking the limit as $\epsilon \to 0$, obtain

$$f_{X_{(1)},\ldots,X_{(n)}}(x_1,\ldots,x_n) = n!\prod_{i=1}^{n} f(x_i). \tag{18}$$

□

Exercise 6.17 (Ross' P 6.44)**.** *Let X_1, X_2, X_3 be independently uniformly distributed over the interval $[0,1]$. Compute the probability that the largest of the three is larger than the sum of the other two.*

Solution. Here is a methodical approach:

1. Write the joint distribution of the order statistics (use uniformity, independence, and the constant 3!).

$$f(x) = 1$$
$$\text{independence} \implies f_{X_1,X_2,X_3}(x_1,x_2,x_3) = \prod_i f(x_i) = 1 \quad (x_i \in [0,1])$$
$$\text{Equation (18)} \implies f_{X_{(1)},X_{(2)},X_{(3)}}(x_1,x_2,x_3) = 3! \cdot 1 = 6 \quad (0 \le x_1 \le x_2 \le x_3 \le 1).$$

2. Define the region of the event properly (use $x_1 < x_2 < x_3$ and $x_1 + x_2 < x_3$).

$$2x_1 \le x_1 + x_2 \le x_3 \le 1 \implies x_1 \le 1/2.$$
$$x_1 + x_2 \le 1 \implies x_2 \le 1 - x_1.$$

Therefore, necessarily, the region of the event is a subset of the region $R = \{(x_1,x_2,x_3) : 0 \le x_1 \le 1/2 \wedge x_1 \le x_2 \le 1 - x_1 \wedge x_1 + x_2 \le x_3 \le 1\}$. This region is consistent with all inequalities, hence $R = \{X_{(3)} \ge X_{(1)} + X_{(2)}\}$.

3. Integrate.

$$P\{X_{(3)} \ge X_{(2)} + X_{(1)}\} = \int_0^{1/2} \int_{x_1}^{1-x_1} \int_{x_1+x_2}^{1} 6 dx_3 dx_2 dx_1$$

$$
\begin{aligned}
&= 6\int_0^{1/2}\int_{x_1}(1-x_1-x_2)dx_2dx_1 \\
&= 6\int_0^{1/2}(2x_1^2-2x_1+1/2)dx_1 \\
&= 6\cdot\frac{1}{12} = \boxed{\frac{1}{2}}
\end{aligned}
$$

□

Exercise 6.18 (Ross' TE 6.11)**.** *Let $X_1, X_2, \dots, X_n$ be i.i.d $Uniform(0,1)$ random variables. Fix a constant $c \in (0,1)$. Define a random variable $N := \min\{n : X_n > c\}$. Is N independent of X_N?*

Solution. Yes, N is independent of X_N.

- **Intuition:** Note that the random variables are independent and not sorted, hence indices are totally arbitrary. Therefore, knowing the smallest index n of a random variable whose value is above c tells you nothing about the value of X_n. The sample $(0, \frac{1+c}{2}, c, c, \dots)$ and the sample $(\frac{1+c}{2}, 0, c, c, \dots)$ are equally likely, though the smallest index changes.
- **Proof:** The random variables N, X_N are independent iff $P\{N = n, X_N \in A\} = P\{N = n\}P\{X_N \in A\}$ for every positive integer n and measurable set $A \subseteq \mathbb{R}$ (we may assume that $A \subseteq (c, 1]$).

$$
\begin{aligned}
P\{N=n\} &= P\{X_1 \le c, X_2 \le c, \dots, X_{n-1} \le c, X_n > c\} \\
&= \left[\prod_{i=1}^{n-1} P\{X_i \le c\}\right] P\{X_n > c\} = c^{n-1}(1-c), \\
P\{X_N \in A\} &= \int_A \frac{1}{1-c}dx = \frac{|A|}{1-c} \\
\implies P\{N=n\}P\{X_N \in A\} &= c^{n-1}(1-c)\frac{|A|}{1-c} = c^{n-1}\,|A|\,. \\
P\{N=n, X_N \in A\} &= P\{X_1 \le c, \dots, X_{n-1} \le c, X_n > c, X_n \in A\} \\
&= \left[\prod_{i=1}^{n-1} P\{X_i \le c\}\right] P\{X_n \in \underbrace{(c,1] \cap A}_{A}\} \\
&= c^{n-1}\int_A dx = c^{n-1}\,|A| \\
&= P\{N=n\}P\{X_N \in A\}.
\end{aligned}
$$

Therefore, N is independent of X_N.

□

Exercise 6.19. *Prove the identity*

$$
P\{X \le s, Y \le t\} = P\{X \le s\} + P\{Y \le t\} + P\{X > s, Y > t\} - 1. \tag{19}
$$

Solution. [proof 1] Start with

$$
\begin{aligned}
P\{X \le s, Y \le t\} &= 1 - P\{\{X > s\} \cup \{Y > t\}\} \\
&= 1 - P\{X > s, Y \le t\} - P\{X \le s, Y > t\} - P\{X > s, Y > t\} \\
\implies -P\{X \le s, Y \le t\} &= P\{X > s, Y \le t\} + P\{X \le s, Y > t\} + P\{X > s, Y > t\} - 1.
\end{aligned}
$$

Add $2P\{X \le s, Y \le t\}$ to both sides. Noting that

$$
\begin{aligned}
P\{X \le s\} &= P\{X \le s, Y \le t\} + P\{X \le s, Y > t\}, \\
P\{Y \le t\} &= P\{X \le s, Y \le t\} + P\{X > s, Y \le t\},
\end{aligned}
$$

we obtain

$$\begin{aligned}
P\{X \le s, Y \le t\} &= (P\{X \le s, Y \le t\} + P\{X > s, Y \le t\}) \\
&\quad + (P\{X \le s, Y > t\} + P\{X \le s, Y \le t\}) + P\{X > s, Y > t\} - 1 \\
&= P\{Y \le t\} + P\{X \le s\} + P\{X > s, Y > t\} - 1,
\end{aligned}$$

as desired. □

Solution. [proof 2] Let $A = \{X \le s\}, B = \{Y \le t\}$. We equivalently want to prove

$$P(A \cap B) = P(A) + P(B) + P(A^{\mathsf{c}} \cap B^{\mathsf{c}}) - 1.$$

Rearranging, we want to prove

$$1 - P(A^{\mathsf{c}} \cap B^{\mathsf{c}}) = P(A) + P(B) - P(A \cap B).$$

Observe the following:

- LHS $= P(A \cup B)$ by the complement property since, by DeMorgan's Law, $A \cup B = (A^{\mathsf{c}} \cap B^{\mathsf{c}})^{\mathsf{c}}$.
- $P(A \cup B) =$ RHS by the simple inclusion-exclusion principle.

Therefore, the identity holds. □

Solution. [proof 3 (visually)]

	$X \le s$	$X > s$
$Y > t$		
$Y \le t$	**1**	

$=$

	$X \le s$	$X > s$
$Y > t$		
$Y \le t$	**1**	**1**

$+$

	$X \le s$	$X > s$
$Y > t$	**1**	
$Y \le t$	**1**	

$+$

	$X \le s$	$X > s$
$Y > t$		**1**
$Y \le t$		

$-$

	$X \le s$	$X > s$
$Y > t$	**1**	**1**
$Y \le t$	**1**	**1**

□

Exercise 6.20 (Ross' TE 6.20)**.** *Let $X_1, X_2, \ldots$ be a sequence of iid continuous random variables. Find*

1. $P\{X_6 > X_1 \mid X_1 = \max(X_1, \ldots, X_5)\}$
2. $P\{X_6 > X_2 \mid X_1 = \max(X_1, \ldots, X_5)\}$

Solution. The ideas are as follows:

1. $\{X_6 > X_1 \mid X_1 = \max(X_1, \ldots, X_5)\} = \{X_6 > \max(X_1, \ldots, X_5)\} = \{X_6 = \max(X_1, \ldots, X_6)\}$. Since the variables are exchangeable, the probability is

$$\boxed{\frac{1}{6}}$$

.

2. $$P\{X_6 > X_2 \mid X_1 = \max(X_1, \ldots, X_5)\} = \frac{P\{\{X_6 > X_1 > \text{else}\} \cup \{X_1 > \text{else}, X_6 > X_2\}\}}{P\{X_1 > \text{first } 5\}}$$
$$= \frac{[4! + (5!/2)]/[6!]}{1/5} = \boxed{\frac{7}{12}}.$$

Explanation: we obtain 1/5 in the denominator since each of $X_1, \ldots, X_5$ is equally likely to be the maximum, and we obtain the numerator as follows:

- There are 6! total permutations of $X_1, \ldots, X_5, X_6$.
- If $X_6 > X_1$, then X_6 is the max of all, X_1 is the second max, and since the remaining 4 rvs are exchangeable, there are 4! permutations such that $X_6 > X_2$.
- If $X_1 > X_6$, then X_1 is the max of all, and since in this case $X_2, X_3, \ldots, X_6$ are exchangeable (all of them are iid rvs that are less than X_1), then there are 5! possible permutations and, by exchangeability again, exactly 5!/2 of them satisfy $X_6 > X_2$.

□

Exercise 6.21 (Green Book: Expected Value of Max and Min)**.** *Let $X_1, X_2, \ldots, X_n$ be iid $Uniform(0, 1)$. Find*

1. *The cdf, pdf, and expected value of $Z = \max\{X_1, X_2, \ldots, X_n\}$.*
2. *The cdf, pdf, and expected value of $Y = \min\{X_1, X_2, \ldots, X_n\}$.*

Solution. Let f, F denote the pdf and cdf of the $Uniform(0, 1)$ respectively.

1. The maximum Z:
   (a) Its cdf:
   $$F_Z(z) = P\{\max\{X_1, X_2, \ldots, X_n\} \le z\} = P\{X_1 \le z, X_2 \le z, \ldots, X_n \le z\}$$
   $$= \prod_{i=1}^{n} P\{X_i \le z\} = [F(z)]^n = \begin{cases} 0 & z < 0 \\ z^n & 0 \le z \le 1 \\ 1 & z > 1. \end{cases}$$
   (b) Its pdf:
   $$f_Z(z) = \frac{d}{dz} F_Z(z) = \begin{cases} nz^{n-1} & 0 \le z \le 1 \\ 0 & \text{otherwise.} \end{cases}$$
   (c) Its expectation:
   $$\mathrm{E}[Z] = \int_0^1 z \cdot nz^{n-1} dz = \frac{n}{n+1}.$$
2. The minimum Y:
   (a) Its cdf:
   $$F_Y(y) = P\{\min\{X_1, X_2, \ldots, X_n\} \le y\} = 1 - P\{\min\{X_1, X_2, \ldots, X_n\} > y\}$$
   $$= 1 - P\{X_1 > y, X_2 > y, \ldots, X_n > y\}$$
   $$= 1 - \prod_{i=1}^{n} P\{X_i > y\} = 1 - \prod_{i=1}^{n} [1 - P\{X_i \le y\}]$$
   $$= 1 - [1 - F(y)]^n = \begin{cases} 0 & y < 0 \\ 1 - (1-y)^n & 0 \le y \le 1 \\ 1 & y > 1. \end{cases}$$

(b) Its pdf:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} n(1-y)^{n-1} & 0 \le y \le 1 \\ 0 & \text{otherwise.} \end{cases}$$

(c) Its expectation:

$$\begin{aligned} \mathrm{E}[Y] &= \int_0^1 y \cdot n(1-y)^{n-1} dy = n \int_0^1 (1-u) u^{n-1} du \\ &= n \int_0^1 \left(u^{n-1} - u^n\right) du \\ &= n \left[\frac{1}{n} - \frac{1}{n+1}\right] = 1 - \frac{n}{n+1} = \frac{1}{n+1} \end{aligned}$$

□

Exercise 6.22 (Green Book: Random Ants)**.** 500 *ants are independently and uniformly at random placed on a 1-foot string. Immediately, each ant starts moving at a 1-foot-per-minute speed in the direction of either of the two string ends with equal probability. When two ants collide, they immediately reverse direction, moving at the same speed. What is the expected time for all ants to reach either end?*

Solution. Let $X_1, X_2, \ldots, X_{500} \sim Uniform(0.1)$ be iid. Observe the following:

- If ant i never collides, then if it moves towards the 0 end, it takes X_i minutes, and if it moves towards the 1 end, it takes $1 - X_i$ minutes, which is also $Uniform(0, 1)$.
- What if a collision happened? Here is the key insight: **we may assume that collisions never happen.** Suppose a collision happened between ants i and j:

  $$(\xrightarrow{i\text{: 1-ft/min}} \cdot \xleftarrow{j\text{: 1-ft/min}}).$$

  Consequently, the ants reversed direction at the same speed post-collision:

  $$(\xleftarrow{i\text{: 1-ft/min}} \cdot \xrightarrow{j\text{: 1-ft/min}}).$$

  Since the ants are exchangeable, we may switch labels:

  $$(\xleftarrow{j\text{: 1-ft/min}} \cdot \xrightarrow{i\text{: 1-ft/min}}).$$

  Effectively, the ants just moved past each other, as if this collision did not happen.
- Now, fix an assignment of directions. It follows from the previous bullets that the movement of each ant is a uniform random variable taking on values between 0 and 1. The expected time it takes for all ants to finish is the expected maximum time it takes one of the 500 ants to finish. That is the expected value of 500 iid $Uniform(0, 1)$ random variables, taking on values in minutes. By Exercise (6.21), that is $\frac{500}{501}$ minutes.
- Since every possible assignment of directions results in the same expected time, by the law of total expectation, the answer is

  $$\boxed{\text{Expected time } = \frac{500}{501} = 0.998 \text{ minutes}}.$$

□

7 Properties of Expectation

Definition 7.1 (Expectation)**.** *Let X be a real-valued random variable.*

Discrete case. *If X has probability mass function p, then the expectation of X is defined by*

$$\mathrm{E}[X] = \sum_x x\, p(x),$$

provided the series is absolutely convergent:

$$\sum_x |x|\, p(x) < \infty.$$

Continuous case. *If X has probability density function f, then the expectation of X is defined by*

$$\mathrm{E}[X] = \int_{-\infty}^{\infty} x f(x)\, dx,$$

provided the improper Riemann integral

$$\int_{-\infty}^{\infty} |x| f(x)\, dx < \infty$$

exists.

Remark 7.1. *The definitions above are included to mention the condition of absolute convergence or absolute integrability.*

Exercise 7.2. *Prove linearity of expectation for n arbitrary random variables.*

Solution. [proof attempt 1] We prove it in the continuous case. Let $X_1, \ldots, X_n$ be arbitrary continuous random variables with joint pdf f. Then,

$$\begin{aligned}
\mathrm{E}\left[\sum_{i=1}^n X_i\right] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty}(x_1 + x_2 + \cdots + x_n) f(x_1, x_2, \ldots x_n) dx_n dx_{n-1} \ldots dx_1 \\
&= \sum_{i=1}^n \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} x_i f(x_1, x_2, \ldots x_n) dx_n dx_{n-1} \ldots dx_1 \\
&= \sum_{i=1}^n \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} x_i f(x_1, x_2, \ldots x_n) dx_n dx_{n-1} \ldots dx_{i+1} dx_{i-1} \ldots dx_1 dx_i,
\end{aligned}$$

where the last step follows using Fubini's theorem. Proceeding, we obtain,

$$\begin{aligned}
\mathrm{E}\left[\sum_{i=1}^n X_i\right] &= \sum_{i=1}^n \int_{-\infty}^{\infty} x_i \int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} f(x_1, x_2, \ldots x_n) dx_n dx_{n-1} \ldots dx_{i+1} dx_{i-1} \ldots dx_1 dx_i \\
&= \sum_{i=1}^n \int_{-\infty}^{\infty} x_i \left[\int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} f(x_1, x_2, \ldots x_n) dx_n dx_{n-1} \ldots dx_{i+1} dx_{i-1} \ldots dx_1\right] dx_i \\
&= \sum_{i=1}^n \int_{-\infty}^{\infty} x_i f_{X_i}(x_i) dx_i \\
&= \sum_{i=1}^n \mathrm{E}[X_i].
\end{aligned}$$

□

Solution. [proof attempt 2] We prove linearity of expectation in the continuous case by strong induction on $n \geq 2$.

First, we prove the linearity of expectation in the binary case when $n = 2$. Let X and Y be arbitrary continuous random variables with joint pdf f. Then,

$$\mathrm{E}[X + Y] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x + y) f(x, y) dx dy$$

$$
\begin{aligned}
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf(x,y)dxdy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} yf(x,y)dxdy \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf(x,y)dydx + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} yf(x,y)dxdy \\
&= \int_{-\infty}^{\infty} x\left[\int_{-\infty}^{\infty} f(x,y)dy\right]dx + \int_{-\infty}^{\infty} y\left[\int_{-\infty}^{\infty} f(x,y)dx\right]dy \\
&= \int_{-\infty}^{\infty} xf_X(x)dx + \int_{-\infty}^{\infty} yf_Y(y)dy \\
&= \mathrm{E}[X] + \mathrm{E}[Y].
\end{aligned}
$$

Now, let $n \geq 3$, and suppose that linearity of expectation holds for every $2 \leq k \leq n-1$ variables. Let $X_1, X_2, \ldots, X_n$ be $n \geq 3$ arbitrary continuous random variables. Thus,

$$
\begin{aligned}
\mathrm{E}\left[\sum_{i=1}^{n} X_i\right] = \mathrm{E}\left[X_n + \sum_{i=1}^{n-1} X_i\right] &= \mathrm{E}[X_n] + \mathrm{E}\left[\sum_{i=1}^{n-1} X_i\right] \\
&= \mathrm{E}[X_n] + \sum_{i=1}^{n-1} \mathrm{E}[X_i] = \sum_{i=1}^{n} \mathrm{E}[X_i].
\end{aligned}
$$

□

Exercise 7.3 (Ross' P 7.6)**.** *A fair die is rolled* 10 *times. Calculate the expected sum of the* 10 *rolls.*

Solution. For each $1 \leq i \leq 10$, let X_i denote the number the die landed on in the i-th roll. By linearity of expectation,

$$
\mathrm{E}\left[\sum_{i=1}^{10} X_i\right] = \sum_{i=1}^{10} \mathrm{E}[X_i] = \sum_{i=1}^{10} 3.5 = 10 \cdot 3.5 = \boxed{35}.
$$

□

Exercise 7.4 (Law of Total Expectation)**.** *State and prove the law of total expectation.*

Solution. The statement is

$$
\mathrm{E}[X] = \mathrm{E}[\mathrm{E}[X \mid Y]]. \tag{20}
$$

Equivalently, if Y is a discrete random variable,

$$
\mathrm{E}[X] = \sum_{y} P\{Y = y\}\mathrm{E}[X \mid Y = y] \tag{21}
$$

and if Y is a continuous random variable,

$$
\mathrm{E}[X] = \int_{-\infty}^{\infty} f_Y(y)\mathrm{E}[X \mid Y = y]\, dy. \tag{22}
$$

To prove the continuous case (assuming X is continuous as well), we start with

$$
\begin{aligned}
\mathrm{E}[X] &= \int_{-\infty}^{\infty} xf_X(x)dx \\
&= \int_{-\infty}^{\infty} x\left[\int_{-\infty}^{\infty} f_{X,Y}(x,y)dy\right]dx \\
&= \int_{-\infty}^{\infty} x\left[\int_{-\infty}^{\infty} f_Y(y)f_{X|Y}(x \mid y)dy\right]dx \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf_Y(y)f_{X|Y}(x \mid y)dydx \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf_Y(y)f_{X|Y}(x \mid y)dxdy \quad \text{(by Fubini's theorem)} \\
&= \int_{-\infty}^{\infty} f_Y(y)\left[\int_{-\infty}^{\infty} xf_{X|Y}(x \mid y)dx\right]dy
\end{aligned}
$$

$$= \int_{-\infty}^{\infty} f_Y(y) \mathrm{E}[X \mid Y = y]\, dy.$$

□

Exercise 7.5 (Ross' P 7.1)**.** *You throw a (fair, six-sided) die and a (fair) coin simultaneously. If you get heads, you win twice the value that the die lands on, and if you get tails, you win half the value the die lands on. Find the expected winning.*

Solution. Let X be the die and Y be the coin. Let Z be the winning value. That is

$$Z = \begin{cases} 2X & \text{if } Y = H \\ \frac{1}{2}X & \text{if } Y = T. \end{cases}$$

We find $\mathrm{E}[Z]$. By the law of total expectation,

$$\begin{aligned}
\mathrm{E}[Z] &= P\{Y = H\}\mathrm{E}[Z \mid Y = H] + P\{Y = T\}\mathrm{E}[Z \mid Y = T] \\
&= \frac{1}{2}\mathrm{E}[2X] + \frac{1}{2}\mathrm{E}\left[\frac{1}{2}X\right] = \frac{5}{4}\mathrm{E}[X] \\
&= \frac{5}{4}\sum_{k=1}^{6} kP\{X = k\} = \frac{5}{4}\sum_{k=1}^{6} k\frac{1}{6} \\
&= \frac{5}{4 \cdot 6}\sum_{k=1}^{6} k = \frac{5}{4 \cdot 6}\frac{6 \cdot 7}{2} = \frac{35}{8} = \boxed{4.375}.
\end{aligned}$$

□

Exercise 7.6 (Variance decomposition formula / Law of Total Variance)**.** *Prove the identity*

$$\mathrm{Var}(X) = \mathrm{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathrm{E}[X \mid Y])$$

Solution. On the LHS, we have

$$\mathrm{Var}(X) = \mathrm{E}\left[X^2\right] - \mathrm{E}[X]^2$$

On the RHS, for the first term, we have

$$\begin{aligned}
\mathrm{E}[\mathrm{Var}(X \mid Y)] &= \mathrm{E}\left[\mathrm{E}\left[X^2 \mid Y\right] - \mathrm{E}[X \mid Y]^2\right] \\
&= \mathrm{E}\left[\mathrm{E}\left[X^2 \mid Y\right]\right] - \mathrm{E}\left[\mathrm{E}[X \mid Y]^2\right] \\
&= \mathrm{E}\left[X^2\right] - \mathrm{E}\left[\mathrm{E}[X \mid Y]^2\right],
\end{aligned}$$

and for the second term, we have

$$\begin{aligned}
\mathrm{Var}(\mathrm{E}[X \mid Y]) &= \mathrm{E}\left[\mathrm{E}[X \mid Y]^2\right] - \mathrm{E}[\mathrm{E}[X \mid Y]]^2 \\
&= \mathrm{E}\left[\mathrm{E}[X \mid Y]^2\right] - \mathrm{E}[X]^2.
\end{aligned}$$

Together, we obtain

$$\begin{aligned}
\mathrm{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathrm{E}[X \mid Y]) &= \left(\mathrm{E}\left[X^2\right] - \mathrm{E}\left[\mathrm{E}[X \mid Y]^2\right]\right) + \left(\mathrm{E}\left[\mathrm{E}[X \mid Y]^2\right] - \mathrm{E}[X]^2\right) \\
&= \mathrm{E}\left[X^2\right] - \mathrm{E}[X]^2 \\
&= \mathrm{Var}(X),
\end{aligned}$$

as desired. □

Remark 7.7. *In other words, with the interpretation that Y determines which group X represents, the law of total variance tells us that*

Total variance = Avg. within-group variance + Variance of group means.

Exercise 7.8. *Given the following dataset*

Group (Y)	*Observations* (X)	*Group Mean* $E[X\|Y]$
A	1, 2, 3	2
B	4, 5, 6	5
C	7, 8, 9	8

compute total variance in two different ways.

Solution. We first compute the relevant values.

x	y	μ_y	μ	$(x-\mu_y)^2$	$(\mu_y-\mu)^2$	$(x-\mu)^2$
1	A	2	5	1	9	16
2	A	2	5	0	9	9
3	A	2	5	1	9	4
4	B	5	5	1	0	1
5	B	5	5	0	0	0
6	B	5	5	1	0	1
7	C	8	5	1	9	4
8	C	8	5	0	9	9
9	C	8	5	1	9	16
sum				6	54	60

Now, on the one side, we have

$$\text{Total Variance} = \frac{1}{9} \cdot 60 = \frac{20}{3},$$

and on the other side, by the law of total variance.

$$\text{Avg. within-group variance} \;+\; \text{Variance of group means} = \frac{6}{9} + \frac{54}{9} = \frac{60}{9} = \frac{20}{3}$$

□

Remark 7.9. *A couple of comments on the previous exercise:*

- *We used the true variance of the empirical distribution (in which we divide by n). If we use the unbiased estimate of the population variance (in which we divide by $n-1$), then in general, the law of total variance does not hold.*
- *Variance of group means depends on group size; if the groups had different sizes, we would need a weighted average.*

Exercise 7.10 (Ross' TE 7.1)**.** *Show that*

$$\mathrm{E}[X] = \arg\min_{a\in\mathbb{R}} \mathrm{E}\left[(X-a)^2\right]$$

Solution. Write

$$\begin{aligned} f(a) &= \mathrm{E}\left[(X-a)^2\right] = \mathrm{E}\left[X^2 - 2aX + a^2\right] \\ &= \mathrm{E}\left[X^2\right] - 2a\mathrm{E}[X] + a^2, \end{aligned}$$

assuming that $\mathrm{E}\left[X^2\right] < \infty$. Taking the first derivative,

$$f'(a) = 2a - 2\mathrm{E}[X] \implies (f'(a) = 0 \iff a = \mathrm{E}[X]),$$

we find that $\mathrm{E}[X]$ is the only local extremum. Taking the second derivative,

$$f''(a) = 2 > 0,$$

we deduce that f is convex, so we conclude that $\mathrm{E}[X]$ is in fact the global minimum. □

Exercise 7.11 (Ross' TE 7.2)**.** *Suppose that X is a continuous random variable with density function f. Show that $\mathrm{E}[|X-a|]$ is minimized when a is equal to the median of F. Hint: Write*

$$\mathrm{E}[|X-a|] = \int |x-a|\, f(x)dx.$$

Now break up the integral into the regions where $x < a$ and where $x > a$ and differentiate.

Solution.

$$\begin{aligned}
g(a) = \mathrm{E}[|X-a|] &= \int_{-\infty}^{\infty} |x-a|\, f(x)dx \\
&= \int_{-\infty}^{a} (a-x)f(x)dx + \int_{a}^{\infty} (x-a)f(x)dx \\
&= a\left[\int_{-\infty}^{a} f(x)dx - \int_{a}^{\infty} f(x)dx\right] + \left[\int_{-\infty}^{a} -xf(x)dx + \int_{\infty}^{a} -xf(x)dx\right] \\
&= a\left[F(a) - (1-F(a))\right] + \left[\int_{-\infty}^{a} -xf(x)dx + \int_{\infty}^{a} -xf(x)dx\right] \\
&= a\left[2F(a)-1\right] + \left[\int_{-\infty}^{a} -xf(x)dx + \int_{\infty}^{a} -xf(x)dx\right] \\
\implies g'(a) &= 2F(a) - 1 + 2af(a) - 2af(a) = 2F(a) - 1 \\
\implies (g'(a) = 0 &\iff F(a) = \frac{1}{2} \iff a \text{ is the median.})
\end{aligned}$$

Note that $g''(a) = 2f(a) \geq 0$, hence g is convex. Therefore, the median is a global minimum. □

Remark 7.12. *Exercises* (7.10) *and* (7.11) *show us two statistical insights:*

- *The **squared** deviation is minimized at the **mean**.*
- *The **absolute** deviation is minimized at the **median**.*

Exercise 7.13 (Green Book: Connecting Noodles)**.** *A bowl has* 100 *noodles in it. You pick two free noodle ends uniformly at random and connect them, and you repeat this process until there are no free noodle ends. Calculate the expected number of noodle loops.*

Solution. For each $n \geq 1$, let X_n denote the number of loops when the bowl initially has n noodles and no loops. If the bowl has 1 noodle, you certainly will pick two ends of the same noodle and connect them to form a single loop; therefore, $\mathrm{E}[X_1] = 1$. Let $n \geq 2$.

- With probability $\frac{n}{\binom{2n}{2}} = \frac{n}{n(2n-1)} = \frac{1}{2n-1}$, you will pick two ends of the same noodle and connect them to form a single loop, in which case you have 1 loop and $n-1$ free noodles.
- Otherwise, you will pick two different noodles and connect them such that they become practically a single noodle, in which case you have 0 loops and $n-1$ free noodles.

Therefore, by the law of total expectation,

$$\mathrm{E}[X_n] = \frac{1}{2n-1}(1 + \mathrm{E}[X_{n-1}]) + (1 - \frac{1}{2n-1})\mathrm{E}[X_{n-1}] = \frac{1}{2n-1} + \mathrm{E}[X_{n-1}].$$

Digging down, we obtain

$$\begin{aligned}
\mathrm{E}[X_n] &= \frac{1}{2n-1} + \mathrm{E}[X_{n-1}] \\
&= \frac{1}{2n-1} + \frac{1}{2(n-1)-1} + \mathrm{E}[X_{n-2}] \\
&= \frac{1}{2n-1} + \cdots + \frac{1}{2(2)-1} + \mathrm{E}[X_1] \\
&= \sum_{k=1}^{n} \frac{1}{2k-1}.
\end{aligned}$$

□

Exercise 7.14 (Green Book: Optimal Hedge Ratio)**.** *You just bought a share of stock A and you want to hedge it by shorting stock B. How many shares of stock B should you short to minimize the variance of the hedged position? Assume that the returns of stock A has variance σ_A^2, the returns of stock B has variance σ_B^2, and the returns of stocks A and B have correlation coefficient ρ.*

Solution. Let R_A, R_B denote the returns of a share of stock A and a share of stock B, respectively. Since you are hedging one share of stock A, your returns from A are R_A, and since you are shorting stock B, your returns from B are $-hR_B$ where h is the number of shares of stock B you short. Therefore, your returns are $R_A - hR_B$. Therefore, we want to minimize $\mathrm{Var}\,(R_A - hR_B)$ with respect to h. We have

$$\begin{aligned}\mathrm{Var}\,(R_A - hR_B) &= \sigma_A^2 + h^2\sigma_B^2 - 2h\mathrm{Cov}\,(R_A, R_B)\\ &= \sigma_A^2 + h^2\sigma_B^2 - 2h\rho\sigma_A\sigma_B.\end{aligned}$$

(This is convex quadratic in terms of h, hence if its local extremum, if it exists, is the unique global minimum.)

Setting the derivative with respect to h to 0, we obtain

$$2h\sigma_B^2 - 2\rho\sigma_A\sigma_B = 0 \implies \boxed{h = \rho\frac{\sigma_A}{\sigma_B}}.$$

□

Remark 7.15. *As an aside, Exercise* (7.14) *involves 3 trading terms:*

- ***to long*** *a stock is to buy now to sell later.*
- ***to short*** *a stock borrow and sell now to buy back later.*
- ***to hedge*** *a stock is to take an offsetting position to reduce risk.*

Exercise 7.16 (Green Book: Correlation of Max and Min)**.** *Let X_1, X_2 be iid $Uniform(0,1)$ random variables. Let $Y = \min(X_1, X_2), Z = \max(X_1, X_2)$. Find the correlation between Y and Z.*

Solution. We find

$$\rho(Y,Z) = \frac{\mathrm{Cov}\,(Y,Z)}{\sqrt{\mathrm{Var}\,(Y)\,\mathrm{Var}\,(Z)}}.$$

Recall:

$$\mathrm{Cov}\,(Y,Z) = \mathrm{E}\,[YZ] - \mathrm{E}\,[Y]\,\mathrm{E}\,[Z]\,,\ \mathrm{Var}\,(Y) = \mathrm{E}\left[Y^2\right] - \mathrm{E}\,[Y]^2\,,\ \mathrm{Var}\,(Z) = \mathrm{E}\left[Z^2\right] - \mathrm{E}\,[Z]^2\,.$$

By Exercise (6.21), $\mathrm{E}\,[Y] = \frac{1}{3}, \mathrm{E}\,[Z] = \frac{2}{3}$. We find $\mathrm{E}\left[Y^2\right], \mathrm{E}\left[Z^2\right], \mathrm{E}\,[YZ]$. Recalling the density functions for max and min, we have

$$\mathrm{E}\left[Y^2\right] = \int_0^1 y^2 \cdot 2(1-y)dy = \frac{1}{6},\ \mathrm{E}\left[Z^2\right] = \int_0^1 z^2 \cdot 2zdz = \frac{1}{2}.$$

Recalling the joint density function for order statistics, we have

$$\mathrm{E}\,[YZ] = \int_0^1\int_y^1 yz \cdot (2)dzdy = \int_0^1 y\frac{1-y^2}{2} \cdot 2dy = \int_0^1 [y - y^3]dy = \frac{1}{4}.$$

Therefore,

$$\mathrm{Cov}\,(Y,Z) = \frac{1}{4} - \frac{1}{3}\cdot\frac{2}{3} = \frac{1}{36},\ \mathrm{Var}\,(Y) = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{18},\ \mathrm{Var}\,(Z) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}.$$

Therefore,

$$\rho(Y,Z) = \frac{1/36}{\sqrt{(1/18)(1/18)}} = \boxed{\frac{1}{2}}.$$

□

Exercise 7.17 (Green Book: Joint Default Probability)**.** *Suppose there is a %50 probability that bond A will default next year and %30 probability that bond B will default next year. What is the range of the probability that at least one bond will default next year? What is the range of the correlation between bond A and bond B?*

Solution. Let A be the event that bond A will default next year, and B be the event that bond B will default next year.

At most, A and B are disjoint, which is possible since the probabilities sum up to no more than 1; in this case

$$P(A \cup B) = P(A) + P(B) = 0.8.$$

At least, one event is a subset of the other; in this case, the smaller event, B, is a subset of the larger event, A, hence

$$P(A \cup B) = P(A) = 0.5.$$

Therefore,

$$0.5 \leq P(A \cup B) \leq 0.8.$$

Now, let $I_A \sim Bernoulli(0.5), I_B \sim Bernoulli(0.3)$ be the indicator variables that bond A will default and that bond B will default. We find

$$\rho(I_A, I_B) = \frac{\mathrm{Cov}\,(I_A, I_B)}{\sqrt{\mathrm{Var}\,(I_A)\,\mathrm{Var}\,(I_B)}}.$$

We have

$$\mathrm{Var}\,(I_A) = 0.5 \cdot 0.5 = 0.25,\ \mathrm{Var}\,(I_B) = 0.3 \cdot 0.7 = 0.21.$$

$$\begin{aligned}\mathrm{Cov}\,(I_A, I_B) &= \mathrm{E}\,[I_A I_B] - \mathrm{E}\,[I_A]\,\mathrm{E}\,[I_B] \\ &= P(AB) - P(A)P(B) = P(AB) - 0.5 \cdot 0.3 = P(AB) - 0.15.\end{aligned}$$

Note:

- If A and B are disjoint, then $P(AB) = 0$.
- If $B \subseteq A$, then $P(AB) = P(B) = 0.3$.

Therefore, $-0.15 = 0 - 0.15 \leq \mathrm{Cov}\,(I_A, I_B) \leq 30 - 0.15 = 0.15$ Therefore,

$$-\sqrt{\frac{3}{7}} = \frac{-0.15}{\sqrt{0.25 \cdot 0.21}} \leq \rho(I_A, I_B) \leq \frac{0.15}{\sqrt{0.25 \cdot 0.21}} = \sqrt{\frac{3}{7}}.$$

□

7.1 Moment Generating Functions (MGFs)

7.1.1 Discrete distributions

RV	MGF $M_X(t)$	Domain
Bernoulli(p)	$pe^t + 1 - p$	$t \in (-\infty, \infty)$
b(n, p)	$(pe^t + 1 - p)^n$	$t \in (-\infty, \infty)$
Geom(p)	$\frac{pe^t}{1-(1-p)e^t}$	$t \in \left(-\infty, \ln\left(\frac{1}{1-p}\right)\right)$
nb(r, p)	$\left(\frac{pe^t}{1-(1-p)e^t}\right)^r$	$t \in \left(-\infty, \ln\left(\frac{1}{1-p}\right)\right)$
Poisson(λ)	$e^{\lambda(e^t-1)}$	$t \in (-\infty, \infty)$
hg(N, M, n)		
Zeta(s)		

Exercise 7.18. *Derive the mgf for the binomial distribution.*

Solution. Sketch:

$$\sum e^{tk}\binom{n}{k}p^k(1-p)^{n-k} = \sum \binom{n}{k}(pe^t)^k(1-p)^{n-k} = (pe^t + 1 - p)^n$$

□

Exercise 7.19. *Derive the mgf for the Bernoulli distribution.*

Solution. Sketch: the bernoulli is binomial with n = 1;

$$pe^t + 1 - p$$

□

Exercise 7.20. *Derive the mgf for the Poisson distribution.*

Solution. Sketch:

$$\sum e^{tk}e^{-\lambda}\frac{\lambda^k}{k!} = e^{\lambda(e^t-1)}\sum e^{-\lambda e^t}\frac{(\lambda e^t)^k}{k!} = e^{\lambda(e^t-1)}$$

□

Exercise 7.21. *Derive the mgf for the geometric distribution.*

Solution. Sketch:

$$\sum_{k\geq 1} e^{tk}(1-p)^{k-1}p = pe^t\sum_{k\geq 1}[(1-p)e^t]^{k-1} = pe^t\sum_{k\geq 0}[(1-p)e^t]^k = \frac{pe^t}{1-(1-p)e^t}$$

□

Exercise 7.22. *Derive the mgf for the negative binomial distribution.*

Solution. Let $X \sim nb(r, p)$. Thus,

$$\begin{aligned} M_X(t) = \mathrm{E}\left[e^{tX}\right] &= \sum_{k=r}^{\infty} e^{tk} \binom{k-1}{r-1} p^r (1-p)^{k-r} \\ &= (pe^t)^r \sum_{k=r}^{\infty} \binom{k-1}{r-1} [(1-p)e^t]^{k-r} \\ &= (pe^t)^r \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} [(1-p)e^t]^{k}. \end{aligned}$$

Note that $\binom{k+r-1}{r-1}$ is the number of solutions to

$$n_1 + n_2 + \cdots + n_r = k, \text{ where } \forall i \in [r], n_i \geq 0.$$

In particular, it's the number of solutions to

$$x^{n_1} x^{n_2} \ldots x^{n_r} = x^{n_1+n_2+\cdots+n_r} = x^k.$$

Therefore, equivalently, with $x = (1-p)e^t$, we have

$$\begin{aligned} M_X(t) &= (pe^t)^r \left[\sum_{k=0}^{\infty} [(1-p)e^t]^k\right]^r \\ &= (pe^t)^r \left[\frac{1}{1-(1-p)e^t}\right]^r \quad ((1-p)e^t \leq 1 \iff 1-p \leq e^{-t} \iff t \leq \ln\left(\frac{1}{1-p}\right)) \\ &= \left(\frac{pe^t}{1-(1-p)e^t}\right)^r, \quad t \in \left(-\infty, \ln\left[\frac{1}{1-p}\right]\right) \end{aligned}$$

□

Exercise 7.23. *Derive the mgf for the hypergeometric distribution.*

Solution. TBC. □

Exercise 7.24. *Derive the mgf for the zeta distribution.*

Solution. TBC. □

7.1.2 Continuous distributions

RV	MGF $M_X(t)$	Domain
Uniform(a,b)	$\begin{cases} \frac{e^{tb}-e^{ta}}{t(b-a)} & t \neq 0, \\ 1 & t = 0. \end{cases}$	$t \in (-\infty, \infty)$
$N(\mu, \sigma^2)$	$e^{\mu t + \frac{1}{2}\sigma^2 t^2}$	$t \in (-\infty, \infty)$
Exp(λ)	$\frac{\lambda}{\lambda - t}$	$t \in (-\infty, \lambda)$
Gamma(α, λ)	$\left(\frac{\lambda}{\lambda-t}\right)^{\alpha}$	$t \in (-\infty, \lambda)$
$\chi^2(n)$	$\left(\frac{1}{1-2t}\right)^{n/2}$	$t \in (-\infty, \frac{1}{2})$
$T(n)$		
$F(n_1, n_2)$		

Exercise 7.25. *Derive the mgf for the uniform distribution.*

Solution. Let $X \sim Uniform(a,b), a < b$. Thus,

$$\begin{aligned} M_X(t) = \mathrm{E}\left[e^{tX}\right] &= \int_a^b \frac{e^{tx}}{b-a} dx \\ &= \frac{1}{b-a} \frac{e^{tx}}{t}\bigg|_a^b \\ &= \frac{e^{tb}-e^{ta}}{t(b-a)}, \quad t \neq 0. \end{aligned}$$

At $t = 0$, set $M_X(0) = \lim_{t\to 0} \mathrm{E}\left[e^{tX}\right]$. Thus,

$$\begin{aligned} M_X(0) &:= \lim_{t\to 0} \frac{e^{tb}-e^{ta}}{t(b-a)} \quad \text{(indeterminate; } 0/0) \\ &=^{L'hopital} \lim_{t\to 0} \frac{be^{tb}-ae^{ta}}{(b-a)} = \frac{b-a}{b-a} = 1. \end{aligned}$$

□

Exercise 7.26. *Derive the mgf for the normal distribution.*

Solution. Let $Z \sim N(0,1)$. Thus,

$$\begin{aligned}
M_Z(t) = \mathrm{E}\left[e^{tZ}\right] &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2) \cdot \exp(tz) dz \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2 + tz) dz \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}(z^2 - 2tz + t^2) + \frac{t^2}{2}) dz \\
&= \exp(\frac{t^2}{2}) \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}(z-t)^2) dz}_{\text{pdf of } N(t,1)} \\
&= \exp(\frac{t^2}{2}), \quad t \in (-\infty, \infty).
\end{aligned}$$

Now, let $X \sim N(\mu, \sigma^2)$. Note that $X \sim \sigma Z + \mu$. Therefore,

$$\begin{aligned}
M_X(t) = \mathrm{E}\left[e^{tX}\right] &= \mathrm{E}\left[e^{t(\sigma Z + \mu)}\right] = \mathrm{E}\left[e^{t\sigma Z} e^{t\mu}\right] \\
&= \mathrm{E}\left[e^{t\sigma Z}\right] \mathrm{E}\left[e^{t\mu}\right] = e^{t\mu} M_Z(t\sigma) = e^{t\mu} e^{\frac{(t\sigma)^2}{2}} \\
&= \exp(t\mu + \frac{1}{2}\sigma^2 t^2), \ t \in (-\infty, \infty).
\end{aligned}$$

□

Exercise 7.27. *Derive the mgf for the gamma distribution.*

Solution. Let $X \sim Gamma(\alpha, \lambda), \alpha > 0, \lambda > 0$. Thus,

$$\begin{aligned}
M_X(t) = \mathrm{E}\left[e^{tX}\right] &= \int_0^{\infty} e^{tx} \cdot \frac{\lambda e^{-\lambda x} (\lambda x)^{\alpha - 1}}{\Gamma(\alpha)} dx \\
&= \int_0^{\infty} \frac{\lambda e^{-(\lambda - t)x} (\lambda x)^{\alpha - 1}}{\Gamma(\alpha)} dx \\
&= \frac{\lambda^\alpha}{(\lambda - t)^\alpha} \underbrace{\int_0^{\infty} \frac{(\lambda - t) e^{-(\lambda - t)} ([\lambda - t]x)^{\alpha - 1}}{\Gamma(\alpha)}}_{\text{pdf of } Gamma(\alpha, \lambda - t), \ \lambda - t > 0} \\
&= \left(\frac{\lambda}{\lambda - t}\right)^\alpha, \ t \in (-\infty, \lambda).
\end{aligned}$$

□

Exercise 7.28. *Derive the mgf for the exponential distribution.*

Solution. Let $X \sim exp(\lambda)$. Thus, $X \sim Gamma(1, \lambda)$. Therefore,

$$M_X(t) = \frac{\lambda}{\lambda - t}, \ t \in (-\infty, \lambda).$$

□

Exercise 7.29. *Derive the mgf for the chi-squared distribution.*

Solution. Let $X \sim \chi^2_n$. Thus, $X \sim Gamma(\frac{n}{2}, \frac{1}{2})$. Therefore,

$$\begin{aligned}
M_X(t) &= \left(\frac{1/2}{1/2 - t}\right)^{n/2}, \ t \in (-\infty, 1/2) \\
&= \left(\frac{1}{1 - 2t}\right)^{n/2}, \ t \in (-\infty, 1/2)
\end{aligned}$$

□

Exercise 7.30. *Derive the mgf for the T distribution.*

Exercise 7.31. *Derive the mgf for the F distribution.*

Exercise 7.32. *Derive the mgf for the beta distribution.*

8 Inequalities and Asymptotics

8.1 Inequalities

Exercise 8.1. *State and prove Markov Inequality. What information does it use? What information does it provide?*

Solution. Markov inequality provides an upper tail bound for nonnegative random variables using only the distribution mean.

- **Statement:** $(P\{X \geq 0\} = 1 \wedge \mathrm{E}[X] < \infty \wedge a > 0) \implies P\{X \geq a\} \leq \frac{\mathrm{E}[X]}{a}$.
- **Proof:** $X \geq aI[X \geq a] \implies \mathrm{E}[X] \geq \mathrm{E}[aI[X \geq a]] \implies \mathrm{E}[X] \geq aP\{X \geq a\}$.

□

Exercise 8.2. *State and prove Chebyshev's Inequality. What information does it use? What information does it provide?*

Solution. Chebyshev's inequality provides a concentration bound on deviation from the mean using both the distribution mean and the variance.

- **Statement:** $(\mu = \mathrm{E}[X] < \infty \wedge \sigma^2 = \mathrm{Var}(X) < \infty \wedge a > 0) \implies P\{|X - \mu| \geq a\} \leq \frac{\sigma^2}{a^2}$.
- **Proof:** $P\{|X - \mu| \geq a\} = P\{(X - \mu)^2 \geq a^2\} \leq \frac{\mathrm{E}\left[(X-\mu)^2\right]}{a^2} = \frac{\sigma^2}{a^2}$.

□

Exercise 8.3 (Cantelli / One-sided Chebyshev's Inequality)**.** *Let X be a random variable with $\mu = \mathrm{E}[X]$ and $\sigma^2 = \mathrm{Var}(X)$. Prove that for every $a > 0$,*

$$\textbf{\textit{Upper Tail Bound:}}\ P\{X \geq \mu + a\} \leq \frac{\sigma^2}{\sigma^2 + a^2} \tag{23}$$

$$\textbf{\textit{Lower Tail Bound:}}\ P\{X \leq \mu - a\} \leq \frac{\sigma^2}{\sigma^2 + a^2} \tag{24}$$

Solution. [Solution developed jointly with Yanshuo Chen and Carwyn Collinsworth in UMD CMSC651 Spring 2026]

Note that for any $b \geq 0$

$$P(X - \mu \geq a) = P(X - \mu + b \geq a + b) \leq P((X - \mu + b)^2 \geq (a + b)^2).$$

Then by Markov inequality,

$$\Pr[(X - \mu + b)^2 \geq (a + b)^2] \leq \frac{E[(X - \mu + b)^2]}{(a + b)^2}$$

Let $Y = X - \mu$, $\mathbb{E}[Y] = 0$ and $Var(Y) = \sigma^2$, we have $\mathbb{E}[(Y + b)^2] = \mathbb{E}[Y^2 + b^2 + 2Yb] = \mathbb{E}[Y^2] + b^2 + 2\mathbb{E}[Y]b = Var(Y) + \mathbb{E}[Y]^2 + b^2 + 2\mathbb{E}[Y]b = \sigma^2 + b^2$. So,

$$\begin{aligned}\Pr[(X - \mu + b)^2 \geq (a + b)^2] &\leq \frac{E[(X - \mu + b)^2]}{(a + b)^2} \\ &= \frac{\sigma^2 + b^2}{(a + b)^2}\end{aligned}$$

treat it as a function f of b, to obtain the bound, we need to minimize the function over all $b \geq 0$.

Differentiating using the quotient rule:

$$f'(b) = \frac{(2b)(a + b)^2 - (\sigma^2 + b^2) \cdot 2(a + b)}{(a + b)^4}.$$

Factoring $2(a + b)$ from the numerator:

$$f'(b) = \frac{2(a + b)\Big(b(a + b) - (\sigma^2 + b^2)\Big)}{(a + b)^4} = \frac{2\Big(ab - \sigma^2\Big)}{(a + b)^3}.$$

For $b \geq 0$ we have $(a+b)^3 > 0$, hence the sign of $f'(b)$ is the sign of $ab - \sigma^2$:

$$f'(b) < 0 \iff b < \frac{\sigma^2}{a}, \qquad f'(b) = 0 \iff b = \frac{\sigma^2}{a}, \qquad f'(b) > 0 \iff b > \frac{\sigma^2}{a}.$$

Therefore f is decreasing on $\left[0, \frac{\sigma^2}{a}\right]$ and increasing on $\left[\frac{\sigma^2}{a}, \infty\right)$. Hence f attains its global minimum at

$$b = \frac{\sigma^2}{a}.$$

Based on this function,

$$\begin{aligned}
\Pr[(X - \mu + \sigma^2/a)^2 \geq (a + \sigma^2/a)^2] &\leq \frac{E[(X - \mu + \sigma^2/a)^2]}{(a + \sigma^2/a)^2} \\
&= \frac{\sigma^2 + (\sigma^2)^2/a^2}{(a + \sigma^2/a)^2} = \frac{\sigma^2(a^2 + \sigma^2)}{(a^2 + \sigma^2)^2} \\
&= \frac{\sigma^2}{a^2 + \sigma^2}
\end{aligned}$$

□

Exercise 8.4. *Let $X_1, \ldots, X_n \sim$ Bernoulli(p) be i.i.d, and let $X = \sum_{i=1}^n X_i$. Compare the Markov bound and the Chebyshev's bound for $P\{X \geq 2np\}$.*

Solution.

$$\textbf{Markov: } P\{X \geq 2np\} \leq \frac{np}{2np} = \frac{1}{2}.$$

$$\textbf{One-sided Chebyshev's: } P\{X \geq 2np\} \leq \frac{np(1-p)}{np(1-p) + n^2p^2} = \frac{1-p}{1-p+np} \leq \frac{1}{np}.$$

If np is large, then Chebyshev's bound can be much smaller. For instance, with a sample of 50 patients each ill independently with probability 1/5, whereas Markov tells us that the probability that the number of ill patients is at least 20 is at most 0.5, Chebyshev's tells us that that probability is actually no more than 0.1. □

Exercise 8.5. *State and prove the general Chernoff bound.*

Solution. Chernoff inequality provides an upper tail bound for any random variable using its moment generating function.

- **Statement:** Let X be a random variable with mgf M_x. Fix $a > 0$.

$$P\{X \geq a\} \leq \inf_{t>0}(e^{-ta} M_X(t)) \tag{25}$$

$$P\{X \leq a\} \leq \inf_{t>0}(e^{ta} M_X(-t)). \tag{26}$$

- **Proof:** The idea is to exponentiate and then apply Markov and then optimize over t. We prove the upper tail bound. For every $t > 0$, we have

$$P\{X \geq a\} = P\{tX \geq ta\} = P\{e^{tX} \geq e^{ta}\} \leq \frac{\mathrm{E}\left[e^{tX}\right]}{e^{ta}} = e^{-ta} M_X(t).$$

Since this holds for every $t > 0$, it follows that

$$P\{X \geq a\} \leq \inf_{t>0}(e^{-ta} M_X(t)).$$

To prove the lower tail bound, observe that

$$P\{X \leq a\} = P\{-tX \geq -ta\} = P\{e^{-tX} \geq e^{-ta}\},$$

and the rest follows similarly.

□

Exercise 8.6 (Chernoff bound: KL-divergence form)**.** *Let $X_1, X_2, \ldots, X_n$ be independent random variables with support $[0,1]$. Let $X = \sum_{i=1}^n X_i$. Assume for each $1 \le i \le n$ that $\mathrm{E}[X_i] = p$, so $\mu = \mathrm{E}[X] = np$. Prove the following bounds:*

$$\textbf{\textit{Upper Tail Bound:}}\ \forall q > p,\ P\{X \ge nq\} \le e^{-n \cdot D_{KL}(q||p)}, \tag{27}$$

$$\textbf{\textit{Lower Tail Bound:}}\ \forall q < p,\ P\{X \le nq\} \le e^{-n \cdot D_{KL}(q||p)}. \tag{28}$$

where the Kullback-Liebler divergence of q from p is given by $D_{KL}(q||p) = q \ln \frac{q}{p} + (1-q) \ln \frac{1-q}{1-p}$.

Solution. We prove the upper tail bound; the lower tail case is similar.

- **Proof Idea:** We apply the Chernoff bound and upper bound the mgf using convexity. Optimizing the bound over t leads to an expression whose minimum has the form of the KL divergence.
- **Proof:** Let $q > p$. Then,

$$\begin{aligned}
P\{X \ge nq\} &= P\{e^{tX} \ge e^{tnq}\} \le e^{-tnq}\mathrm{E}\left[e^{tX}\right] \\
e^{tX} = \prod_{i=1}^n e^{tX_i} &\implies \mathrm{E}\left[e^{tX}\right] = \mathrm{E}\left[\prod_{i=1}^n e^{tX_i}\right] \\
X_i\text{'s independent} &\implies e^{tX_i}\text{'s independent} \implies \mathrm{E}\left[e^{tX}\right] = \prod_{i=1}^n \mathrm{E}\left[e^{tX_i}\right] \\
e \text{ is convex } \wedge X_i \in [0,1] &\implies e^{tX_i} \le X_i e^t + (1 - X_i) \\
&\implies \mathrm{E}\left[e^{tX}\right] \le (pe^t + 1 - p)^n \\
&\implies P\{X \ge nq\} \le e^{-tnq}(pe^t + 1 - p)^n.
\end{aligned}$$

Since this holds for every $t > 0$, it follows that

$$P\{X \ge nq\} \le \inf_{t>0} f(t) \text{ where } f(t) = e^{-tnq}(pe^t + 1 - p)^n.$$

Minimizing $f(t)$ is equivalent to minimizing

$$\phi(t) = \ln f(t) = n \ln(pe^t + 1 - p) - tnq.$$

Taking the derivative and setting to 0,

$$\begin{aligned}
0 = \phi'(t) &= \frac{npe^t}{pe^t + 1 - p} - nq \\
&\implies nq(pe^t + 1 - p) = npe^t \\
&\implies e^t(qp - p) = q(p-1) \\
&\implies e^t = \frac{q(p-1)}{p(q-1)},\ t = \ln(\frac{q(p-1)}{p(q-1)}).
\end{aligned}$$

Since

$$\phi''(t) = \frac{npe^t[1-p]}{(pe^t + 1 - p)^2} > 0,$$

the function $\phi(t)$ is strictly convex, hence the extremum we obtained is the unique global minimum. Therefore,

$$\begin{aligned}
\min \phi(t) &= n \ln(p\frac{q(p-1)}{p(q-1)} + 1 - p) - nq \ln(\frac{q(p-1)}{p(q-1)}) \\
&= -n \left(\underbrace{(1-q) \ln \frac{1-q}{1-p} + q \ln \frac{q}{p}}_{D_{KL}(q||p)} \right) \text{ (after nontrivial algebra)} \\
\implies \min f(t) &= e^{-nD_{KL}(q||p)}.
\end{aligned}$$

Therefore,

$$P\{X \le nq\} \le e^{-nD_{KL}(q||p)}.$$

□

Remark 8.7. *It can be shown that* $D_{KL}(q||p) \geq 0$.

Exercise 8.8 (Chernoff bound: multiplicative form)**.** *Let* $X_1, X_2, \ldots, X_n$ *be independent random variables with support* $[0,1]$. *Let* $X = \sum_{i=1}^n X_i$. *Assume for each* $1 \leq i \leq n$ *that* $\mathrm{E}[X_i] = p$, *so* $\mu = \mathrm{E}[X] = np$. *Let* $\delta > 0$. *Prove the following bounds:*

$$\textbf{\textit{Upper Tail Bound:}}\ P\{X \geq (1+\delta)\mu\} \leq e^{-\mu[(1+\delta)\ln(1+\delta)-\delta]} \tag{29}$$

$$\textbf{\textit{Lower Tail Bound:}}\ P\{X \leq (1-\delta)\mu\} \leq e^{-\mu[(1-\delta)\ln(1-\delta)+\delta]} \tag{30}$$

Exercise 8.9 (Chernoff bound: user-friendly multiplicative form)**.** *Let* $X_1, X_2, \ldots, X_n$ *be independent random variables with support* $[0,1]$. *Let* $X = \sum_{i=1}^n X_i$. *Assume for each* $1 \leq i \leq n$ *that* $\mathrm{E}[X_i] = p$, *so* $\mu = \mathrm{E}[X] = np$. *Let* $\delta \in [0,1]$. *Prove the following bounds:*

$$\textbf{\textit{Upper Tail Bound:}}\ P\{X \geq (1+\delta)\mu\} \leq e^{-\mu\delta^2/3} \tag{31}$$

$$\textbf{\textit{Lower Tail Bound:}}\ P\{X \leq (1-\delta)\mu\} \leq e^{-\mu\delta^2/2} \tag{32}$$

Exercise 8.10. *State and prove Jensen's Inequality for discrete and continuous random variables.*

Lemma 8.11. *Let* I *be a countable set of real numbers. Let* $\lambda_x \geq 0$ *for each* $x \in I$ *such that* $\sum_{x \in I} \lambda_x = 1$. *Let* $f : \mathbb{R} \to \mathbb{R}$. *Then,*

$$f \text{ convex} \implies f\left(\sum_{x \in I} \lambda_x x\right) \leq \sum_{x \in I} f(x)\lambda_x, \tag{33}$$

$$f \text{ concave} \implies f\left(\sum_{x \in I} \lambda_x x\right) \geq \sum_{x \in I} f(x)\lambda_x \tag{34}$$

Proof. Focus on the convex case; the other is symmetric. Use strong induction on n. When $n = 2$, the inequality holds by definition. Assume the inequality holds for all $2 \leq k \leq n-1$. We prove it holds for n:

$$\begin{aligned}
f(\sum_{i=1}^n \lambda_i x_i) &= f(\sum_{i=1}^{n-2} \lambda_i x_i + (\lambda_n + \lambda_{n-1})(\frac{\lambda_{n-1}}{\lambda_{n-1}+\lambda_n} x_{n-1} + \frac{\lambda_n}{\lambda_{n-1}+\lambda_n} x_n)) \\
&\leq \sum_{i=1}^{n-2} \lambda_i f(x_i) + (\lambda_{n-1} + \lambda_n) f(\frac{\lambda_{n-1}}{\lambda_{n-1}+\lambda_n} x_{n-1} + \frac{\lambda_n}{\lambda_{n-1}+\lambda_n} x_n) \\
&\leq \sum_{i=1}^{n-2} \lambda_i f(x_i) + (\lambda_{n-1} + \lambda_n)(\frac{\lambda_{n-1}}{\lambda_{n-1}+\lambda_n} f(x_{n-1}) + \frac{\lambda_n}{\lambda_{n-1}+\lambda_n} f(x_n)) \\
&= \sum_{i=1}^{n-2} \lambda_i f(x_i) + \lambda_{n-1} f(x_{n-1}) + \lambda_n f(x_n) \\
&= \sum_{i=1}^{n} \lambda_i f(x_i).
\end{aligned}$$

Taking the limit as $n \to \infty$, we obtain

$$f\left(\sum_{i=1}^{\infty} \lambda_i x_i\right) \leq \sum_{i=1}^{\infty} \lambda_i f(x_i),$$

hence the inequality holds for every countable set I. □

Lemma 8.12. *Let* $A \subseteq \mathbb{R}$ *be a continuous set. Let* $\lambda_x \geq 0$ *for each* $x \in A$ *such that* $\int_A \lambda_x dx = 1$. *Let* $f : \mathbb{R} \to \mathbb{R}$ *be continuous on* A. *Then,*

$$f \text{ convex} \implies f\left(\int_A \lambda_x x dx\right) \leq \int_A f(x)\lambda_x dx, \tag{35}$$

$$f \text{ concave} \implies f\left(\int_A \lambda_x x dx\right) \geq \int_A f(x)\lambda_x dx. \tag{36}$$

Proof. Assume the discrete version holds. Focus on the convex case. Use the definition of the integral, then apply the discrete version (you may need to normalize the weights for them to actually add up to 1), then use continuity to manipulate the limit, as follows:

$$\begin{aligned}
\int_A f(x)\lambda_x dx &= \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\lambda_{x_i}(x_i - x_{i-1}) \\
&= \lim_{n\to\infty} \left[\left(\sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1}) \right) \left(\sum_{i=1}^{n} f(x_i) \frac{\lambda_{x_i}(x_i - x_{i-1})}{\sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1})} \right) \right] \\
&\geq \lim_{n\to\infty} \left[\left(\sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1}) \right) f \left(\sum_{i=1}^{n} x_i \frac{\lambda_{x_i}(x_i - x_{i-1})}{\sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1})} \right) \right] \\
&= \left[\lim_{n\to\infty} \sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1}) \right] \lim_{n\to\infty} f \left(\sum_{i=1}^{n} x_i \frac{\lambda_{x_i}(x_i - x_{i-1})}{\sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1})} \right) \\
&= \left[\lim_{n\to\infty} \sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1}) \right] f \left(\lim_{n\to\infty} \sum_{i=1}^{n} x_i \frac{\lambda_{x_i}(x_i - x_{i-1})}{\sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1})} \right) \\
&= \left[\lim_{n\to\infty} \sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1}) \right] f \left(\lim_{n\to\infty} \frac{\sum_{i=1}^{n} x_i \lambda_{x_i}(x_i - x_{i-1})}{\sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1})} \right) \\
&= \left[\lim_{n\to\infty} \sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1}) \right] f \left(\frac{\lim_{n\to\infty} \sum_{i=1}^{n} x_i \lambda_{x_i}(x_i - x_{i-1})}{\lim_{n\to\infty} \sum_{i=1}^{n} \lambda_{x_i}(x_i - x_{i-1})} \right) \\
&= \left[\int_A \lambda_x dx \right] f \left(\frac{\int_A \lambda_x x dx}{\int_A \lambda_x dx} \right) \\
&= f \left(\int_A \lambda_x x dx \right).
\end{aligned}$$

□

Solution. Jensen's inequality provides an upper (or lower) bound for the image of a convex combination of x_i's under a convex (or concave) function f by the corresponding convex combination of $f(x_i)$'s.

- **Statement:** Let X be a discrete/continuous random variable, and let $f : \mathbb{R} \to \mathbb{R}$.

$$f \text{ convex} \implies \mathrm{E}[f(x)] \geq f(\mathrm{E}[X]), \tag{37}$$
$$f \text{ concave} \implies \mathrm{E}[f(x)] \leq f(\mathrm{E}[X]). \tag{38}$$

- **Proof:** Idea: Focus on the convex case; the concave case is symmetric. Note that probabilities are nonnegative and sum up to 1.
  - For the discrete case: expectation is a convex combination in summation form; use Lemma (8.11).
  - For the continuous case: expectation is a convex combination in integral form; use Lemma (8.12).

□

Exercise 8.13. *Prove that* $\mathrm{Var}(X) \geq 0$ *for every discrete/continuous random variable* X*.*

Solution.

$$\begin{aligned}
0 \leq \mathrm{Var}(X) = \mathrm{E}\left[X^2\right] - \mathrm{E}[X]^2 &\iff \mathrm{E}\left[X^2\right] \geq \mathrm{E}[X]^2 \\
&\impliedby \text{Jensen's inequality since } f(x) = x^2 \text{ is convex on } \mathbb{R}.
\end{aligned}$$

□

Exercise 8.14. *Prove the AM-GM inequality.*

Solution. Let $x_1, x_2, \ldots, x_n$ be positive real numbers. Then,

$$\text{AM} \geq \text{GM} \iff \frac{1}{n}\sum_{k=1}^{n} x_k \geq \left[\prod_{k=1}^{n} x_k\right]^{1/n} \iff \ln\left(\frac{1}{n}\sum_{k=1}^{n} x_k\right) \geq \ln\left(\left[\prod_{k=1}^{n} x_k\right]^{1/n}\right)$$

$$\iff \ln\left(\frac{1}{n}\sum_{k=1}^{n} x_k\right) \geq \frac{1}{n}\sum_{k=1}^{n} \ln(x_k) \impliedby \text{Jensen's inequality since ln is concave on } (0,\infty).$$

□

8.2 Convergence modes

Exercise 8.15 (Deterministic sequence converging to a constant)**.** *Let $(X_n)_{n\in\mathbb{Z}^+}$ be a sequence of random variables such that with probability 1, $X_n = \frac{1}{n}$. Let X be a random variable such that with probability 1, $X = 0$. Does the sequence converge to X? In what sense?*

Solution. Let's test the three modes of convergence.

- **Convergence in distribution:** Note the distributions

$$F_{X_n}(x) = \begin{cases} 0 & x < \frac{1}{n} \\ 1 & x \geq \frac{1}{n} \end{cases}, \quad F_X(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}.$$

  Let's consider the continuity points of F_X: $x \in \mathbb{R} \setminus \{0\}$. If at each continuity point we have $\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$, then convergence in distribution holds. Let $x \in \mathbb{R} \setminus \{0\}$.

  - If $x < 0$, then $F_X(x) = 0$, and for each $n \geq 1$, since $x < 0 < \frac{1}{n}$, then $F_{X_n}(x) = 0$; it follows that $\lim_{n\to\infty} F_{X_n}(x) = 0 = F_X(x)$.
  - If $x > 0$, then $F_X(x) = 1$, and by the Archimedean property $\exists N \in \mathbb{Z}^+, N > \frac{1}{x}$, so $\frac{1}{N} < x$, so for each $n \in \mathbb{Z}^+$ such that $n > N$ so that $\frac{1}{n} < \frac{1}{N} < x$, we have $F_{X_n}(x) = 1$; it follows that $\lim_{n\to\infty} F_{X_n}(x) = 1 = F_X(x)$.

  Therefore, $X_n \xrightarrow{d} X$.

- **Almost sure convergence:** Define

$$\begin{aligned} &\Omega = \text{set of all possible outcomes}, \\ &\Omega_0 = \{\omega \in \Omega : X(\omega) \neq 0\}, \\ &\Omega_n = \{w \in \Omega : X_n(\omega) \neq 1/n\} \ (\forall n \geq 1), \\ &\Omega' = \bigcup_{n=0}^{\infty} \Omega_n. \end{aligned}$$

  We know for each $n \geq 0$ that $P(\Omega_n) = 0$. By Boole's inequality, $P(\Omega') \leq \sum_{n=0}^{\infty} P(\Omega_n) = 0$. Let $\Omega'' = \left\{\omega \in \Omega : X(\omega) = 0 \text{ and } \forall n \in \mathbb{Z}^+, X_n(\omega) = \frac{1}{n}\right\}$.

  Observe that $\Omega'' = \Omega \setminus \Omega'$, so $P(\Omega'') = 1 - P(\Omega') = 1$.

  Now, for each $\omega \in \Omega''$, we have $\lim_{n\to\infty} X_n(\omega) = \lim_{n\to\infty} \frac{1}{n} = 0 = X(\omega)$. Therefore, $P\{\lim_{n\to\infty} X_n = X\} \geq P(\Omega'') = 1$. Therefore, $X_n \xrightarrow{a.s.} X$.

- **Convergence in probability:** Since we already have almost sure convergence, then because convergence in probability is weaker, it holds as well that $X_n \xrightarrow{p} X$.

Answer: $\boxed{\textbf{Yes, almost surely}}$. □

Remark 8.16. *In Exercise* (8.15)*, proving convergence in distribution is not necessary since it follows from almost sure convergence.*

Remark 8.17. *Along the way in solving Exercise* (8.15)*, we used Boole's inequality to prove the following:*

A countable union of sets with probability zero has probability zero.

Exercise 8.18 (Convergence in probability is preserved under continuous transformation.)**.** *Prove that if h is a continuous function, then*

$$X_n \xrightarrow{p} X \implies h(X_n) \xrightarrow{p} h(X).$$

Solution. Let $\epsilon > 0$. We prove that

$$P\{|h(X_n) - h(x)|\} \geq \epsilon\} \to 0 \text{ as } n \to \infty.$$

Equivalently, we prove that

$$P\{|h(X_n) - h(x)|\} < \epsilon\} \to 1 \text{ as } n \to \infty.$$

By continuity of h, $\exists \delta > 0$ such that $\forall z, |z - x| < \delta \implies |h(z) - h(x)| < \epsilon$. Consequently, $P\{|h(X_n) - h(x)|\} < \epsilon \mid |X_n - X| < \delta\} = 1$.

By the law of total probability,

$$\begin{aligned} P\{|h(X_n) - h(x)|\} < \epsilon\} &= P\{|h(X_n) - h(x)|\} < \epsilon \mid |X_n - X| < \delta\} P\{|X_n - X| < \delta\} \\ &\quad + P\{|h(X_n) - h(x)|\} < \epsilon \mid |X_n - X| \geq \delta\} P\{|X_n - X| \geq \delta\} \\ &\geq P\{|h(X_n) - h(x)|\} < \epsilon \mid |X_n - X| < \delta\} P\{|X_n - X| < \delta\} \\ &= P\{|X_n - X| < \delta\} \to 1 \text{ as } n \to \infty. \end{aligned}$$

□

Exercise 8.19 (Convergence in distribution is preserved under continuous transformation.)**.** *Prove that if h is a continuous function, then*

$$X_n \xrightarrow{d} X \implies h(X_n) \xrightarrow{d} h(X).$$

Lemma 8.20 (A Portmanteau result)**.**

$$X_n \xrightarrow{d} X \iff \forall g \textit{ continuous and bounded, } \lim_{n\to\infty} \mathrm{E}\left[g(X_n)\right] = \mathrm{E}\left[g(X)\right]$$

Proof. ADVANCED! REQUIRES MEASURE THEORY. □

Solution. In this proof, we require that Lemma (8.20), whose proof is out of scope, holds.

1. Assume $X_n \xrightarrow{d} X$ and h is continuous.
2. Let g be continuous and bounded.
3. Observe that the function $g \circ h$ is continuous and bounded.
4. By the Portmanteau result in Lemma (8.20), since $X_n \xrightarrow{d} X$,
$$\lim_{n\to\infty} \mathrm{E}\left[(g \circ h)(X_n)\right] = \mathrm{E}\left[(g \circ h)(X)\right].$$
5. Equivalently,
$$\lim_{n\to\infty} \mathrm{E}\left[g(h(X_n))\right] = \mathrm{E}\left[g(h(X))\right]$$
6. Since g is an arbitrary continuous and bounded function, it follows from (5), by the Portmanteau result in the lemma, that $h(X_n) \xrightarrow{d} h(X)$.

□

Exercise 8.21 (Almost sure convergence is preserved under continuous transformation.)**.** *Prove that if h is a continuous function, then*

$$X_n \xrightarrow{a.s.} X \implies h(X_n) \xrightarrow{a.s.} h(X).$$

Solution. This proof is easy. Assume $X_n \xrightarrow{a.s.} X$ and h is continuous. Thus, for some set Ω' of possible outcomes such that $P(\Omega)' = 1$, we have $\forall \omega \in \Omega', \lim_{n\to\infty} X_n(\omega) = X(\omega)$. By continuity, $\forall \omega \in \Omega'$.

$$\lim_{n\to\infty} h(X_n)(\omega) = \lim_{n\to\infty} h(X_n(\omega)) = h(\lim_{n\to\infty} X_n(\omega)) = h(X(\omega)) = h(X)(\omega).$$

Therefore, $h(X_n) \xrightarrow{a.s.} h(X)$. □

Exercise 8.22 (Partial means of i.i.d unit uniform variables converge in probability to 1/2.)**.** *Let $X_1, X_2, \ldots$ be a sequence of iid $Uniform(0,1)$ random variables. Let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$. Prove that $\bar{X}_n \xrightarrow{p} \frac{1}{2}$.*

Solution. Let $\epsilon > 0$. We prove $\bar{X}_n \to \frac{1}{2}$ using the definition. That is, we prove

$$P\left\{\left|\bar{X}_n - \frac{1}{2}\right| \geq \epsilon\right\} = P\left\{\left|n\bar{X}_n - \frac{n}{2}\right| \geq n\epsilon\right\} \to 0 \text{ as } n \to \infty.$$

Note that

$$\mathrm{E}\left[n\bar{X}_n\right] = \sum_i \mathrm{E}\left[X_i\right] = \frac{n}{2}, \ \mathrm{Var}\left(n\bar{X}_n\right) = \sum_i \mathrm{Var}\left(X_i\right) = n\frac{(1-0)^2}{12} = \frac{n}{12}.$$

Therefore, by Chebyshev's inequality, we obtain

$$P\left\{\left|n\bar{X}_n - \frac{n}{2}\right| \geq n\epsilon\right\} \leq \frac{n/12}{n^2\epsilon^2} = \frac{1}{n}(12\epsilon^2)^{-1} \to 0 \text{ as } n \to \infty.$$

□

8.3 Laws of Large Numbers and the Central Limit Theorem

Exercise 8.23 (WLLN)**.** *State and prove the weak law of large numbers. What convergence mode is involved?*

Solution. The weak law of large numbers tells us that sample mean converges in probability to population mean.

- **Statement:** Let $X_1, X_2, \ldots$ be a sequence of i.i.d random variables with finite mean μ and finite variance σ^2. Let $\bar{X}_n = \frac{1}{n}\sum_{k=1}^n X_k$ for each $n \geq 1$. Then, $\bar{X}_n \xrightarrow{p} \mu$. That is, $\forall \epsilon > 0$,

$$P\left\{\left|\bar{X}_n - \mu\right| \geq \epsilon\right\} \to 0 \text{ as } n \to \infty.$$

- **Proof:** Note

$$\begin{aligned}
\mathrm{E}\left[n\bar{X}_n\right] &= \mathrm{E}\left[\sum_{k=1}^n X_k\right] = \sum_{k=1}^n \mathrm{E}\left[X_k\right] = \sum_{k=1}^n \mu = n\mu, \\
\mathrm{Var}\left(n\bar{X}_n\right) &= \mathrm{Var}\left(\sum_{k=1}^n X_k\right) = \sum_{k=1}^n \mathrm{Var}\left(X_k\right) \ \text{(by independence)} \\
&= \sum_{k=1}^n \sigma^2 = n\sigma^2.
\end{aligned}$$

  Thus, by Chebyshev's inequality,

$$P\left\{\left|\bar{X}_n - \mu\right| \geq \epsilon\right\} = P\left\{\left|n\bar{X}_n - n\mu\right| \geq n\epsilon\right\} \leq \frac{n\sigma^2}{n^2\epsilon^2} = \frac{1}{n}\frac{\sigma^2}{\epsilon^2} \to 0 \text{ as } n \to \infty.$$

□

Exercise 8.24 (CLT)**.** *State and prove the central limit theorem. What convergence mode is involved?*

Solution. The central limit theorem tells us that for any distribution, the sample mean centered (at the population mean) and scaled (by the sample mean standard deviation) converges in distribution to the standard normal random variable.

- **Statement:** Let $X_1, X_2, \ldots$ be a sequence of i.i.d random variables with finite mean μ and finite variance σ^2. Let $\bar{X}_n = \frac{1}{n}\sum_{k=1}^{n} X_k$. Then,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

- **Proof:** Approach: Reduce to the simple case ($\mu = 1, \sigma^2 = 1$); replace cdfs with mgfs, take natural log, then find the limit with double L'Hopital.

  First, we prove CLT when $\mu = 0$ and $\sigma^2 = 1$. Thus, we want

$$\sqrt{n}\bar{X}_n = \sum_{i=1}^{n} \frac{X_i}{\sqrt{n}} \xrightarrow{d} N(0,1).$$

  1. Instead of proving

$$\lim_{n\to\infty} F_{\sqrt{n}\bar{X}_n}(x) = \Phi(x),$$

     because mgfs uniquely determine cdfs, it suffices to prove

$$\lim_{n\to\infty} M_{\sqrt{n}\bar{X}_n}(t) = M_{N(0,1)}(t) = e^{\frac{t^2}{2}}.$$

  2. Due to the iid condition,

$$\begin{aligned} M_{\sqrt{n}\bar{X}_n}(t) &= M_{\sum_{i=1}^{n} \frac{X_i}{\sqrt{n}}}(t) \\ &= [M_{\frac{X_1}{\sqrt{n}}}(t)]^n. \end{aligned}$$

  3. Define

$$L(t) = \ln(M_{\frac{X_1}{\sqrt{n}}}(t)).$$

     Now, instead of proving,

$$\lim_{n\to\infty} [M_{\frac{X_1}{\sqrt{n}}}(t)]^n = e^{t^2/2},$$

     by continuity of the natural log, it suffices to prove

$$\lim_{n\to\infty} nL(t) = t^2/2.$$

  4. Note

$$\begin{aligned} M_{\frac{X_1}{\sqrt{n}}}(t) &= \mathrm{E}\left[e^{t\frac{X_1}{\sqrt{n}}}\right] \\ &= \mathrm{E}\left[e^{\frac{t}{\sqrt{n}}X_1}\right] \\ &= M_{X_1}(t/\sqrt{n}). \end{aligned}$$

  5. Now,

$$\begin{aligned} \lim_{n\to\infty} nL(t) &= \frac{L(t)}{n^{-1}} \\ &= \frac{\ln(M_{X_1}(t/\sqrt{n}))}{n^{-1}}. \end{aligned}$$

     In the denominator, $n^{-1} \to 0$. In the numerator,

$$M_{X_1}(t/\sqrt{n}) \to M_{X_1}(0) = 1,$$

     so

$$\ln(M_{X_1}(t/\sqrt{n})) \to \ln(1) = 0.$$

     Therefore, the limit is in the form $0/0$, hence L'Hopital applies.

6. In the numerator,

$$\begin{aligned}\frac{d}{dn}\ln(M_{X_1}(t/\sqrt{n})) &= \frac{\frac{d}{dn}M_{X_1}(t/\sqrt{n})}{M_{X_1}(t/\sqrt{n})} \\ &= \frac{M'_{X_1}(t/\sqrt{n})\frac{d}{dn}(tn^{-1/2})}{M_{X_1}(t/\sqrt{n})} \\ &= \frac{(-t/2)n^{-3/2}M'_{X_1}(t/\sqrt{n})}{M_{X_1}(t/\sqrt{n})}.\end{aligned}$$

7. By L'Hopital,

$$\begin{aligned}\lim_{n\to\infty} nL(t) &= \lim_{n\to\infty}\frac{(-t/2)n^{-3/2}M'_{X_1}(t/\sqrt{n})}{-n^{-2}M_{X_1}(t/\sqrt{n})} \\ &= \frac{t}{2}\lim_{n\to\infty}\frac{M'_{X_1}(t/\sqrt{n})}{n^{-1/2}M_{X_1}(t/\sqrt{n})}.\end{aligned}$$

8. In the numerator,

$$M'_{X_1}(t/\sqrt{n}) \to M'_{X_1}(0) = \mathrm{E}\left[X_1\right] = \mu = 0.$$

   In the denominator, since

$$n^{-1/2} \to 0 \quad \text{and} \quad M_{X_1}(t/\sqrt{n}) \to 1,$$

   the denominator goes 0. Therefore, the limit is in the form 0/0, hence L'Hopital applies again.

9. By L'Hopital,

$$\begin{aligned}\lim_{n\to\infty} nL(t) &= \frac{t}{2}\frac{-\frac{t}{2}n^{-\frac{3}{2}}M''_{X_1}(t/\sqrt{n})}{-\frac{1}{2}n^{-\frac{3}{2}}M_{X_1}(t/\sqrt{n}) + n^{-\frac{1}{2}}M'_{X_1}(t/\sqrt{n})(-\frac{t}{2}n^{-\frac{3}{2}})} \\ &= \frac{t^2}{2}\lim_{n\to\infty}\frac{M''_{X_1}(t/\sqrt{n})}{M_{X_1}(t/\sqrt{n}) + tn^{-\frac{1}{2}}M'_{X_1}(t/\sqrt{n})}.\end{aligned}$$

10. In the numerator,

$$M''_{X_1}(t/\sqrt{n}) \to M''_{X_1}(0) = \mathrm{E}\left[X^2\right] = \sigma^2 + \mu^2 = 1.$$

   In the denominator,

$$M_{X_1}(t/\sqrt{n}) \to 1 \quad \text{and} \quad tn^{-\frac{1}{2}}M'_{X_1}(t/\sqrt{n}) \to 0,$$

   so the denominator goes to 1. Therefore, the limit is 1.

11. Therefore,

$$\lim_{n\to\infty} nL(t) = t^2/2,$$

   as desired.

As for the general case when $\mu \neq 0$ or $\sigma^2 \neq 1$:

1. Let

$$X_i^* = \frac{X_i - \mu}{\sigma}$$

   for each i; note that

$$\mathrm{E}\left[X_i^*\right] = 0, \quad \mathrm{Var}\left(X_i^*\right) = 1.$$

2. Let

$$\bar{X}_n^* = \frac{1}{n}\sum_{i=1}^{n}\frac{X_i - \mu}{\sigma}.$$

3. Note that

$$\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) = \sqrt{n}\bar{X}_n^*.$$

4. Following the proof for the simple case,

$$\sqrt{n}\bar{X}_n^* \xrightarrow{d} N(0,1),$$

as desired.

□

Remark 8.25. *An equivalent form of the CLT is*

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N\left(0, \sigma^2\right).$$

Exercise 8.26 (SLLN)**.** *State and prove the strong law of large numbers. What convergence mode is involved?*

Solution. The strong law of large numbers tells us that sample mean converges almost surely to population mean.

- **Statement:** Let $X_1, X_2, \ldots$ be a sequence of iid random variables with finite mean μ and finite variance σ^2. For every $n \geq 1$, let $\bar{X}_n = \frac{1}{n}\sum_{k=1}^n X_k$. Then, for all possible outcomes ω except for a set of outcomes with probability zero,

$$P\left\{\lim_{n\to\infty} \bar{X}_n(\omega) = \mu\right\} = 1.$$

- **Proof: TBC.**

□

Exercise 8.27 (Ross' TE 8.8)**.** *Explain why a gamma distribution with parameters (t, λ) has an approximately normal distribution when t is large.*

Solution. Let $X \sim Gamma(t, \lambda)$. Note that we can write X as $X = \sum_{i=1}^t X_i = t\left(\frac{1}{t}\sum_{i=1}^t X_i\right) = t\bar{X}_t$ where X_i's are iid $Exp(\lambda)$. Let $\mu = \mathrm{E}\left[X_1\right] = 1/\lambda$, $\sigma^2 = \mathrm{Var}\left(X_1\right) = 1/\lambda^2$. By CLT, $\bar{X}_t \xrightarrow{d} N(\mu, \sigma^2/t)$, so $t\bar{X}_t \xrightarrow{d} tN(\mu, \sigma^2/t) = N(t\mu, t\sigma^2) = N(t/\lambda, t/\lambda^2)$. □

8.4 Slutsky's Theorem

Exercise 8.28 (Slutsky's theorem)**.** *State and prove Slutsky's theorem.*

Solution. We state and prove.

Statement: Let $(X_n)_{n\geq 1}$ and $(Y_n)_{n\geq 1}$ be two sequences of random variables. Let X be a random variable and c be a constant. Prove the following:

If

$$X_n \xrightarrow{d} X \text{ and } Y_n \xrightarrow{p} c,$$

then

1. $X_n + Y_n \xrightarrow{d} X + c$
2. $X_n Y_n \xrightarrow{d} cX$
3. $X_n / Y_n \xrightarrow{d} X/c$ if $c \neq 0$.

Proof: TBC. □

Exercise 8.29. *Let $X_1, X_2, \ldots$ be a sequence of iid random variables with finite mean μ and finite variance σ^2. Let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$. Find the asymptotic distributions of*

1. $\sqrt{n}(\bar{X}_n - \mu) + \bar{X}_n$,

2. $\bar{X}_n \cdot \sqrt{n}(\bar{X}_n - \mu)$.

Solution. By CLT,

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

We know (by WLLN) that

$$\bar{X}_n \xrightarrow{p} \mu.$$

Therefore, by Slutsky's theorem,

$$\sqrt{n}(\bar{X}_n - \mu) + \bar{X}_n \xrightarrow{d} N(0, \sigma^2) + \mu = N(\mu, \sigma^2),$$
$$\bar{X}_n \cdot \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mu \cdot N(0, \sigma^2) = N(0, \mu^2\sigma^2).$$

□

Exercise 8.30. *Let $X_1, X_2, \ldots$ be a sequence of iid random variables. We know by CLT that*

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1).$$

Suppose we don't know the population standard deviation σ, and we consider to use the sample standard deviation S_n instead. What is the asymptotic distribuiton in this case?

Solution. By CLT, the numerator converges in probability to $N(0, \sigma^2)$. Knowing that the sample standard deviation converges in probability to σ, if $\sigma \neq 0$, then by Slutsky's

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \xrightarrow{d} \frac{N(0, \sigma^2)}{\sigma} = N(0, 1).$$

Therefore, asymptotically, we can use the sample standard deviation instead of the population standard deviation for the same asymptotic behavior. □

Exercise 8.31. *Let $X_1, X_2, \ldots$ be an arbitrary sequence of iid random variables. Let $\bar{X}_n$ denote the sample mean. Find the asymptotic distribution of*

$$T_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{1 + \bar{X}_n^2}.$$

Solution. Here is the proof outline:

1. By CLT,
$$A_n = \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

2. Knowing that the sample mean is a consistent estimator of the population mean,
$$\bar{X}_n \xrightarrow{p} \mu.$$

3. Because convergence in probability is preserved under continuous mappings,
$$B_n = 1 + \bar{X}_n^2 \xrightarrow{p} 1 + \mu^2 \neq 0.$$

4. Therefore, by Slutsky's,
$$T_n = \frac{A_n}{B_n} \xrightarrow{p} \frac{N(0, \sigma^2)}{1 + \mu^2} = N\left(0, \frac{\sigma^2}{(1 + \mu^2)^2}\right)$$

□

Exercise 8.32. *Let $X_1, X_2, \ldots$ be a sequence of iid random variables. Find the asymptotic distribution of T_n:*

$$T_n = \frac{\textit{estimation error of mean at CLT scale}}{\textit{root mean square (RMS) of observations}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sqrt{\frac{1}{n}\sum_{i=1}^n X_i^2}}.$$

Solution. By CLT,

$$A_n = \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

Note

$$\begin{aligned}
\sum_{i=1}^{n}(X_i - \bar{X}_n)^2 &= \sum(X_i^2 - 2X_i\bar{X}_n + \bar{X}_n^2) = \sum X_i^2 - 2\bar{X}_n \sum X_i + n\bar{X}_n^2 \\
&= \sum X_i^2 - 2\bar{X}_n(n\bar{X}_n) + n\bar{X}_n^2 = \sum X_i^2 - n\bar{X}_n^2 \\
\implies \frac{1}{n}\sum X_i^2 &= \frac{1}{n}\sum(X_i - \bar{X}_n)^2 + \bar{X}_n^2 = \frac{n-1}{n}S_n^2 + \bar{X}_n^2.
\end{aligned}$$

Note:

1. $\bar{X}_n \xrightarrow{p} \mu \implies \bar{X}_n^2 \xrightarrow{p} \mu^2$.
2. Also, we know $S_n^2 \xrightarrow{p} \sigma^2$. Since $\lim_{n\to\infty} \frac{n-1}{n} = 1$, then $\frac{n-1}{n} \xrightarrow{p} 1$. Therefore, $\frac{n-1}{n}S_n^2 \xrightarrow{p} 1 \cdot \sigma^2 = \sigma^2$.
3. Therefore, $\frac{n-1}{n}S_n^2 + \bar{X}_n^2 \xrightarrow{p} \sigma^2 + \mu^2$
4. Therefore,

$$B_n = \sqrt{\frac{1}{n}\sum X_i^2} \xrightarrow{p} \sqrt{\sigma^2 + \mu^2}.$$

   Therefore, by Slutsky's,

$$T_n = \frac{A_n}{B_n} \xrightarrow{d} \frac{N(0, \sigma^2)}{\sqrt{\sigma^2 + \mu^2}} = N\left(0, \frac{\sigma^2}{\sigma^2 + \mu^2}\right).$$

□

Remark 8.33. *In solving the previous problem, we used the conservation of the convergence in probability of* <u>*random vectors*</u> *:*

$$X_n \xrightarrow{p} X \wedge Y_n \xrightarrow{p} Y \implies (X_n, Y_n) \xrightarrow{p} (X, Y) \implies \forall g \text{ continuous, } g(X_n, Y_n) \xrightarrow{p} g(X, Y).$$

Thus, we justified

$$\begin{aligned}
X_n \xrightarrow{p} Y \wedge Y_n \xrightarrow{p} Y &\implies X_n + Y_n \xrightarrow{p} X + Y, \\
X_n \xrightarrow{p} Y \wedge Y_n \xrightarrow{p} Y &\implies X_nY_n \xrightarrow{p} XY.
\end{aligned}$$